

BOSCH  ROBERT BOSCH (AUSTRALIA) PTY. LTD. A.B.N. 48 004 315 628	ENGINEERING SPECIFICATION FOA Barra Body Electronics Module High and Low Series	F005 VP 0058 Page 1 of 110
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	Department	Date	Signature
Drawn	EGE3-JEV	23/9/02	
Checked	EGE3-MS&RW	23/9/02	 P.W.
Approved	EGE3	23/9/02	
Approved	STF	23/9/02	

1. ALTERATION LIST

Issue	ECR Number	Alteration	By	Valid From	Checked
8	V001119	Sect 3: Add comment: "This specification shall be read in conjunction with the Engineering Specification for the FoA Barra Alarm Body Electronics Module High and Low Series, F005VP0354. Any modifications to the Engineering Specification for the FoA Barra Body Electronics Module High and Low Series, F005VP0058 will apply to F005VP0354 by default. F005VP0354 only details additional requirements specific to the FoA Barra Alarm Body Electronics Module." Sect 7.2.5: Change test method from SDS EC-0082 to using RBAU setup which involves a Testometric force gauge machine to determine the rate of movement on the all 5 connectors. Change the Insertion force from 70N to 130N with the comment that "Insertion force applied is without secondary locking feature depressed and with positive audible engagement feedback". Sect 8.15.2: Change Factory settings for Auto Headlamp turn on frequency from 340Hz to 300Hz and changed note 3 to the following: "Setting selections are (0x07, 0x55) after the final tester at RBAU and (0x04, 0x00) if read by WDS using \$22 command on PID \$7141at FoA. This corresponds to input-captured values of 0x1A1 for the headlamps to turn ON, and 0x341 to turn OFF (highlighted in the above table in grey)." Sect 8.12.2.2: Change after table, WriteDataByCommon Identifier to Write MemoryBlock service \$3B instead of \$2E. Appendix 10.5.3: State Encoded Diagnostic PID, E115 for Last received transmitter ID Code (TIC) Reference Number, add note that \$2F command is not supported for Barra BEM/Low Series. Appendix 10.5.5: ASCII and BCD Diagnostic PID, E231 for Module Serial Number 1 (BCD) change Data Bit State Description: Year,Month,Month,Digit1,Digit2,Digit3,Digit4,Digit5 to Digit1,Digit2,Digit3,Digit4,Digit5,Digit6,Digit7,Digit8.	EGE3-JEV	23/9/02	EGE3-MS
7	V001062	Updated to reflect Version #13 software to support mask Sect 8.3: Change Nominal A/D realised value for Detent 4 from \$79 to \$71, Detent 5 from \$7A to \$72. Add Note "Since the mechanical design of the lever combination switch has detents 4 and 5 next to each other, there is a risk that detents 4 and 5 will not be distinguishable in production eg detent 5 may always appear as detent 4". Sect 8.7: Update to include Software Requirement Specification of Power Window Control (Issue 3 dated 2/8/02). In summary, power window supply will not activate if driver door is manually unlocked, previously would activate. Sect 8.13.8: Update to include Software Requirement Specification of Intermediate Plant Mode (Issue 3 dated 18/7/02). In summary Intermediate Plant Mode can only be entered via WDS if BEM is in Plant Mode.	EGE3-JEV	7/8/02	EGE3-JEV

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6	V000979	Updated to reflect Version #12 software to support mask Sect 8.9: Add the following "The data byte transmitted is a result of a vote of two data bytes out of the last three. At least two data bytes that are within 5°C of each other will be accepted with the last data byte of the two transmitted via CAN accordingly. If however, all three data bytes are outside 5°C of each other, than at worst case the previous correct data will be used. For example, upon power up the first data byte will be \$FF" Sect 8.12.4: Change default from Two Stage Central RF Unlock to one stage RF Central Unlock as programmed into Eeprom. Sect 8.20: Added complete list of configurable options detailing which function is enabled/disabled at factory setting whereas previously only a few were defined (eg, 1 stage/2 stage, Taxi Mode, Police Mode, etc) Sect 10.5.3: CAN message 726, State Encoded PID C901 change comment for Battery Saver from 30 minutes to 15 minutes	EGE3-JEV	19/6/02	EGE3-JEV
5	V000948	Updated to reflect Version #11 software to support mask Sect 6.1.1&6.1.2: Add note: " <u>Danger:</u> Do not operate or store the Barra Remote Key above 80°C. The Lithium Battery contains flammable materials such as Lithium and organic solvent which could cause it to ignite, explode or become damaged." Sect 8.3: Update Wiper Dwell Pot settings to include dwell potentiometer timing when shorted. Also adjusted the Nominal A/D realised value for all Detent positions to provide as much tolerance as possible relative to the actual potentiometer resistance including its own tolerance. Sect 8.7: Remove T41 and T42 and replace with T09 and T10 Sect 8.9: Change data for Cabin Temperature at 24.5°C from expected CAN data of 87D to 7D. Change A/D data from 62.5°C-75°C with correct values. Sect 8.10: Solar Sensor table clarified to clearly define out of range data as being above A/D of \$F4 and below A/D of \$09. Equation updated. Sect 8.12.3: Update Software Requirement of Central Door Locking functions (Issue 6 dated 29/4/02) by updating manual central locking fuction such that when pressing soldier button down while driver door is open, all doors unlock rather than drivers door unlock. Added the following note with regards to doors unlock whenever a crash bit is detected via CAN "All door ajar switches may return erroneous information during & after a crash. Once a crash unlock has occurred, then subsequent door switch inputs shall be ignored including the manual Central Locking switch bit via CAN, until the next ignition switch cycle occurs from off to out of Ignition and Accessories off state." Sect 8.12.4: The following statement has been removed "The software includes measures to reduce wear on the driver's door lock motor by not unlocking it when it is already unlocked." due to all doors unlocked if drivers door is already unlocked. Sect 8.15: Update Software Requirement of Auto Headlamps (Issue 11 dated 19/4/02) by Headlamp OFF diagnostics not being performed at Ignition OFF and changing Headlamp buzzer to Headlamp buzzer bit since BEM transfers data via CAN to Cluster (Cluster drives the buzzer). Sect 8.16.6: Remove the following statement in for Dealer DST Key Teach mode since it is not valid since WDS and CAN are operational in both Ignition off and on positions: "This statement is not valid if Diagnostics is operational only during Ignition ON." Add the statement, "Dealer teach mode can only be activated during a time window of T49 just after Ignition is turned to the off position". Sect 8.21: Change T06, Front wiper motor delay after washer motor on from 0.45sec to 0.48sec. Sect 10.5.1: For PID 7123 added note that Ignition switch is in Accessories or Ignition position Sect 10.5.2: For PID A442 all states added note that Ignition switch is in Accessories or Ignition position and additional note that that reading is independent of Intermittent switch position. Also provided more detail that for state 01 that wiper switch in off , low speed or high speed positions	EGE3-JEV	21/5/2002	EGE3-JEV

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		Sect 10.5.3: For PID C125 state 11, added note that "The \$2E command is only valid if sent within 2 minutes of ignition switch being turned to the off position." Sect 10.5.4: For PID A452 and A453 added note that Ignition switch is in Accessories or Ignition position Sect 10.6: For DTC's 9250, 9251, 9253, 9259, 9261, E197 added note that Ignition switch is in the Ignition position. For DTC's 9348, 9349, 9446, 9458, A891 added note that Ignition switch is in Accessories or Ignition position. For DTC 9693 added note that headlamp switch must be in Auto position. For DTC's A072, A073, A074, A075 added note "when BEM is sending seat memory data". For DTC's A929, A930 added note "when BEM is challenging the transponder" Sect 10.7: Calibration Offset changed from ± 7 to ± 6. Figure 1A : Update Twilight frequencies from Dark 390 Hz to 370Hz, Bright 50Hz to 65Hz.. Figure 1A&1B: Add note "<u>Danger</u>: Do not operate Barra Remote Key above 80°C. The Lithium Battery contains flammable materials that may cause it to ignite or explode".			
4	V000889	Updated to reflect Version #10 software to support 'O' series level. Changes to sections: 3: Add definition SWS being Sumitomo Wiring Systems 3.3.1: FoA reference specifications list updated to latest level by changing: - the Network Communication Specification Area Network (500k) application specific requirements for Barra Vehicle Communication Networks from Draft 1.4 dated 18/12/01 to Version 1.4 dated 12/2/02. - Barra CAN message list from 3.0 dated 18/1/02 to 4.0 dated 27/2/02. 4.2: Added Note defining the expected operational range for all functions (9 to 16V) except immobilisation (8 to 24V). 7.6.1: Add to the Narrowband Electric Field, a concession note from FoA/Rod Heubner that "Radiated Emission Specification exceedances at 64MHz and 256MHz are not co-incident with any land mobile and are accepted at $<16\text{dBuV}$ & $<25\text{dBuV}$ respectively." 8.11: Update Software Requirement of Courtesy Lamp, Police Mode and Taxi Mode (Issue 7 dated 28/3/02) 8.12.2.2: Any references to E265 must be identified as a Proposal. 8.12.4: For Remote passenger doors unlock when holding the remote Door Unlock button after the drivers door has unlocked for <u>2 sec</u> with two stage unlocking active. Update table to include the 2 sec. 8.14: Update Software Requirement of Memory Seat Module (Issue 4 dated 28/3/02) 8.15: Update Software Requirement of Auto Headlamps (Issue 10 dated 28/3/02) 8.21: Timing parameters changed: - T05, Front Washer motor on time changed from 1.2sec $\pm 20\%$ to 1.25sec $\pm 10\%$. - T06, Front Wiper Motor Delay after washer motor on changed from 0.4sec $\pm 5\%$ to 0.45sec $\pm 15\%$ - T35, changed from Auto Headlamp relay coil on delay time to delayed turn on time - T36, changed from Auto Headlamp relay coil off delay time to delayed turn off time - T37, changed from Auto Headlamp Ignition off turn off time to Auto headlamp delayed turn off time with Headlamps on, Ign off, driver's door open - T38, changed from Auto Headlamp Ignition off turn off time (driver's door opened) to Auto headlamp delayed turn off time with Headlamps on, Ign off, driver's door closed. Appendix 10.5.4: For PID Number B902, Bit0 remove Message Number #6F6 since this is not a periodic command. Appendix 11: Update CAN Message List (Issue 4.0 dated 27/2/02)	EGE3-JEV	4/4/02	EGE3-JEV

Issue	ECR Number	Alteration	By	Valid From	Checked
3	-	Complete update for additional CP level functions. Changes to sections: 3.3: FoA reference specifications list updated to latest level. 3.4: Added Cabin temperature sensor Fan Motor Trigger (BBEM/H). 3.5& 3.6: Block diagram updated to reflect Analog Clock Negative Feed, Boot/Tailgate relocated to be scanned continuously, CTS fan to be microprocessor driven rather than direct from Ignition, CAN messages updated. 4.2: Updated table to reflect new connector pin number designations and removed Battery voltage sense specification. 5.1: Updated with Analog Clock negative supply added and CTS fan drive removed. Adjusted current sense to fit convention, adjusted to new connector pin number designations, adjusted characteristics to suit hardware changes. 5.2: To be measured relative to analog ground return as defined. Adjusted current sense to fit convention. 5.3: Removed Battery Voltage Sense, adjusted nominal currents, adjusted table to new connector pin number designations, added Transceiver transmit data feedback input for diagnostic purposes. 5.4: Adjusted table to new connector pin number designations, adjusted nominal currents to suit hardware. 5.5: Added CTS fan motor drive characteristics and adjusted table to new connector pin number designations including adjustment to characteristics 5.6: Added Analog clock negative feed and adjusted table to new connector pin number designations including adjustment to characteristics. Adjusted the Quiescent Current to take into account RF Receiver poll rate from 16ms on 48ms off to 24ms on 42ms off which increases the overall Quiescent current of RF Receiver from 1.5mA to 2.5mA thus increasing the overall BEM Quiescent current to 4.5mA. 6.1: Test Matrix adjusted to reduce PV1 test time by locating Vibration and Thermal Shock resistance after Life Test (for future PV tests this will be relocated back to its original state). Dust test was relocated to the Mechanical test group. Insertion force testing was performed on all test parts, rather than testing an additional 24 parts per series. 7.1.4: Powered Thermal Cycle updated to reflect FoA Environmental Specification, 00.00EA-D11 adding 3 additional test days. Figure 3 was updated accordingly. 7.2.3: Locating screw maximum torque incorrect, changed from 92.1Nm to 93.1Nm. 7.3.1: Update Sinewave noise to clarify voltage requirements for additional frequencies. 8.7: Update to include Software Requirement Specification of Power Window Control (Issue 2 dated 18/9/01) 8.8: Update to include Positive supply to Analog Clock 8.9: Update to include Software Requirement Specification of Trigger for CTS (Issue 2 dated 18/10/01). Previously Calibration Factor signed value (8bit) now 4 bit (low nibble). 8.11: Update to include Software Requirement Specification of Courtesy Lamps, Police Mode and Taxi Mode (Issue 6 dated 22/1/02). 8.11.3.5: Updated to include Battery Disable effects in Accessories and Start positions. 8.12.2.2: Update to include Software Requirement of E265 functions (Issue 3 dated 3/10/01) 8.12 except 8.12.2.2: Update to include Software Requirement of Central Door Locking including Hazard Lamp Flash (Issue 5 dated 22/1/02)	EGE3-JEV	12/2/02	EGE3-MS&RW

Issue	ECR Number	Alteration	By	Valid From	Checked
3	-	8.13.8: Update to include Software Requirement of Intermediate Plant Mode (Issue 2 dated 12/9/01) 8.13: Update to include Software Requirement for RF Receiver Related Functions with 8 Remote Key Training including additional modifications such as changing the poll on time from 12ms to 16ms (Issue 2 dated 9/8/01) 8.14: Update Software Requirement for Memory Seat and Priority Key Control (Issue 3 dated 3/8/01). 8.14.1: Bit 0 is updated to include Master Key or Plant Mode 0: Update Software Requirement of Auto Headlamps (Issue 9 dated 11/9/01) 8.15.3.5: Updated the Warning Buzzer bit 54 to bit 18 as per CAN message list. References to CAN message list is being made. 8.16: Update to include Software Requirement Specification of Immobilisation including 8 transponder training (Issue 2 dated 9/8/01) 8.18: Update to include Software Requirement Specification of the CAN communication system (Issue 2 dated 11/2/02) 8.18.8: Updated CAN default strategies to include additional priority keys and Buzzer Warning. Also \$D900, DTC logging will only occur when this bit is set. 8.21: Update Timing parameters, Battery Saver timeout (normal mode), T11 from 30 mins to 15 mins; Police battery saver blackout mode switch hold time, T17 from 10 secs to Door Trigger Disable Mode switch hold time of 5 secs; Added Duration of buzzer for enabling/disabling Door Trigger Disable Mode of T55=2.5sec; Added Duration of courtesy lamp pulse of T56=0.5 secs; Added CTS trigger high duration of T57=25.4 sec; Added CTS trigger off duration of T58=0.2sec; Update Hazard lamp activation time for door lock activation, T20 from 0.8 secs to 0.5 secs; Update Hazard lamp activation time for boot release T21 from 1.6 secs to 1.0 sec; Remove Hazard lamp activation time before horn warning, T22; Update Automatic central door re-lock time from 45 secs to 40 secs; Update Time slice used fro BBEM to Seat Module communication from T33 of 0.02 secs to 0.04 sec; Update Time delay for transition T34 from 0.01 secs to 0.02 secs; Added minimum inter-frame period, T53=0.2 secs; Added delay from RK unlock to send data frame T54=0.2secs. Fig 1A & 1B: Update Test Circuit to reflect new pin numbers, analog clock negative supply, simulation of Solar/Twilight Sensor and adding Barra RF Key. Appendix 10: Update Diagnostics Part II Specification (Issue Issue 9 dated 8/2/02) Appendix 11: Update CAN Message List (Issue 3.0 dated 18/1/02)			

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2	-	Complete update for CP level. Changes to sections: 3: Removed references to AP2 level parts. 3.1: Updated special characteristics (dimensions). 3.2: Reference tables added, specific SDS references removed. Change matrix removed. 3.3: Functional overview simplified and functional classes added. 3.4, 3.5: Block diagrams updated. 4: Environmental maximum ratings added, electrical maximum ratings updated. 5.1: Fixed feed, source and returns table added. 5.2...5.5: Functions, electrical data and pin numbers updated for all sections. 5.6: Standby current now identified to individual circuits, functions updated, pin numbers added. 6: DV test information removed to control plan; notes on PV testing, statistical analysis, re-validation, lots and records simplified; commercial notes removed; PV test matrices updated; in process testing removed. 7: States and classifications updated. 7.1: Reduced life, receiver sensitivity tests removed; most tests updated to new method. 7.2: Connector insertion force, audible noise tests added; other tests updated to new method. 7.3: Over current, short circuit tests removed; all other tests updated to new method. 7.5: Conducted emissions time domain test added; frequency domain test updated to new method. 7.6: Radiated emissions test updated to new method. 8: Debouncing notes added. 8.6: Low washer fluid delay added. 8.9: Cabin temperature data now transmitted in °C (coded) with tables and graph added. 8.10: Sunload Sensor Module interface description added added including tables and graphs 8.13: Low series door ajar now as per high series. Boot ajar valid at all times. 8.14: Boot release time not stored in EEPROM. 8.16: Comment added regarding driver's door unlock activation when already unlocked. 8.22: Auto headlamp description simplified and modified using Australian Arrow Sensor 8.23: Normal operation – PCM enabling updated. 8.27: Auto headlamp settings added to configurable options table. 8.28: T25, T33 fixed, T50 added. 9: Sequence tables updated to Barra level, frequency spectrum deleted. Figures 1A, 1B: Updated to Barra level. Figures 2A...2G: All updated. Receiver sensitivity setup deleted. Transient waveforms and emission level line deleted. Figure 5: ESD diagrams updated. Figure 6: Mounting position updated. Appendix A: Diagnostic Specification Part II added Appendix B: CAN Message List 2.6 added	EGE23-JEV	16/7/01	EGE23-SC
1	-	AP2 Specification		28-11-00	

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3. GENERAL

This specification details the electrical characteristics and performance of the Robert Bosch Australia (RBAU) Barra Body Electronics Module (BBEM). The BBEM combines a transponder challenge/response immobilisation system with body electronic functions. Low series and high series variants are available to suit all levels of vehicles. The BBEM is intended to be mounted inside the passenger compartment as part of the Central Display Unit (CDU) which is to be supplied by Sumitomo Wiring Systems (SWS).

Unless otherwise specified all references to BEM in this document should be understood as referring to BBEM. High and Low Series Modules may also be identified as BEM/H and BEM/L or High and Low respectively.

The workmanship performed and parts used in the manufacture and assembly of each unit shall be equivalent to the approved samples.

Functionality and/or Durability of the BEM is guaranteed only for the loads and input characteristics that are used in the production validation process. Bosch is to be notified of any change to wiring harness, loads and/or sensors that are related to the BEM. In those cases, Bosch can advise what if any effect these changes would have on the BEM.

Connector pin numbers are identified in the wiring diagram.

Unless otherwise specified, all temperature tolerances shall be $\pm 5^{\circ}\text{C}$, the ambient temperature shall be 23°C (T_{TEST}), relative humidity shall be $45\% \pm 15\%$ and supply voltage shall be 13.5 ± 0.5 Volts DC for all tests in this specification. Currents, frequency, time, dimensions, resistance, inductance, capacitance shall be $\pm 10\%$ unless otherwise specified.

All currents are defined as follows: Positive (+) for currents into BEM
 Negative (-) for currents out of BEM

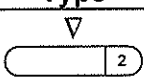
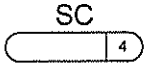
The following definitions shall apply throughout this specification:

- N/A Not applicable.
- VIL Voltage input low state.
- VIH Voltage input high state.
- VOL Voltage output low state (for solid state outputs).
- VOH Voltage output high state (for solid state outputs).
- VOUS Voltage output unenergised state (for switched outputs).
- VOES Voltage output energised state (for switched outputs).
- ∇ Critical Characteristic (Ford)
- SC Significant Characteristic (Ford)
- SDS System Design Specification (Ford)
- DS Design Specification (Ford)
- ES Engineering Specification (Ford)
- MRDB Master Reference Database (Ford)
- WDR Worldwide Design Standard (Ford)
- FPDS Ford Product Development System
- SWS Sumitomo Wiring Systems

This specification shall be read in conjunction with the Engineering Specification for the FoA Barra Alarm Body Electronics Module High and Low Series, F005VP0354. Any modifications to the Engineering Specification for the FoA Barra Body Electronics Module High and Low Series, F005VP0058 will apply to F005VP0354 by default. F005VP0354 only details additional requirements specific to the FoA Barra Alarm Body Electronics Module.

3.1 Critical and Significant Characteristics

The following list is not exhaustive and is awaiting final confirmation of all special characteristics.

Type	Characteristic	Applicability	See Also
	Power window activation time once ignition turned off and door opened (Reference T10 < 45sec).	All	Control plan, sales drawing, section 8.7
	Mounting dimensions: Pitch between mounting holes, Distances between locating pins and mounting holes	All	Control plan, sales drawing

Note:

-  Ford critical characteristic.
 Ford significant characteristic.
 RBAU legal obligation requirement.
 RBAU customer fitment requirement.

3.2 Part and Document References

Description	High Series	Low Series
RBAU sales drawing	F005 VS 0006 Sheet 1	
RBAU, SWS and Ford part numbers	F005 VS 0006 Sheet 1	
RBAU vehicle wiring diagram	F005 VS 0006 Sheet 2 & Sheet 3	
RBAU product brief	PB1600	PB1604
RBAU system FMEA	SFMEA1600	
RBAU design FMEA	DFMEA1600	
RBAU control plan	CP1600	
RBAU Barra Receiver Module sales drawing (refer for all BRFM details)	F005 VS 0007	
RBAU Barra Transceiver Ignition Lock Assembly sales drawing (refer for all BTILA details)	F005 VS 0001	
RBAU Barra Cabin Temperature Sensor sales drawing (refer for all BCTS details)	F005 CS 0003	

Note: For full SDS Compliance see Product Brief

3.3 Ford Reference Specifications

Identified Ford references are as follows. Inclusion in this list does not imply that all aspects are followed.

Spec Number	Version Number	Revision Date
Spec Description		

3.3.1 Ford CAN Specifications:

—	Version 1.4	12 February 2002
1. Network Communication Specification Controller Area Network (500k). Application Specific Requirements for Barra Vehicle Communication Networks		
ISO-11898	—	15 November 1993
2. —		
ES-XS4T-12K529-CB	2000.1	4 January 2000
3. Ford CAN specification, Vehicle Controller Area Network (CAN) Implementation Requirements		
ES-XS4T-12K529-DB	2000.1	—
4. HS-CAN – ECU Test Procedure Specification		
MS-XS4F-3458-AVA1	—	—
5. Drawing and parts list of Ford CAN Physical layer (Uses TJA1050 in lieu of 82C250)		
—	4.0	27 February 2002
6. Barra CAN message list		

3.3.2 Ford Diagnostic Specifications:

—	2001.0	16 February 2001
1. Global Diagnostic Specification Part One		
—	2001.0	9 February 2001
2. CAN Generic Diagnostic Specification		
—	2001.0	9 March 2001
3. Module Programming and Configuration Design Specification		
—	1.5	1 October 1997
4. Keyword Protocol 2000. Implementation of Diagnostic Services Recommended Practice		
—	—	11 April 2001
5. MRDB-Master Reference Database (Excel version)		
ISO 15765-2r	3.0	30 November 1999
6. Road Vehicles – Diagnostics on CAN - Part 2: Network Layer Services		
—	2001.0	9 April 2001
7. CAN Generic Diagnostic Specification – Electronic Control Unit Test Procedure Specification		

3.3.3 Ford SDS Requirements (deviations from these SDS requirements are identified in Product Brief PB1600):

CPSC: 18.04.00	8	6 December 1999
1. Generic Body Module SDS		
CPSC: 19.10.00	7	12 February 1999
2. Generic Body E/E Feature Function SDS		
CPSC: 19.02.00	4	23 March 1998
3. Remote Keyless Entry sub-system SDS		
CPSC: 19.01.00	5	5 November 1998
4. Passive Anti-Theft System SDS		

3.3.4 Ford EMC and Environmental Specifications:

ES-XW7T-1A278-AB	B	30 April 1999
1. Electrical Component EMC Requirements and Test Procedures		
WDR 00.00EA-D11	—	May 1999
2. Environmental Requirements – (Non EMC) – Vehicle Electrical and Electronic Systems		

3.3.5 Ford Miscellaneous Specifications:

WSS-99P-9999A1	—	24 January 2000
1. Substance use restrictions		
—	Release 1.3	9 August 1996
2. Remote Keyless Entry Protocol		
—	Preliminary 2	15 March 2001
3. Barra EEC/BEM Immobilisation Code Communications – Confidential		

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3.4 Functional Overview

Symbols in bold define the functional class for each function. These classes are defined in ES-XW7T-1A278-AB and shown in section 7.

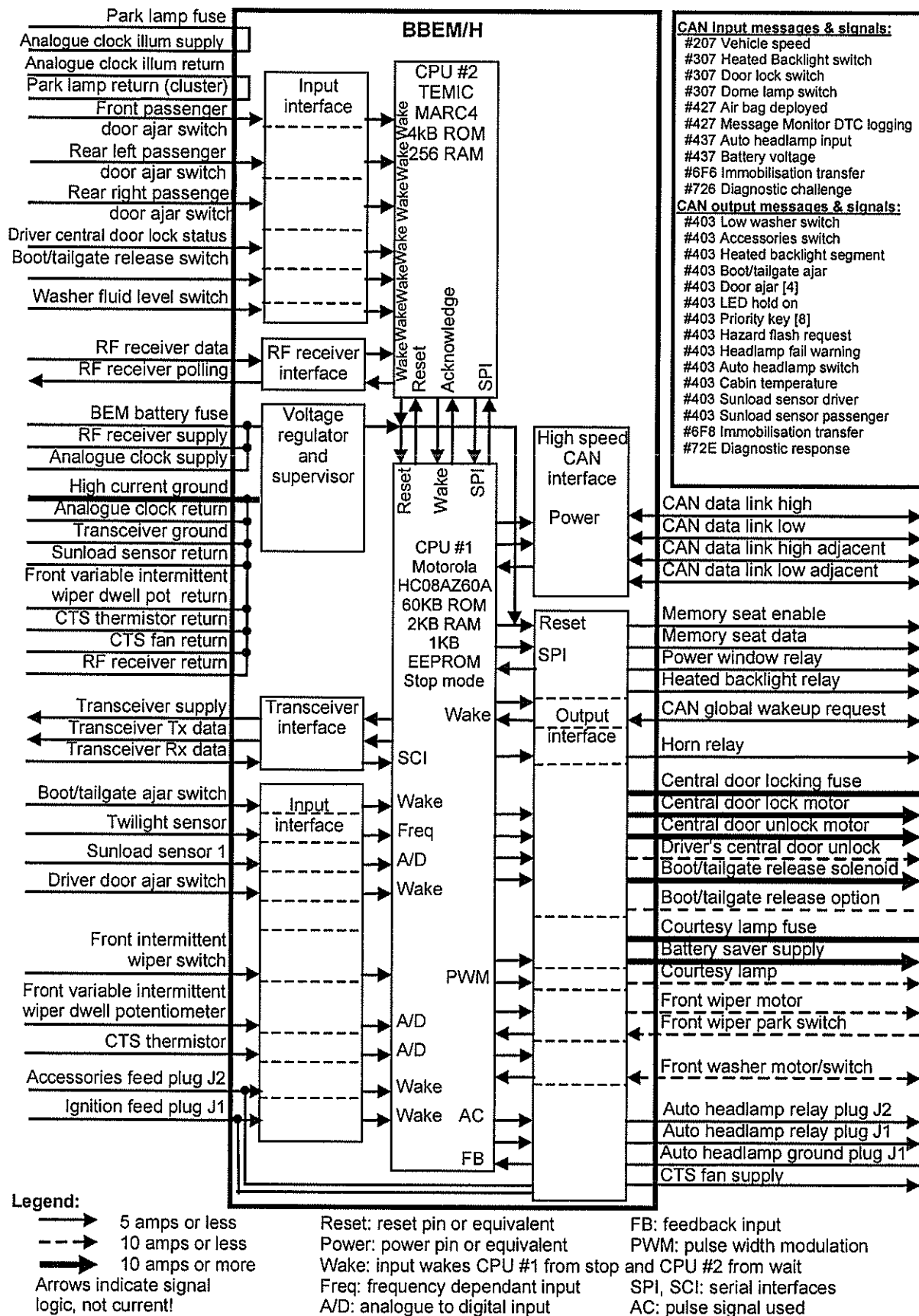
The Barra Body Electronics Module for Low Series performs the following functions:

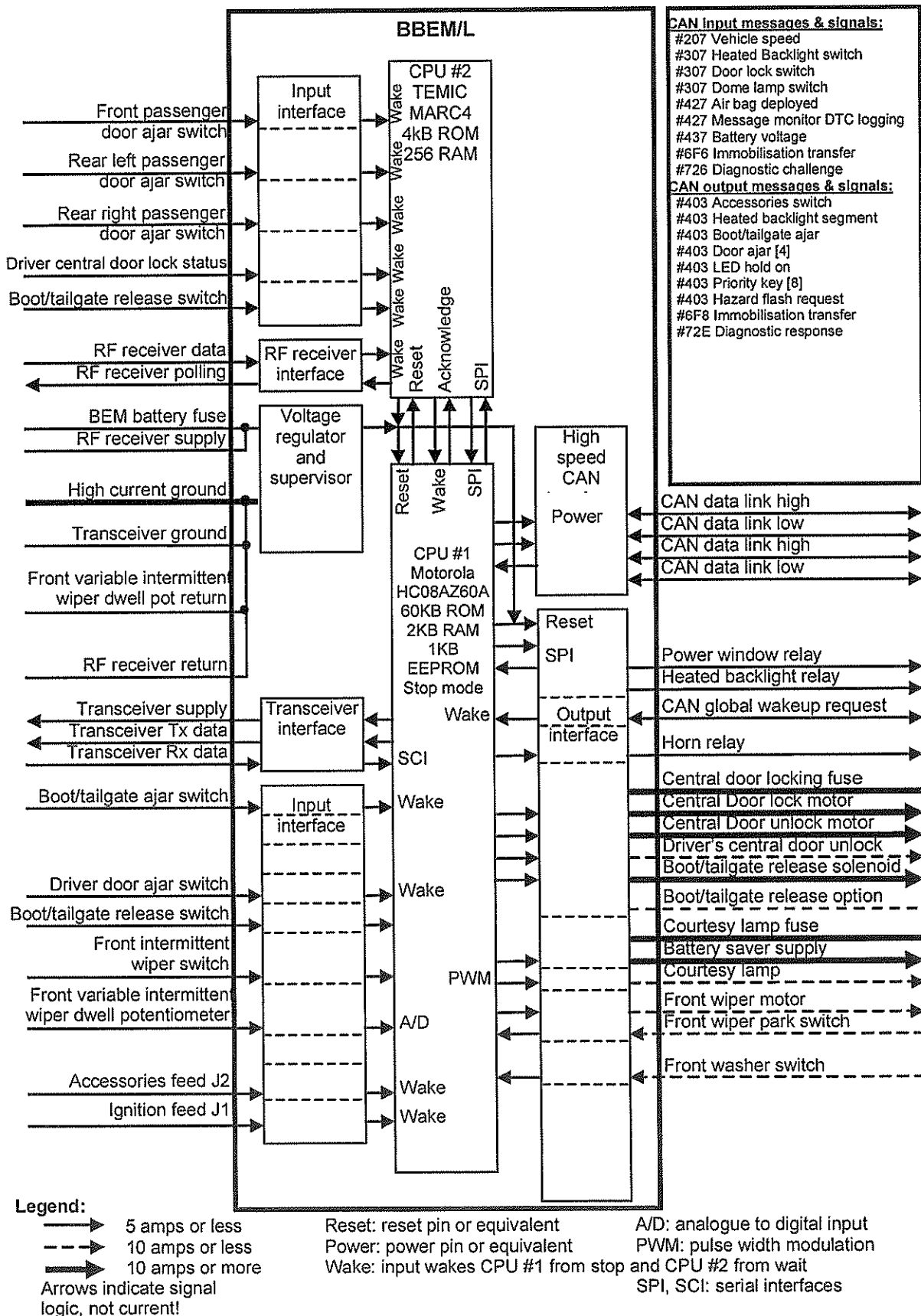
- **C (B1 for EMC)** Transponder challenge response immobilisation
- **A** Ford Shark Protocol RF receiver interface
- **A** Remote central locking with selectable one or two stage unlock and hazard acknowledgment¹
- **A** Remote control for illuminated entry
- **A** Remote boot or tailgate release with hazard acknowledgment
- **A** Door unlocked horn and hazard warning
- **A** DST Key in lock horn and hazard warning
- **A** Remote panic alarm using horn and hazards
- **A** Automatic door re-lock timer for remote entry
- **B** Manual control for dome lamp, illuminated entry delay and illuminated exit delay
- **B** Manual central locking control
- **B** Manual boot or tailgate release with security time out and speed dependant disable
- **A** Priority key signalling via CAN
- **A** Door ajar and boot ajar signalling via CAN
- **B** Battery saver for dome, boot, glovebox lamps and Audio One Hour Radio Play
- **A** Automatic dome lamp illumination triggered by ignition DST key removal or ignition off with software enable facility
- **B** Heated backlight (rear demist) with automatic time out
- **B** Front intermittent wiper dwell timer
- **A** Either variable dwell with speed dependency, or fixed dwell; selected by automatic hardware detection
- **B** Front wipe after wash timer with fixed number of wipes
- **A** Low noise relay for wiper functions
- **B** Power window supply control
- **A** Accessories line status via CAN
- **B1** 500kb/s CAN interface with bi-directional global wake-up signal
- **B** Ford diagnostic interface via CAN

The Barra Body Electronics Module for High Series performs all the above low series functions and the following additional functions:

- **C** Automatic headlamp control with twilight sensing, delayed turn-off and failsafe strategies
- **A** Front washer motor control with minimum wash time and wipe after wash
- **A** Low washer fluid indication via CAN
- **A** Analogue clock interface
- **A** Cabin temperature sensor interface and signal via CAN
- **A** Sun-load sensor interface and signal via CAN
- **A** Memory seat interface for priority key information
- **A** Cabin temperature sensor Fan Motor Trigger

¹ Hazard acknowledgements and warnings are signalled via CAN

3.5 Block Diagram – High Series


3.6 Block Diagram – Low Series


4. MAXIMUM RATINGS

4.1 Environmental

Parameter	Conditions	Minimum	Maximum	Unit
Temperature range	Operating	-20	85	°C
Temperature range	Storage	-20	105	°C
Relative humidity	—	0	95	%
Non-random vibration	Frequency range from 5Hz to 200Hz	—	4	G

4.2 Electrical

Parameter	Conditions	Voltage (V)		Pin Numbers
		Min	Max	
Operating voltage, Vs, performance as per section 5	—	9	16	BEM Battery Fuse (BBF) =J3/11 Courtesy Lamp Fuse (CLF) =J3/6 Central Door Locking Fuse (CDLF) =J5/3
Operating function, for class B, B1 and C functions (See Note)	Switched outputs may not activate	8	16	BBF, CLF, CDLF
Short term over voltage, Vov (See Note)	t ≤ 60 sec t ≤ 60 min	—	24 19	BBF, CLF, CDLF
Short term reverse battery voltage	t ≤ 60 sec	—	-14	BBF, CLF, CDLF
Inductive load switching transient	See section 7.3.2	-150	150	BBF, CLF, CDLF
Alternator load dump transient	See section 7.3.4	—	45	BBF, CLF, CDLF
Electrostatic discharge voltage	See section 7.3.8	-10,000	10,000	All

Note: All functions except Immobilisation shall be fully operational between 9V and 16V.
Immobilisation shall be fully operational between 8V and 24V.

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5. INTERFACE CHARACTERISTICS

5.1 Fixed Feed, Source and Return Characteristics

Circuit	Connected To	Nominal Operating Current (mA)	Maximum Voltage Drop At Nominal Current (mV)	Maximum Rated Current for 5 sec (A)	Pin Number	
					High	Low
BEM battery fuse	—	+4,000	—	+10	J3/11	J3/11
Courtesy lamp fuse	—	+12,000	—	+30	J3/6	J3/6
Central door locking fuse	—	+30,000 ①	—	+60	J5/3	J5/3
High current ground	—	-70,000 ②	—	-100	J5/4	J5/4
Auto headlamp ground (Plug J1)	—	-0.5	—	-2 ③	J1/6	—
Accessories feed (Plug J2)	—	+200	—	—	J2/12	J2/12
Ignition feed (Plug J1)	—	+200	—	+10	J1/10	J1/10
Park Lamp Fuse	Analogue clock Illumination supply	+20	30	+10	J1/2	—
Analogue clock illumination supply	Park lamp Fuse	-20	30	-10	J4/10	—
Park Lamp Return (driver)	Analogue clock Illumination return	-20	30	-10	J1/1	—
Analogue clock illumination return	Park lamp Return (driver)	+20	30	+10	J4/12	—
RF receiver supply	BEM battery fuse	-10	30	-10	J4/1	J4/1
RF receiver return	High current ground	+10	30	+10	J4/4	J4/4
Analogue clock supply	BEM battery fuse	-2	30	-10	J4/5	—
Analogue clock return	High current ground	+2	30	+10	J4/6	—
Transceiver ground	High current ground	+200	30	+5	J1/3	J1/3
Sunload sensor return	High current ground	+10	30	+5	J2/6	—
Front variable intermittent wiper dwell potentiometer return	High current ground	+2	30	+5	J2/8	J2/8
CTS return	High current ground	+2	30	+10	J4/9	—
CTS fan return	High current ground	+50	30	+10	J4/11	—

① Central Locking (all motors) and Boot are activated

② Central Locking (all motors), Wiper, Washer, Battery Saver, Courtesy Lamp and Boot are activated

③ Internally limited.

5.2 Analogue Input Voltages and Currents

Input	Resistance (Ohms)	Voltage VIN (Volts)		Resolved A/D Value		Nominal Current At VIN (mA)	Pin Number	
		Min	Max	Min	Max		High	Low
Front variable intermittent wiper dwell pot (reference to Front var int wiper dwell pot return)	0.0	0.0	0.2	\$00	\$0C	-1.10	J2/19	J2/19
	4700	2.4	2.6	\$79	\$8C	-0.50		
Cabin temperature sensor thermistor (reference to Cabin temp sensor thermistor return)	0.0	0.0	0.2	\$00	\$0C	-1.10	J4/8	—
	4700	2.4	2.6	\$79	\$8C	-0.50		
Sunload sensor (reference to Sunload sensor return)	—	0.0	0.2	\$00	\$0C	-10.00	J2/17	—
	—	4.8	5.0	\$F3	\$FF	-0.11		

5.3 Digital Input Voltages and Currents

Input	Voltage VIL (V)		Nominal Current At VIL (mA)	Voltage VIH (V)		Nominal Current At VIH (mA)	Pin Number	
	Min	Max		Min	Max		High	Low
Accessories feed (Plug J2)	0.0	2.2	0.0	8.0	Vs	+41	J2/12	—
	0.0	2.2	0.0	8.0	Vs	+1	—	J2/12
Ignition feed (Plug J1)	0.0	2.2	0.0	8.0	Vs	+21	J1/10	—
	0.0	2.2	0.0	8.0	Vs	+1	—	J1/10
RF receiver receive data	0.0	Vs-6.8	-0.14	Vs-1.0	Vs	—	J4/2	J4/2
Front driver door ajar switch	0.0	Vs-6.8	-11	Vs-1.0	Vs	—	J1/14	J1/14
Front passenger door ajar switch	0.0	Vs-6.8	-11	Vs-1.0	Vs	—	J1/13	J1/13
Rear right passenger door ajar switch	0.0	Vs-6.8	-11	Vs-1.0	Vs	—	J1/11	J1/11
Rear left passenger door ajar switch	0.0	Vs-6.8	-11	Vs-1.0	Vs	—	J1/12	J1/12
Boot/tailgate release switch	0.0	Vs-6.8	-11	Vs-1.0	Vs	—	J1/9	J1/9
Boot/tailgate ajar switch	0.0	Vs-6.8	-0.07	Vs-1.0	Vs	—	J1/17	J1/17
Front wiper park switch feedback	0.0	1.5	0.0	3.5	Vs	+0.06	J3/1	J3/1
CAN global wake-up request	0.0	Vs-6.8	-3.0	Vs-1.0	Vs	—	J2/21	J2/21
CAN data link low (high speed)	0.5	1.75	—	2.0	3.0	-2.0...+2.5	J2/15	J2/15
							J2/16	J2/16
CAN data link high (high speed)	2.0	3.0	-2.0...+2.5 ①	3.0	4.25	—	J2/13	J2/13
							J2/14	J2/14
Driver central door lock status	0.0	Vs-6.8	-1.4, -11 ①	Vs-1.0	Vs	—	J2/24	J2/24
Front intermittent wiper switch	0.0	Vs-6.8	0.0, -11 ①	Vs-1.0	Vs	—	J1/19	J1/19
Front wash motor switch	0.0	Vs-6.8	0.0, +5000 ①	Vs-1.0	Vs	—	J3/10	—
	0.0	Vs-6.8	0.0, -11 ①	Vs-1.0	Vs	—	—	J3/10
Washer fluid level switch	0.0	Vs-6.8	0.0, -11 ①	Vs-1.0	Vs	—	J1/20	—
Transceiver receive data	0.0	Vs-6.8	0.0, -11 ①	Vs-1.0	Vs	—	J1/7	J1/7
Transceiver transmit data feedback	0.0	1.5	0.0	3.5	Vs	+0.6	J1/8	J1/8
Twilight sensor (Plug J1)	0.0	Vs-6.8	0.0, -11 ①	Vs-1.0	Vs	—	J1/18	—

① Where there are two VIL figures, the first is applicable in the quiescent state, such as ignition off.

5.4 Switched Output Voltages

If enabled, switched outputs will not function if an over-voltage state is detected

Output	Unenergised Voltage V _{OUS} (V)		Nominal Current at V _{OUS} (A)	Energised Voltage V _{OES} (V)		Nominal Current at V _{OES} (A)	Maximum Current (A)	Pin Number	
	Min	Max		Min	Max			High	Low
Battery saver supply	Open circuit		—	Vs-0.3	Vs	-12	-30	J3/3	J3/3
Front wiper motor	VJ3/1	VJ3/1+0.3	+7.0	0.0	0.3	+7.0	+40	J3/2	J3/2
Front wiper park switch	—	—	-7.0	—	—	0.0	-40	J3/1	J3/1
Central door unlock motors	0.0	0.3	+20	Vs-0.3	Vs	-20	±60*	J5/1	J5/1
Central door lock motors	0.0	0.3	+20	Vs-0.3	Vs	-20	±60*	J5/2	J5/2
Driver door unlock motor	0.0	0.3	+7	Vs-0.3	Vs	-7	±60*	J3/9	J3/9
Boot/tailgate release solenoid	VJ3/4	VJ3/4+0.3	±10	Vs-0.3	Vs	-20	-60*	J3/7	J3/7
Boot/tailgate release option	—	—	±10	—	—	0.0	+10	J3/4	J3/4
Front wash motor/switch	Open circuit		—	0.0	0.3	+5.0	+40*	J3/10	—

* Duration of exposure is limited by the normal operating durations of these outputs (section 8).

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5.5 Solid State Output Voltages And Currents

Solid state outputs may shut down in case of voltage > Vov

Output	Voltage VOL (V)		Nominal Current at VOL (mA)	Voltage VOH (V)		Nominal Current at VOH (mA)	Maximum Current (A)	Pin Number	
	Min	Max		Min	Max			High	Low
Heated backlight relay	0.0	2.0	+200	Vs-1.0	Vs	+0.001	+1.0*	J2/10	J2/10
Horn relay	0.0	2.0	+200	Vs-1.0	Vs	+0.001	+1.0*	J2/9	J2/9
Power window relay	0.0	2.0	+200	Vs-1.0	Vs	+0.001	+1.0*	J2/11	J2/11
Memory seat enable	0.0	2.0	+200	Vs-1.0	Vs	+0.001	+1.0*	J2/23	—
Memory seat data	0.0	2.0	+200	Vs-1.0	Vs	+0.001	+1.0*	J2/22	—
Transceiver transmit data	0.0	0.3	+11	Vs-0.3	Vs	+0.001	+1.0*	J1/8	J1/8
Transceiver supply	0.0	1.0	-0.01	Vs-2.0	Vs	-200	-2.4*	J1/4	J1/4
Courtesy (Dome) lamp	0.0	1.5	+7000	Vs-1.0	Vs	+0.02	+60*	J3/12	J3/12
Auto headlamp relay (plug J1)	0.0	1.5	+500	Vs-1.0	Vs	+0.005	+2.0*	J1/6	—
Auto headlamp relay (plug J2)	0.0	1.5	+500	Vs-1.0	Vs	+0.005	+2.0*	J2/1	—
RF receiver polling ①	0.0	0.5	+0.14	Vs-0.3	Vs	+0.001	+0.01	J4/3	J4/3
CTS fan supply ①	0.0	0.5	-0.01	Vs-0.3	Vs	-40	+0.01	J4/7	—
CAN global wake-up request	0.0	0.5	+6	Vs-1.0	Vs	+0.001	+1.0*	J2/21	J2/21
CAN data link high (high speed)	2.0	3.0	-2.0...+2.5	3.0	4.25	—	-0.1*	J2/13 J2/14	J2/13 J2/14
CAN data link low (high speed)	0.5	1.75	—	2.0	3.0	-2.0...+2.5	+0.1*	J2/15 J2/16	J2/15 J2/16

① No short circuit protection since external load is located within CDU

* Internally limited.

5.6 Battery Supply Standby Current

The BEM is considered to be in standby mode when all of the listed conditions are met for a period of T51 and currents apply at 25°C with Vs = 12.0V. The table accounts for all currents entering the BEM. Currents leaving the BEM (including grounds) are not indicated. Where specific voltage levels are indicated, deviations from these levels may increase current drain, some examples of this are indicated in brackets. Wake inputs are identified in the block diagram (see section 3.5, 3.6). The RF Receiver and BEM must be contained within a shielded enclosure to guarantee that no spurious RF data is detected by the BEM.

Pin/Function	Status/Conditions	Maximum Current Into Module (mA)	Pin Number	
			High	Low
Accessories feed (plug J2)	Accessories off (VIL = 0V)	—	J2/12	J2/12
Ignition feed (plug J1)	Ignition off (VIL = 0V)	—	J1/10	J1/10
RF receiver receive data	No RF data detected	—	J4/2	J4/2
Front driver door ajar switch	Door closed (VIH = Vs)	—	J1/14	J1/14
Front passenger door ajar switch	Door closed (VIH = Vs)	—	J1/13	J1/13
Rear right passenger door ajar switch	Door closed (VIH = Vs)	—	J1/11	J1/11
Rear left passenger door ajar switch	Door closed (VIH = Vs)	—	J1/12	J1/12
Boot/tailgate ajar switch	Boot closed (VIH = Vs)	—	J1/17	J1/17
Driver central door lock status	Door locked (VIH = Vs)	—	J2/24	J2/24
Front intermittent wiper switch	Switch not powered	—	J1/19	J1/19
Front wiper park switch	Switch not powered	—	J3/1	J3/1
Front wash motor/switch	Switch not powered	—	J3/10	J3/10
Washer fluid level switch	Switch not powered	—	J1/20	—
Boot/tailgate release switch	Switch not pressed (VIH = Vs)	—	J1/9	J1/9
Transceiver receive data	Transceiver not powered	—	J1/7	J1/7
Twilight sensor	Sensor not powered	—	J1/18	—

Pin/Function	Status/Conditions	Maximum Current Into Module (mA)	Pin Number	
			High	Low
CAN global wake-up request	No wake signal ($V_{IH} = V_S$)	+0.001	J2/21	J2/21
CAN data link high (high speed)	Bus not active	—	J2/13	J2/13
	(Bus active)	(+0.25)	J2/14	J2/14
CAN data link low (high speed)	Bus not active	—	J2/15	J2/15
	(Bus active)	(+0.25)	J2/16	J2/16
Front intermittent wiper dwell pot	Potentiometer not powered	—	J2/19	J2/19
Cabin temperature sensor thermistor	Sensor not powered	—	J4/8	—
Sunload sensor	Sensor not powered	—	J2/17	—
Heated backlight relay	Load not powered	—	J2/10	J2/10
Horn relay	Horn off (V_{OH})	+0.001	J2/9	J2/9
Power window relay	Windows off (V_{OH})	+0.001	J2/11	J2/11
Courtesy lamp	Battery saver not powered	—	J3/12	J3/12
Memory seat enable	No data (V_{OH})	+0.001	J2/23	—
Memory seat data	No data (V_{OH})	+0.001	J2/22	—
Auto headlamp relay plug J1	Headlamps off (V_{OH})	+0.005	J1/6	—
Auto headlamp relay plug J2	Headlamps off (V_{OH})	+0.005	J2/1	—
Transceiver supply	Transceiver not powered (V_{OL})	—	J1/4	J1/4
Transceiver transmit data	Transceiver not powered	—	J1/8	J1/8
RF receiver polling (24ms on, 40ms off)	No RF data detected	+2.5	J4/3	J4/3
Battery saver supply	Battery saver off (V_{OUS})	—	J3/3	J3/3
Front wiper motor	Circuit not powered	—	J3/2	J3/2
Front wiper park switch	Circuit not powered	—	J3/1	J3/1
Central door unlock motor	No activity (V_{OUS})	—	J5/1	J5/1
Central door lock motor	No activity (V_{OUS})	—	J5/2	J5/2
Driver door unlock motor	No activity (V_{OUS})	—	J3/9	J3/9
Boot/tailgate release solenoid	No unlocking (V_{OUS})	—	J3/7	J3/7
Boot/tailgate release option	Current not attributed to BEM	—	J3/4	J3/4
Analogue clock illumination supply	Clock disconnected	—	J4/10	—
Analogue clock illumination return	Clock disconnected	—	J4/12	—
RF receiver supply	Current supplied by BEM	—	J4/1	J4/1
Analogue clock supply	Clock disconnected	—	J4/5	—
CTS fan supply	Sensor not powered	—	J4/7	—
Analogue clock return	Clock disconnected	—	J4/6	—
Transceiver ground	Transceiver not powered	—	J1/3	J1/3
Sunload sensor return	Sensor not powered	—	J2/6	—
Front intermittent wiper dwell return	Potentiometer not powered	—	J2/8	J2/8
CTS return	Sensor not powered	—	J4/9	—
CTS fan return	Sensor not powered	—	J4/11	—
RF receiver return	Current supplied by BEM	—	J4/4	J4/4

Total Standby Current:

BEM battery fuse	All above conditions met	+4.5	J3/11	J3/11
Courtesy lamp fuse	All above conditions met	+0.05	J3/6	J3/6
Central door locking fuse	All above conditions met	+0.2	J5/3	J5/3

6. PRODUCT VALIDATION AND IN-PROCESS TEST REQUIREMENTS

The requirements for design verification (DV), design validation (also called production validation, or PV) and in-process testing are contained within the control plan (see section 3.2). Recording and Record Retention shall conform to QS9000 Procedures. Further details for PV testing are contained below.

Sample quantities have been established to satisfy defined reliability requirements. The notation R.X@.YC is interpreted as the minimum reliability (probability of success) equal to X at a confidence level equal to Y, eg: R.90@.70C means a minimum reliability of 90% at a confidence level 70%. R.X@.YC may be found by using Ford #112-130 Reliability Slide Rule.

All Production Validation Tests must be performed prior to production, or otherwise as approved by Ford PEO and Ford Supplier Quality Assurance (SQA).

For design and process changes following production validation, RBAU and the Ford Product Engineering Office (PEO) will jointly determine which potential changes would have significant impact on the product's function, performance, durability or appearance; and what re-validation requirements are necessary to prove the design or process change. Such changes require prior engineering approval and the attendant Control Plan changes.

6.1 Production Validation Test Matrix

The production validation test matrix shall be completed satisfactorily with parts from production tooling (and processes where possible) before Production Part Approval Process (PPAP) approval and authorisation for shipment of production parts can be effected. Tests are to be conducted in the order shown. All failures must be immediately reported to Ford Product Engineering. RBAU and Ford shall jointly determine what steps are necessary to complete the production validation of the module.

The BEM shall have failed when any of the following occur:

- Failure to meet the nominated performance requirement during the test.
- Failure to pass the nominated functional tests procedure following the test.
- Module or connector(s) have been damaged as a result of the test.

6.1.1 Test Matrix – High Series

Sect	Test	Sample Numbers				Test Stage	
		1-12 (12)	13-18 (6)	19-21 (3)	22-24 (3)	PV1	PV2
—	Initial functional test on final tester ^P	X	X	X	X	✓	✓
3.1	Dimensional check as per sales drawing				X	✓	
7.1.1	Low temperature exposure ^V (Passive)	X				✓	
7.1.2	Low temperature operation ^V (Active)	X				✓	
7.1.3	High temperature exposure ^V (Passive)	X				✓	
7.1.4	Powered thermal cycle ^V (Active)	X				✓	
7.2.2	Audible Noise ^V (Inactive)	X				✓	
7.1.6	Humidity with temperature cycling ^V (Active)	X				✓	
7.1.7	High temperature, high humidity ^V (Active)	X				✓	
7.1.8	Life test ^V (Active)	X				✓	
7.2.1	Vibration ^V (Active)	X				✓	
7.1.5	Thermal shock resistance ^V (Passive)	X				✓	
7.4.2	Radiated immunity, parallel wire ^V (Active)		X			✓	
7.3.1	Conducted immunity, sinewave noise ^V (Active)		X			✓	
7.3.2	Conducted immunity, inductive switching ^V (Active)		X			✓	
7.3.3	Conducted immunity, power cycling ^V (Active)		X			✓	
7.3.4	Conducted immunity, load dump ^V (Active)		X			✓	
7.3.5	Conducted immunity, ground shift ^V (Active)		X			✓	
7.3.6	Conducted immunity, power dropout ^V (Active)		X			✓	
7.3.7	Conducted immunity, DC voltage stress ^V (Active)		X			✓	
7.3.8	Conducted immunity, ESD ^P (Active)		X			✓	
7.4.1	Radiated immunity, stripline method ^V (Active)			X		✓	
7.6.1	Radiated emissions, electric field ^V (Active)			X		✓	✓
7.5.1	Conducted emissions, time domain ^V (Active)			X		✓	
7.5.2	Conducted emissions, frequency domain ^P (Active)			X		✓	
7.2.6	Drop test ^V (Passive)				X	✓	
7.2.3	Locating screws maximum torque ^V (Passive)				X	✓	
7.2.4	Connector lead & lock pull out force ^V (Passive)				X	✓	
7.1.9	Dust test ^P (Passive)				X	✓	
7.2.5	Connector insertion force ^V (Passive)	X	X	X	X	✓	

- Notes:** P Parts to be visually inspected and functionally tested using the production final tester with test results recorded, including error logs (diagnostic trouble codes) after each test marked with ^P.
- V Parts to be visually inspected and functionally tested after each test marked with ^V.
- X Referenced samples are to undergo this test.
- ✓ Indicates whether test is applicable for PV1 and/or PV2 level tests.
- PV1 Samples with all production level components with use of OTP/flash microcomputer is permissible. The order of environmental tests were modified to reduce test time. In future, Thermal Shock resistance, followed by Vibration shall be done immediately after Powered thermal cycle tests.
- PV2 Samples with all production level components including mask microcomputer.

Danger: Do not operate or store the Barra Remote Key above 80°C. The Lithium Battery contains flammable materials such as Lithium and organic solvent which could cause it to ignite, explode or become damaged.

6.1.2 Test Matrix – Low Series

Sect	Test	Sample Numbers				Test Stage	
		1-12 (12)	13-18 (6)	19-21 (3)	22-24 (3)	PV1	PV2
—	Initial functional test on final tester ^P	X	X	X	X	✓	✓
3.1	Dimensional check as per sales drawing				X	✓	
7.1.1	Low temperature exposure ^V (Passive)	X				✓	
7.1.2	Low temperature operation ^V (Active)	X				✓	
7.1.3	High temperature exposure ^V (Passive)	X				✓	
7.1.4	Powered thermal cycle ^V (Active)	X				✓	
7.2.2	Audible Noise ^V (Inactive)	X				✓	
7.1.6	Humidity with temperature cycling ^V (Active)	X				✓	
7.1.7	High temperature, high humidity ^V (Active)	X				✓	
7.1.8	Life test ^V (Active)	X				✓	
7.2.1	Vibration ^V (Active)	X				✓	
7.1.5	Thermal shock resistance ^V (Passive)	X				✓	
7.4.2	Radiated immunity, parallel wire ^V (Active)		X			✓	
7.3.1	Conducted immunity, sinewave noise ^V (Active)		X			✓	
7.3.2	Conducted immunity, inductive switching ^V (Active)		X			✓	
7.3.3	Conducted immunity, power cycling ^V (Active)		X			✓	
7.3.4	Conducted immunity, load dump ^V (Active)		X			✓	
7.3.5	Conducted immunity, ground shift ^V (Active)		X			✓	
7.3.6	Conducted immunity, power dropout ^V (Active)		X			✓	
7.3.7	Conducted immunity, DC voltage stress ^V (Active)		X			✓	
7.3.8	Conducted immunity, ESD ^P (Active)		X			✓	
7.4.1	Radiated immunity, stripline method ^V (Active)			X		✓	
7.6.1	Radiated emissions, electric field ^V (Active)			X		✓	✓
7.5.1	Conducted emissions, time domain ^V (Active)			X		✓	
7.5.2	Conducted emissions, frequency domain ^P (Active)			X		✓	
7.2.6	Drop test ^V (Passive)				X	✓	
7.2.3	Locating screws maximum torque ^V (Passive)				X	✓	
7.2.4	Connector lead & lock pull out force ^V (Passive)				X	✓	
7.1.9	Dust test ^P (Passive)				X	✓	
7.2.5	Connector insertion force ^V (Passive)	X	X	X	X	✓	

Notes: P Parts to be visually inspected and functionally tested using the production final tester with test results recorded, including error logs (diagnostic trouble codes) after each test marked with ^P.

V Parts to be visually inspected and functionally tested after each test marked with ^V.

X Referenced samples are to undergo this test.

✓ Indicates whether test is applicable for PV1 and/or PV2 level tests.

PV1 Samples with all production level components with use of OTP/flash microcomputer is permissible. The order of environmental tests were modified to reduce test time. In future, Thermal Shock resistance, followed by Vibration shall be done immediately after Powered thermal cycle tests.

PV2 Samples with all production level components including mask microcomputer.

Danger: Do not operate or store the Barra Remote Key above 80°C. The Lithium Battery contains flammable materials such as Lithium and organic solvent which could cause it to ignite, explode or become damaged.

7. DURABILITY TESTING

For all active tests, reference is made to the test circuits shown in figures 1A and 1B.
Operating states described in the following tests are defined as follows. Tests may employ more than one state.

Active indicates that the BEM is powered up and is being exercised by the referenced test sequence.

Inactive indicates that the BEM is powered up but is not being exercised.

Passive indicates that the BEM is not connected.

The following tests refer to functional classes, as defined in ES-XW7T-1A278-AB part 1.3. Please refer to section 3.4 for class assignments to specific BEM functions. For reference, these are:

Functional class C: Any function that is essential to the operation of control of the vehicle. (Certain Class C functions which may experience an unintentional change that may surprise the vehicle operator and which cannot be remedied safely and immediately (eg: air bag deployment, base steering loss, base braking loss, engine stalls or surges) may be subject to more stringent requirements.)

Functional class B1: Any function that has the potential to impair a Class C function.

Functional class B: Any function that enhances, but is not essential to, the operation or control of the vehicle.

Functional class A: Any function that provides a convenience.

The following tests refer to performance classifications, as defined in ES-XW7T-1A278-AB part 1.3. Tests for non-EMC attributes are also rated to these classifications. For reference, these are:

Performance class I: The function shall operate as designed (or meet specified limits) during and after exposure to a disturbance.

Performance class II: The function may deviate from designed performance during exposure to a disturbance, but shall not affect safe operation of the vehicle. The function will automatically return to normal after the disturbance is removed. No effect on permanent or temporary memory is allowed. For the immobilisation system, mobilisation must occur once the disturbance is removed, according to normal operation.

Performance class III: The function may deviate from designed performance during exposure to a disturbance but shall not affect safe operation of the vehicle. Simple operator action (eg: cycle power or ignition switch, reprogram presets) may be required to return the function to normal after the disturbance is removed. No effect on permanent type memory is allowed.

7.1 Environmental

7.1.1 Low Temperature Exposure (Passive)

Description	Test Conditions
Test reference source	WDR 00.00EA-D11, requirement 4.5.1
<i>Relaxed minimum exposure temperature</i> (T ₅ in WDR 00.00EA-D11)	-20 ± 2 °C
Performance classification	Passive
Total test time, excluding setup	72 hours

7.1.2 Low Temperature Operation (Active)

Description	Test Conditions
Test reference source	WDR 00.00EA-D11, requirement 4.5.2
Test circuit	Figures 1A and 1B
Test sequence	As per section 9.1
<i>Relaxed minimum operation temperature</i> (T ₁ in WDR 00.00EA-D11)	-20 ± 2 °C
Performance classification	Class I
Total test time, excluding setup	72 hours

7.1.3 High Temperature Exposure (Passive)

Description	Test Conditions
Test reference source	WDR 00.00EA-D11, requirement 4.5.3(a)
<i>Increased maximum exposure temperature</i> (T ₆ in WDR 00.00EA-D11)	105 ± 2 °C
<i>Reduced exposure time</i>	24 hours
Performance classification	Passive
Total test time, excluding setup	24 hours

7.1.4 Powered Thermal Cycle (Active/Passive)

Description	Test Conditions
Test reference source	WDR 00.00EA-D11, requirement 4.5.5
Test circuit	Figures 1A and 1B
Test sequence	As per section 9.1
Temperature Sequence	Figure 3
<i>Increased maximum operating temperature</i> , T _{MAX} (T ₂ in WDR 00.00EA-D11)	85 °C
<i>Decreased minimum operating temperature</i> , T _{MIN} (T ₁ in WDR 00.00EA-D11)	-20 °C
Temperature Transition Time	0.875°C/minute
Hold Time, T _{HOLD1} at Minimum Temperature	2 hours
Hold Time, T _{HOLD2} at Maximum Temperature	2 hours
Operating, T _{ACTIVE}	5 hours
Non-Operating T _{PASSIVE}	3 hours
Approximate Test Duration, T _D (1 Cycle)	8 hours
Number of Cycles	30
Performance classification	Class I
Total test time, excluding setup	240 hours

7.1.5 Thermal Shock Resistance (Passive)

Description	Test Conditions
Test reference source	WDR 00.00EA-D11, requirement 4.5.6
<i>Relaxed minimum exposure temperature</i> (T ₅ in WDR 00.00EA-D11)	-20 ± 2 °C
<i>Increased maximum exposure temperature</i> (T ₆ in WDR 00.00EA-D11)	105 ± 2 °C
Dwell time, t, based on nominal mass	2 hours
Performance classification	(Passive)
Total test time, excluding setup	25 hours

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7.1.6 Humidity with Temperature Cycling (Active & Passive)

Description	Test Conditions
Test reference source	WDR 00.00EA-D11, requirement 4.5.8
Test circuit	Figures 1A and 1B
Test sequence (operating portion of cycle)	As per section 9.1
Mounting Position	Figure 6
Performance classification	Class I (Active), Passive (Inactive)
Special Requirements	Allow any condensation to dry before functional check following the test.
Total test time, excluding setup	120 hours

7.1.7 High Temperature, High Humidity (Active & Passive)

Description	Test Conditions	
	Part 1	Part 2
Test reference source	None	
Test circuit	Figures 1A and 1B	
Test sequence	Not applicable	Section 9.1
Relative humidity	93 to 98 %	
Ambient temperature	37 ± 3 °C	
Test duration, TD (1 cycle = Part 1 + Part 2)	12 hours	12 hours
Number of cycles	4	
Mounting position	Figure 6	
Performance classification	Passive (Inactive)	Class I
Special requirements	- Check all functions operational before starting Passive Test at the end of each cycle. - Allow any condensation to dry before functional check following the test.	
Total test time, excluding setup	96 hours	

7.1.8 Life Test With Temperature Cycling (Active & Passive)

Descriptions	Test Conditions
Test reference source	None
Test circuit	Figures 1A and 1B
Test sequence	As per section 9.1
Temperature sequence	Figure 4
Maximum temperature transition time	1°C/minute
Number of cycles	20
Performance classification	Class I (Active), Passive (Inactive)
Total test time, excluding setup	1000 hours

7.1.9 Dust (Passive)

Descriptions	Test Conditions
Test reference source	WDR 00.00EA-D11, requirement 4.5.10
Applicable exposure	100mg/m ³ for 5 hours
Performance classification	Passive (Inactive)
Total test time, excluding setup	5 hours

7.2 Mechanical

7.2.1 Vibration (Active)

Description	Test Conditions
Test reference source	WDR 00.00EA-D11, requirement 4.6.1
Test circuit	Figures 1A and 1B
Test sequence	As per section 9.1
Vibration classification number	I (mild)
Vibration method	B (mounted via audio module)
Mounting Position	Figure 6
Performance classification	Class I (If RF receiver performance is interfered with by electromagnetic fields created by the vibration test stand, then this shall be recorded in the report but not considered a failure.)
Special requirements	It is acceptable to vibrate the BEM with the peripherals (TILA, BRM and CTS) stationary and repeat by vibrating the peripheral(s) with the BEM stationary.
Total test time, excluding setup	18 hours

7.2.2 Audible Noise (Inactive)

Description	Test Conditions
Test reference source	WDR 00.00EA-D11, requirement 4.6.2
Vibration method	B (mounted via audio module)
Mounting Position	Figure 6
Special Requirements	- It is acceptable to vibrate the BEM with the peripherals (TILA, BRM and CTS) stationary and repeat by vibrating the peripheral(s) with the BEM stationary. - Test for each axis may be performed following vibration test for the same axis.
Performance classification	Class I (no noise observed throughout test.)
Total test time, excluding setup	1 hour

7.2.3 Locating Screws Maximum Torque (Passive)

Description	Test Conditions
Test reference source	None
Minimum torque at point of failure	93.1 Nmm
Performance classification	Passive (Inactive)
Total test time, excluding setup	10 minutes per unit

7.2.4 Connector Lead & Lock Pull Out Force (Passive)

Description	Test Conditions
Test reference source	WDR 00.00EA-D11, requirement 4.6.4 'Lead/lock pull test'
Connector plug and terminal details	Refer to sales drawing
Performance classification	Passive (Inactive)
Special Requirements	No damage or cracking to solder joints, connector/housing mating connector and terminals
Total test time, excluding setup	72 minutes per unit

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7.2.5 Connector Insertion Force (Passive)

Description	Test Conditions
Test reference source	Using Testometric force gauge machine to determine the rate of movement on the following connectors: • J1: Rate of 100mm/min • J2: Rate of 100mm/min • J3: Rate of 25mm/min • J4: Rate of 20mm/min • J5: Rate of 50mm/min
Connector plug and terminal details	Refer to sales drawing
Maximum permissible force	130N
Performance classification	Passive (Inactive)
Special requirements	• Test is only valid for the 'first time' insertion of plugs into module • Insertion force applied is without secondary locking feature depressed and with positive audible engagement feedback
Total test time, excluding setup	2 minutes per unit

7.2.6 Drop Test (Passive)

Description	Test Conditions	
	Part 1	Part 2
Drop distance	1 metre	—
Drop surface	Concrete or steel	
Number of sides to be dropped	6 sides	6 sides 45° to a bench surface
Performance classification	Passive (Inactive)	
Special requirements	• Damage to housing is acceptable. • Connectors and pins must not be damaged and mating connector must be able to make to allow functional test. • Report all damage observed.	• No damage to mounting capability.
Total test time	1 minute per BEM	

7.3.2 Inductive Switching Transients (Active)

Description	Test Conditions
Test reference source	ES-XW7T-1A278-AB Part 3 Section CI 220. See also ESC Component EMC Test Plan.
Test circuit	Figures 1A and 1B
Test setup	See also figure 2A
Test sequence	Section 9.2
Total test time, excluding setup	220 seconds per unit

7.3.3 Power Cycling (Active)

Description	Test Conditions
Test reference source	ES-XW7T-1A278-AB Part 3 Section CI 230. See also ESC Component EMC Test Plan.
Test Circuit	Figures 1A and 1B
Test Setup	See also figure 2B
Applicable tests	CI 230-A, CI 230-C
Test sequence, CI 230-A	Use default switch positions as per test circuit, and apply waveform to Accessories and Ignition lines.
Test sequence, CI 230-C	Section 9.1; test waveform to coincide with 'start' step sequence.
Total test time, excluding setup	240 seconds per unit

7.3.4 Load Dump (Active)

Description	Test Conditions
Test reference source	ES-XW7T-1A278-AB Part 3 Section CI 240. See also ESC Component EMC Test Plan.
Temperature soak before test	80 ± 3 °C for t =30 minutes
Test circuit	Figures 1A and 1B
Test setup	See also figure 2A
Test sequence	Use default switch positions as per test circuit, except: - S1: Ign - S21: Auto (high series only)
Special requirements	- BEM shall be tested without a transponder in order to force a Re-Challenge between TILA and BEM. This allows TILA to be continuously active during transient testing. - Check TILA functionality with transponder after test.
Total test time, excluding setup	100 seconds per unit

7.3.5 Ground Shift (Active)

Description	Test Conditions
Test reference source	ES-XW7T-1A278-AB Part 2 Section CI 250. See also ESC Component EMC Test Plan.
Test circuit	Figures 1A and 1B
Test Setup	See also figure 2D
Applicable tests	Only point 1 testing is applicable
Test sequence	Section 9.2
Minimum dwell time for CI 250-B	5 cycles of test sequence
Total test time, excluding setup	5 minutes per unit

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7.3.6 Power Dropout and Battery Recovery (Active)

Description	Test Conditions
Test reference source	ES-XW7T-1A278-AB Part 3 Section CI 260. See also ESC Component EMC Test Plan.
Test circuit	Figures 1A and 1B
Test setup	See also figure 2B for CI 260-A through CI 260-E See also figure 2C for CI 260-F
Test sequence	Section 9.1
Total test time, excluding setup	12 minutes per unit

7.3.7 DC Voltage Overstress (Active)

Description	Test Conditions
Test reference source	ES-XW7T-1A278-AB Part 3 Section CI 270. See also ESC Component EMC Test Plan.
Temperature soak before test	80 ± 3 °C for t = 30 minutes
Test circuit	Figures 1A and 1B
Test time for CI 270-C	1 minute
Test sequence, CI 270-A	Use default switch positions as per test circuit, except: • S1: Off • S21: Auto (high series only)
Test sequence, CI 270-B, CI 270-C	Use default switch positions as per test circuit, except: • S1: Ign • S21: Auto (high series only)
Special requirements	• BEM shall be tested without a transponder in order to force a Re-Challenge between TILA and BEM. This allows TILA to be continuously active during transient testing. • Check TILA functionality with transponder after test.
Total test time, excluding setup	62 minutes per unit

7.3.8 Electrostatic Discharge (Passive/Active)

Description	Test Conditions
Test reference source	ES-XW7T-1A278-AB Part 3 Section CI 280. See also ESC Component EMC Test Plan.
Applicable tests	CI 280-A, CI 280-C, CI 280-D
Test set-up for CI 280-A	Figure 5A
Test set-up for CI 280-C, CI 280-D	Figure 5B
Test sequence	Use default switch positions as per test circuit, except: • S1: Ign • S21: Auto (high series only)
BEM connector pins not required to be tested (based on clause 3.2.1.3 regarding multiple pins on same circuit)	• Ground: J1/3, J2/6, J2/7, J2/8, J4/4, J4/6, J4/9, J4/11 • CAN Hi: J2/14, • CAN Lo: J2/16, • Battery: J4/1, J4/5, • Analog Clock Feed: J4/10, J4/12
Total test time, excluding setup	4 hours per BEM

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7.4 Radiated Immunity

7.4.1 Radiated Immunity Stripline Method (Active)

Description	Test Conditions
Test reference source	ES-XW7T-1A278-AB Part 2 Sections RI 111. See also ESC Component EMC Test Plan.
Frequency range covered by RI 111	10kHz to 1000MHz
Test circuit	Figures 1A and 1B
Test setup	See also ESC Component EMC Test Plan and figure 2F
Test sequence	Section 9.2
Minimum dwell time	1 cycle of test sequence
Minimum field strength level	Level 3
Additional deviations from specified test levels	Due to limitations of the digital signature transponder, it is acceptable if the transponder immobilisation function fails above 100 V/m.
Special requirements	If malfunction occurs, decrease field strength and report threshold achieved. All deviations from class I to be noted.
Total Test Time	4 hours per BEM, including peripherals

7.4.2 Parallel Wire Miscellaneous Noise (Active)

Description	Test Conditions
Test reference source	ES-XW7T-1A278-AB Part 2 Section RI 130. See also ESC Component EMC Test Plan.
Test circuit	Figures 1A and 1B
Test setup	See also figure 2E
Special conditions	Twisted pair combinations to be as per wiring diagram (see section 3.2)
Test sequence	Section 9.2
Approximate total test time excluding setup	12 minutes per unit

7.5 Conducted Emissions

7.5.1 Time Domain (Active)

Description	Test Conditions
Test reference source	ES-XW7T-1A278-AB Part 2 Section CE 410. See also ESC Component EMC Test Plan.
Test Circuit	Figures 1A and 1B
Test sequence	Use default switch positions as per test circuit, except: - S1: Ign – ignition switch - S14: On – intermittent wipers - S21: Auto – auto headlamps (high series only)
Total Test Time	2.6 hours per BEM and TILA

7.5.2 Frequency Domain (Active)

Description	Test Conditions
Test reference source	ES-XW7T-1A278-AB Part 2 Section CE 420. See also ESC Component EMC Test Plan.
Test Circuit	Figures 1A and 1B
Test sequence	Use default switch positions as per test circuit, except: • S1: Ign – ignition switch • S14: On – intermittent wipers • S21: Auto – auto headlamps (high series only)
Special requirements	• If malfunction occurs replace inductive load to a resistive load and repeat test.
Total Test Time	2.6 hours per BEM and TILA

7.6 Radiated Emissions**7.6.1 Narrowband Electric Field (Active)**

Description	Test Conditions
Test reference source	ES-XW7T-1A278-AB Part 2 Section RE 310. See also ESC Component EMC Test Plan.
Test setup	See also figure 2G
Test sequence	Use default switch positions as per test circuit, except: • S1: Ign – ignition switch • S8: Toggle driver door ajar at regular intervals to activate dome lamp PWM • S14: On – intermittent wipers • S15: Short – wiper dwell • S21: Auto – auto headlamps (high series only)
Upper test frequency limit	1000MHz
Radiated emissions measurement limit	CISPR 25 Class 3
Special requirements	• If malfunction occurs replace inductive load to a resistive load and repeat test.
Official Concession (Rod Heubner 19/12/01)	• Radiated Emission Specification exceedances at 64MHz and 256MHz are not co-incident with any land mobile and are accepted at <16dBuV & <25dBuV respectively.
Total test time	5.3 hours per BEM with peripherals

8. FUNCTIONAL DESCRIPTION

General references for section 8 are:

- Test circuit: figures 1A and 1B.
- Timing parameters: section 8.21.

All digital inputs sensed by the BEM are debounced by detecting four consecutive reads at 8...10ms intervals, except for the wiper park switch where only two consecutive reads are used. All analogue inputs are read at 10ms intervals, with the most recent reading having a weighting of 25% of the filtered value. Note that the following description assumes that no faults in the system are present unless otherwise stated. Functionality may be influenced by such faults, particularly when relating to the accessories and ignition inputs, which are used to determine the ignition switch status.

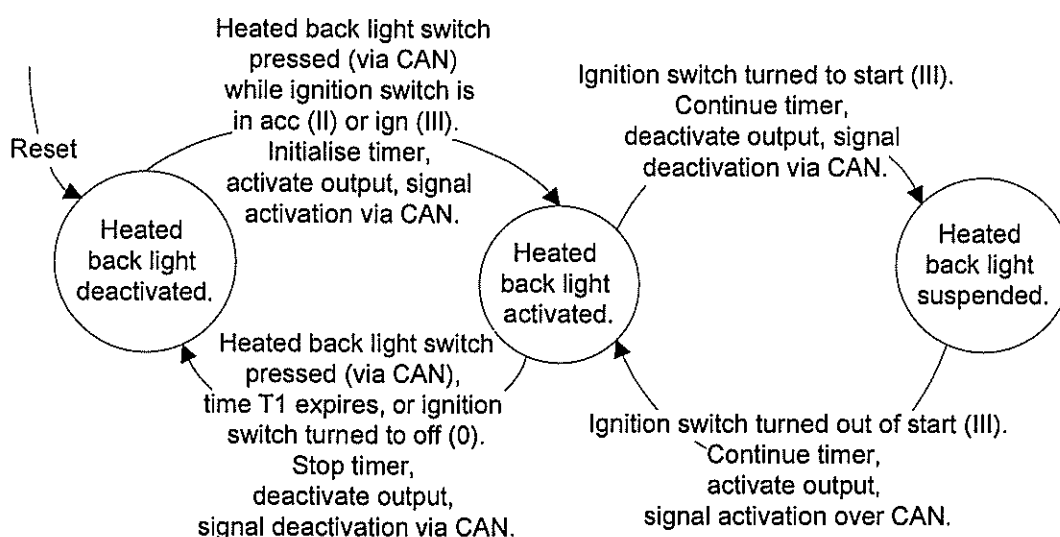
8.1 Accessories Line Status

The BEM reports the status of the accessories line over the CAN bus, subject to the conditions for CAN message transmission, section 8.18. The accessories line is high when the ignition switch (S01) is in the accessories or ignition positions.

8.2 Heated Backlight Timer

The heated backlight relay may be activated (VOL) with the ignition switch (S01) in the accessories or ignition position. Activation is achieved when the pressing of the heated backlight switch (S22, momentary) is reported via the CAN bus. The relay output will deactivate (VOH) when time T01 elapses, the heated backlight switch is reported as being pressed again, or the ignition switch is turned to the off position. While the DST ignition key is being held in the start position, the heated backlight relay is temporarily deactivated, and will resume the previous state when the DST ignition key is released from start.

The operational status of the heated backlight is indicated via the CAN bus. Special efforts have been incorporated to minimise the response time of this message.

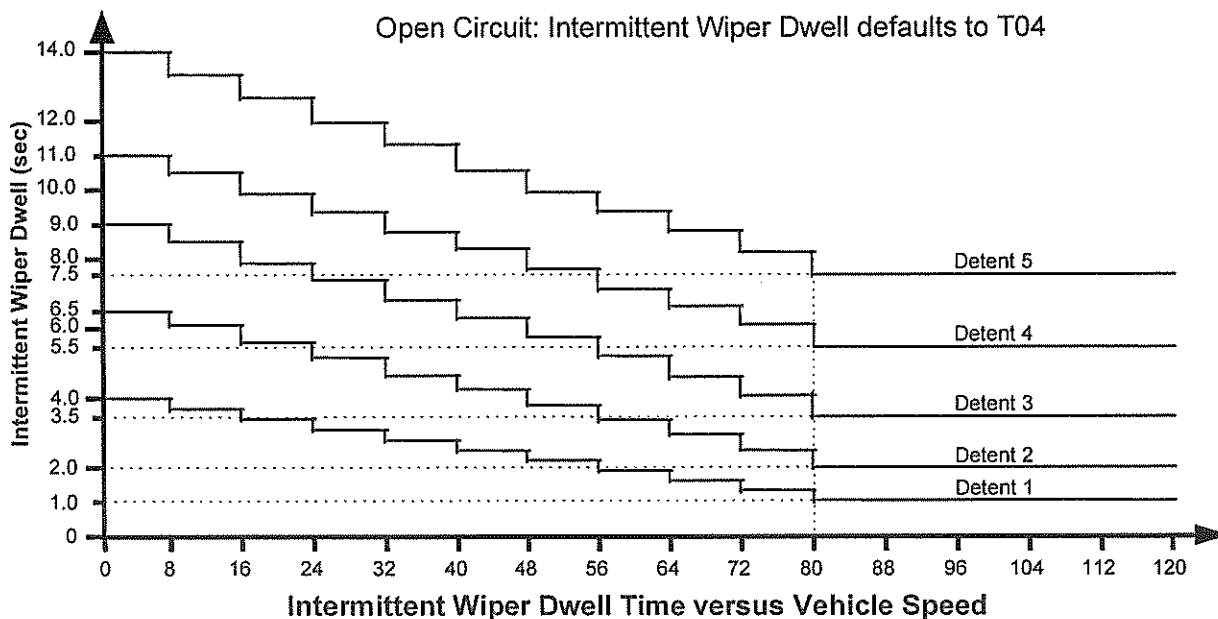


State diagram for heated backlight control

For any discrepancies, the functional description supercedes the above state diagram

8.3 Front Intermittent Wiper Timer with Variable Dwell, Speed Dependency, and Wipe on Take-off

Switching the front intermittent wiper switch (S14) to the intermittent position while the ignition switch (S01) is in the accessories or ignition position will enable the intermittent wipe function. While enabled, the front wiper motor output will be activated briefly, (T02 @ VOES) to trigger a single wipe followed by a delay before the next wipe is initiated. If the wiper dwell potentiometer is disconnected (S14 open), then the delay between wipes shall be of a fixed duration (T04 @ VOUS). If, however, the vehicle is fitted with a wiper dwell potentiometer (S14), then the delay (T03 @ VOUS) is dependent on the potentiometer position. With the ignition switch in the ignition position, the dwell time is also affected by vehicle speed, reducing as speed increases, as per the figure and table below. To reduce customer annoyance due to regular clicking sounds, a silent relay has been used in the BEM's circuitry. When the vehicle accelerates to 8 km/h (as reported via CAN), with the wiper switch in the intermittent position, and the ignition switch in the ignition position, then the next wipe will be brought forward to implement a wipe on take-off function. It shall occur a short time after the previous wipe (T08), or immediately if this time has already elapsed. If the wiper travel time is longer than the time expected such as in the case where the Detent is in position 1, travelling at 80km/h, the wiper park switch feedback is monitored, allowing the next wipe to occur immediately with minimal delay. A 3 second timeout has been implemented just in case Wiper Park switch feedback fails. When no vehicle speed data is sent via CAN, the BEM will assume 0 km/h. The BEM's circuitry allows the wipers to reach the park position at the end of an action, provided that the accessories line is providing power to the wiper motor.



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Intermittent Wiper Dwell Time versus Vehicle Speed

Detent Position	Short to GND	1	2	3	4	5	Open Circuit
Nominal A/D Realised Value	\$00 to \$09	\$0A to \$22	\$23 to \$4C	\$4D to \$65	\$66 to \$71 (See Note)	\$72 to \$B9 (See Note)	\$C0 to \$FF
Speed (km/h)	Front Intermittent Wiper Off Time (seconds)						
0	T03 = 4.1	T03 = 4.1	T03 = 6.6	T03 = 9.1	T03 = 11.1	T03 = 14.1	T04 = 7.0
8	T03 = 3.8	T03 = 3.8	T03 = 6.2	T03 = 8.6	T03 = 10.6	T03 = 13.5	T04 = 7.0
16	T03 = 3.5	T03 = 3.5	T03 = 5.7	T03 = 8.0	T03 = 10.0	T03 = 12.8	T04 = 7.0
24	T03 = 3.2	T03 = 3.2	T03 = 5.3	T03 = 7.5	T03 = 9.5	T03 = 12.2	T04 = 7.0
32	T03 = 2.9	T03 = 2.9	T03 = 4.8	T03 = 6.9	T03 = 8.9	T03 = 11.5	T04 = 7.0
40	T03 = 2.6	T03 = 2.6	T03 = 4.4	T03 = 6.4	T03 = 8.4	T03 = 10.9	T04 = 7.0
48	T03 = 2.3	T03 = 2.3	T03 = 3.9	T03 = 5.8	T03 = 7.8	T03 = 10.2	T04 = 7.0
56	T03 = 2.0	T03 = 2.0	T03 = 3.5	T03 = 5.3	T03 = 7.3	T03 = 9.6	T04 = 7.0
64	T03 = 1.7	T03 = 1.7	T03 = 3.0	T03 = 4.7	T03 = 6.7	T03 = 8.9	T04 = 7.0
72	T03 = 1.4	T03 = 1.4	T03 = 2.6	T03 = 4.2	T03 = 6.2	T03 = 8.3	T04 = 7.0
80 +	T03 = 1.1	T03 = 1.1	T03 = 2.1	T03 = 3.6	T03 = 5.6	T03 = 7.6	T04 = 7.0

Note: Since the mechanical design of the lever combination switch has detents 4 and 5 next to each other, there is a risk that detents 4 and 5 will not be distinguishable in production eg detent 5 may always appear as detent 4.

8.4 Front Wipe After Wash Control (Low Series)

By pressing the front wash switch (S17, momentary) to the wash position, with the ignition switch (S01) in the accessories or ignition position, the wipers will automatically operate (VOES) after a short delay, T06. They will continue to do so for the period that the washers are active. When the washers are turned off, the wipers will perform three wipes, including the one already in progress, by monitoring the feedback from the wiper park switch (S16) and then stop (VOUS). In the event of failure of the feedback, the wipers will stop after a time approximately equivalent to three wipes (T07) following deactivation of the washer motor. Switching the ignition switch to the off or start positions will abort any current wash function. Operation of the intermittent wipe function is suspended while a wash function is in progress.

8.5 Front Washer Motor Control with Wipe After Wash (High Series)

The front washer motor is activated (VOES) by pressing the front wash switch (S17, momentary) to the wash position, with the ignition switch (S01) in the accessories or ignition position. The line is held by the BEM in the activated state for T05. The operator can obtain wash for longer than this time by continuing to hold the wash switch. After the washer motor has been running for T06, the wipers will automatically operate (VOES), and will continue to do so for the period that the washers are active. When the washers turn off, the wipers will perform three wipes, including the one already in progress, by monitoring the feedback from the wiper park switch (S16) and then stop (VOUS). In the event of failure of the feedback, the wipers will stop after a time approximately equivalent to three wipes (T07) following deactivation of the washer motor. Switching the ignition switch to the off or start positions will abort any current wash function. Operation of the intermittent wipe function is suspended while a wash function is in progress.

8.6 Washer Fluid Level (High Series)

With the ignition switch (S01) in the Accessories, Ignition or Start position, the status of the washer fluid bottle water level is updated via the CAN bus. The lower washer switch bit is set when the switch (S18) is in the low position. A change in state must be observed for T50 before the change is transmitted over CAN. With the ignition switch in the off position, the BEM will send a 0 via CAN.

Issue Number:	8	Valid from:	23 September 2002	Print Date:	23 September 2002
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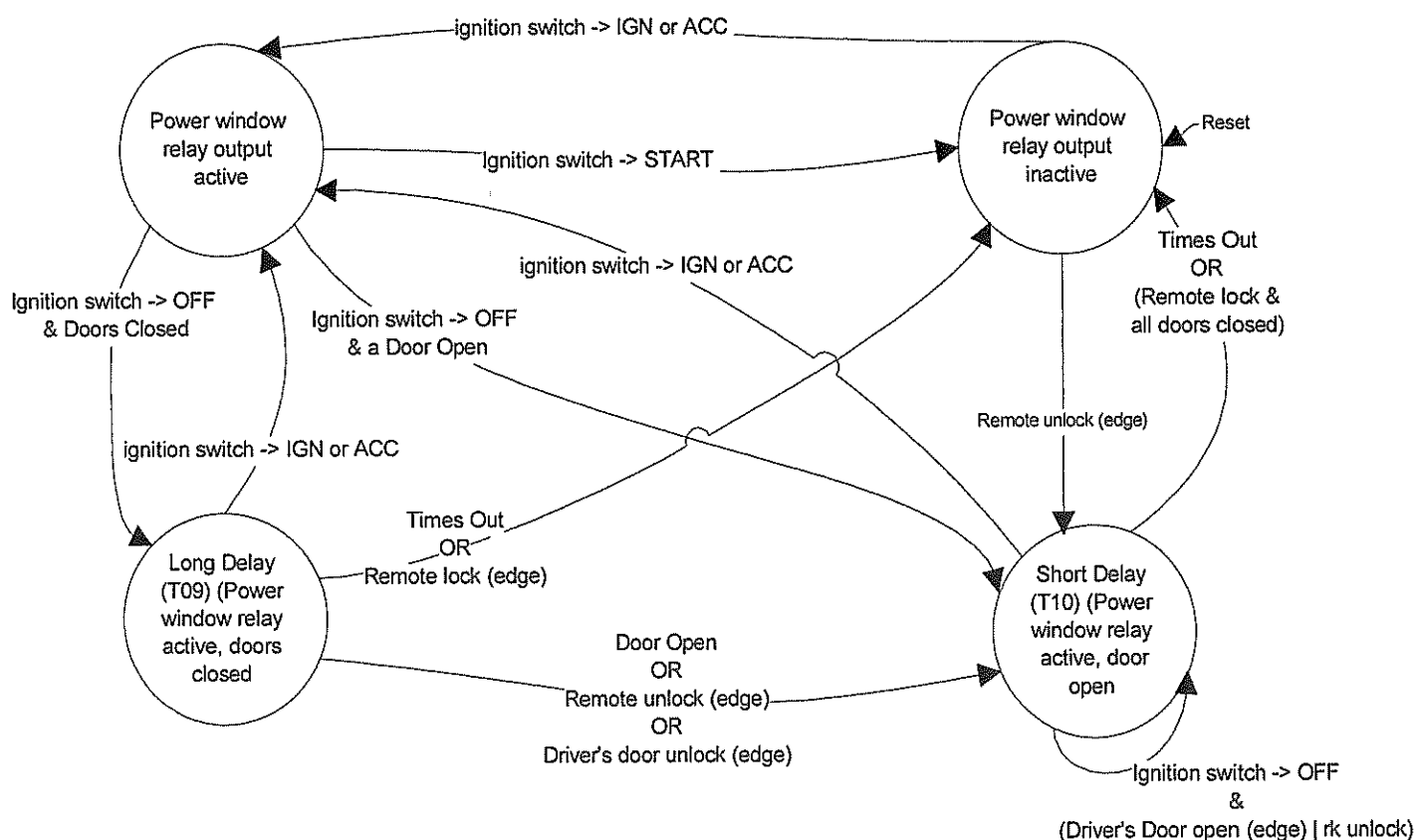
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8.7 Power Window Control

With the ignition switch (S01) in the accessories or ignition position, the power window relay output is activated (VOL). Switching the ignition switch to the start position will deactivate the power window relay (VOH). Switching the Ignition switch to the off position will allow the power window relay output to remain active for T09 before deactivating. However if during this time, a door is opened (S08, S09, S10 or S11), then the time is reduced to a shorter period (T10), starting from when the door was opened, to ensure compliance with Australian Design Rule 42.

Power window relay output is also activated for remote unlock and stays active for T10 before deactivating. If during this time (T10) the drivers door is opened then the power window relay output remains active for another T10 (from when door is opened) before deactivating.

Power window relay output is also deactivated for all doors closed and remote lock requested.



State diagram for power window control

For any discrepancies, the functional description supercedes the above state diagram

From	Activation Method	To
Reset		Inactive Mode
Inactive Mode	Ignition switch-> IGN or ACC	Active Mode
Inactive Mode	Remote unlock (edge)	Short exit delay Mode
Active Mode	Ignition switch -> START	Inactive Mode
Active Mode	(Ignition switch -> OFF) & (doors closed)	Long exit delay Mode
Active Mode	(Ignition switch -> OFF) & (a door open)	Short exit delay Mode
Long exit delay Mode	Ignition switch-> IGN or ACC	Active Mode
Long exit delay Mode	Times Out (T09)	Inactive Mode
Long exit delay Mode	Remote lock (edge)	Inactive Mode
Long exit delay Mode	A Door open	Short exit delay Mode
Long exit delay Mode	Remote unlock (edge)	Short exit delay Mode
Long exit delay Mode	Driver's door unlock (edge)	Short exit delay Mode
Short exit delay Mode	Ignition switch-> (IGN or ACC)	Active Mode
Short exit delay Mode	Times Out (T10)	Inactive Mode
Short exit delay Mode	Remote lock & all doors closed	Inactive Mode
Short exit delay Mode	(Ignition switch -> OFF) & (Driver's door open(edge))	Short exit delay Mode
Short exit delay Mode	(Ignition switch -> OFF) & (remote unlock (edge))	Short exit delay Mode

8.8 Analogue Clock Interface (High Series)

The BEM provides battery power to the analogue clock module at all times, and connects the clock's illumination line to the graphics illumination signal from the cluster (S19 and S25).

8.9 Cabin Temperature Module Interface (High Series)

With the ignition switch (S01) in the ignition or start positions, the BEM provides power to the fan of the cabin temperature module. With the ignition switch in the ignition position, the BEM reads the analogue temperature sensor signal from the module, transforms it into degrees Centigrade and transmits the data over the CAN bus for benefit of the climate control system. The data byte transmitted is a result of a vote of two data bytes out of the last three. At least two data bytes that are within 5°C of each other will be accepted with the last data byte of the two transmitted via CAN accordingly. If however, all three data bytes are outside 5°C of each other, than at worst case the previous correct data will be used. For example, upon power up the first data byte will be \$FF.

Invalid data is signalled with the Ignition switch in off or accessories position or when a valid temperature reading is not available due to a fault, however relevant DTC's will be logged in this instance (See Section 10).

The Calibration factor (CF) is programmed into the BEM by using a diagnostic command "WriteDataByLocalIdentifier". Likewise reading the Calibration factor (CF) programmed into can be done using the "ReadDataByLocalIdentifier" (See Section 10.7).

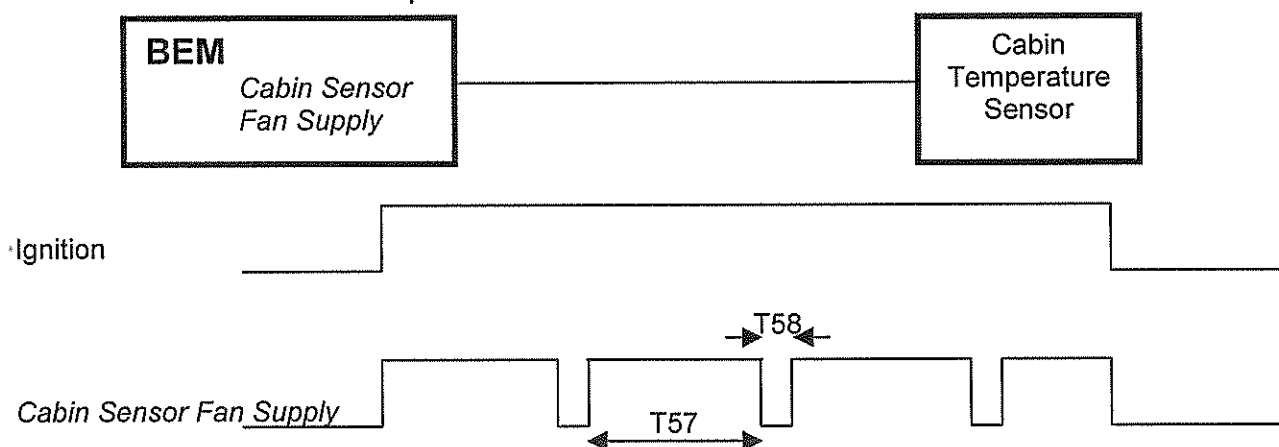
The formula to convert the CAN reading into the actual temperature reading is:

$$TTEMP = (TCAN + CF - 80) \div 2 \quad \text{for } \$27 \leq TCAN \leq \$EF \text{ } (-20.0^{\circ}\text{C} \leq +74.5^{\circ}\text{C})$$

$$\text{For } \$-7 \leq \text{Calibration Factor (CF)} \leq +7$$

$$TTEMP = \text{Invalid data} \quad \text{for } TCAN = \$FF$$

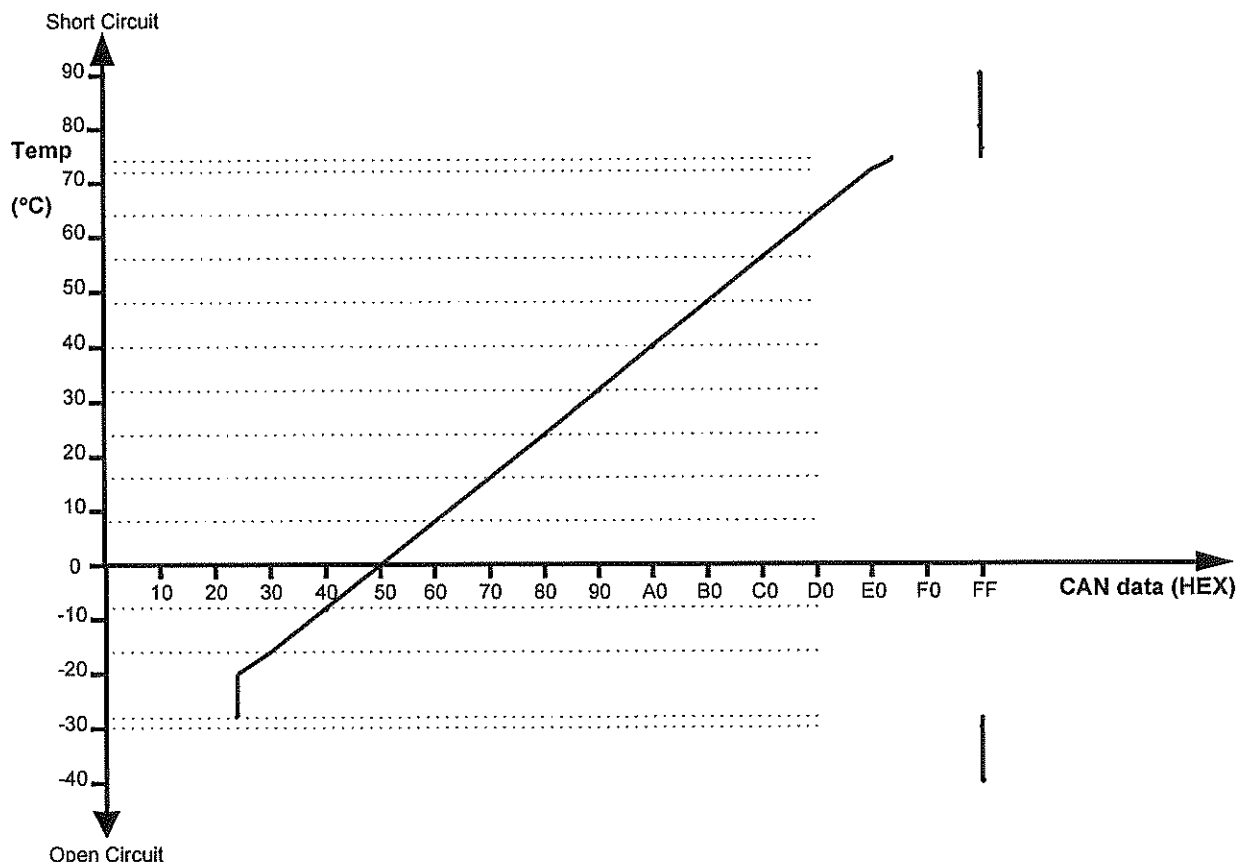
When ignition switch (S01) is in the on position, cabin temperature sensor unit (fan) is triggered by output as per timing diagram. When ignition switch (S01) is in the off position, output *Cabin Sensor Fan Supply* will be low. The T58 off Pulses are required to restart the motor if stalled.



Note: See CTS specification for tolerances of system

TTEMP (°C)	A/D (Hex)	TCAN (Hex)	TTEMP (°C)	A/D (Hex)	TCAN (Hex)	TTEMP (°C)	A/D (Hex)	TCAN (Hex)	TTEMP (°C)	A/D (Hex)	TCAN (Hex)
-29.0	EB	FF	-15.5	DF	31	-7.5	D3	41	-1.5	C7	4D
-27.5	EA	27	-14.5	DE	33	-7.0	D2	42	-1.0	C6	4E
-26.0	E9	27	-14.0	DD	34	-6.5	D1	43	-0.5	C5	4F
-24.5	E8	27	-13.0	DC	36	-6.0	D0	44	0.0	C4	50
-23.5	E7	27	-12.5	DB	37	-5.5	CF	45	0.5	C3	51
-22.0	E6	27	-11.5	DA	39	-5.0	CE	46	1.0	C2-C1	52
-21.0	E5	27	-11.0	D9	3A	-4.5	CD	47	1.5	C0	53
-20.0	E4	28	-10.5	D8	3B	-4.0	CC	48	2.0	BF	54
-19.0	E3	2A	-10.0	D7	3C	-3.5	CB	49	2.5	BE	55
-18.0	E2	2C	-9.5	D6	3D	-3.0	CA	4A	3.0	BD-BC	56
-17.0	E1	2E	-8.5	D5	3F	-2.5	C9	4B	3.5	BB	57
-16.0	E0	30	-8.0	D4	40	-2.0	C8	4C	4.0	BA	58

TTEMP (°C)	A/D (Hex)	TCAN (Hex)	TTEMP (°C)	A/D (Hex)	TCAN (Hex)	TTEMP (°C)	A/D (Hex)	TCAN (Hex)	TTEMP (°C)	A/D (Hex)	TCAN (Hex)
4.5	B9-B8	59	21.0	88	7A	37.5	57	9B	54.0	33	BC
5.0	B7	5A	21.5	86	7B	38.0	56	9C	54.5	32	BD
5.5	B6	5B	22.0	85	7C	38.5	55	9D	55.5	31	BF
6.0	B5	5C	22.5	83	7D	39.0	53	9E	56.0	30	C0
6.5	B3	5D	23.0	82	7E	39.5	52	9F	56.5	2F	C1
7.0	B2	5E	23.5	80	7F	40.0	51	A0	57.0	2E	C2
7.5	B1-B0	5F	24.0	7F	80	40.5	50	A1	57.5	2D	C3
8.0	AF	60	24.5	7D	81	41.0	4E	A2	58.5	2C	C5
8.5	AE	61	25.0	7C	82	41.5	4D	A3	59.0	2B	C6
9.0	AC	62	25.5	7A	83	42.0	4C	A4	59.5	2A	C7
9.5	AB	63	26.0	78	84	42.5	4B	A5	60.5	29	C9
10.0	AA	64	26.5	77	85	43.0	4A	A6	61.0	28	CA
10.5	A8	65	27.0	75	86	43.5	48	A7	62.0	27	CC
11.0	A7	66	27.5	74	87	44.0	47	A8	62.5	25	CD
11.5	A5	67	28.0	72	88	44.5	46	A9	63.5	24	CF
12.0	A4	68	28.5	71	89	45.0	45	AA	64.0	23	D0
12.5	A2	69	29.0	6F	8A	45.5	44	AB	65.0	22	D2
13.0	A1-A0	6A	29.5	6E	8B	46.0	43	AC	65.5	21	D3
13.5	9F	6B	30.0	6C	8C	46.5	42	AD	66.5	20	D5
14.0	9E	6C	30.5	6B	8D	47.0	41	AE	67.5	1F	D7
14.5	9C	6D	31.0	69	8E	47.5	40	AF	68.5	1E	D9
15.0	9B	6E	31.5	68	8F	48.0	3F	B0	69.0	1D	DA
15.5	99	6F	32.0	67	90	48.5	3E	B1	70.0	1C	DC
16.0	98	70	32.5	65	91	49.0	3D	B2	71.0	1B	DE
16.5	96	71	33.0	64	92	49.5	3C	B3	72.0	1A	E0
17.0	95	72	33.5	62	93	50.0	3B	B4	73.5	19	E3
17.5	93	73	34.0	61-60	94	50.5	3A	B5	74.5	18	EF
18.0	91	74	34.5	5F	95	51.0	39	B6	75	17	FF
18.5	90	75	35.0	5E	96	51.5	38	B7			
19.0	8E	76	35.5	5D	97	52.0	37	B8			
19.5	8D	77	36.0	5B	98	52.5	36	B9			
20.0	8B	78	36.5	5A	99	53.0	35	BA			
20.5	8A	79	37.0	59	9A	53.5	34	BB			


Cabin Temperature versus CAN data

8.10 Sunload Sensor Module Interface (High Series)

Activation of the battery saver supply (section 8.11.2) also provides power to the sunload/twilight sensor module. With the ignition switch (S01) in the ignition position and the battery saver supply active, the BEM reads the single sunload sensor signal and transmits the raw data representing driver and passenger sunload readings over the CAN bus for benefit of the climate control system. Please refer to section 0 for description of the auto headlamp system including the use of the twilight signal. Invalid data is signalled with the Ignition switch in off or accessories on position or when a valid sunload reading is not available due to a fault. With Ignition on a fault condition will always default headlamp relays on if in Auto position.

The Input voltage, V_{SUNLOAD} , is the voltage seen at the connector pin of the BEM.

The formula to calculate the A/D value depending on the regulator voltage is:

$$\text{A/D} = (256 \times V_{\text{SUNLOAD}} / V_{\text{REGULATOR}}) - 1)$$

$$V_{\text{SUNLOAD}} = \text{Invalid data, TCAN} = \$\text{FF}$$

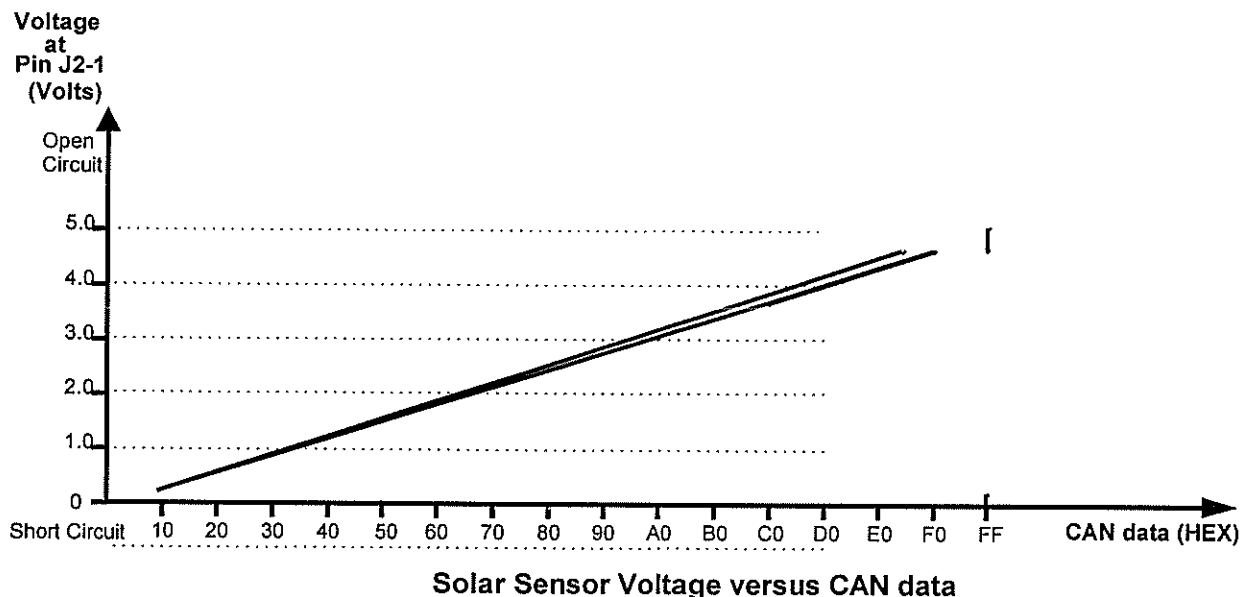
$$\text{for } (V_{\text{SUNLOAD}} > 0.2\text{V and } V_{\text{SUNLOAD}} < 4.7\text{V})$$

$$\text{for } (4.9 \leq V_{\text{REGULATOR}} \leq 5.1)$$

Note: Sunload Sensor Tolerances is $\pm 2.5\%$ (does not consider additional $\pm 2.5\%$ due to Sun Sensor Module)

V_{SUNLOAD} (Volts)	A/D (Hex)	TCAN (Hex)	V_{SUNLOAD} (Volts)	A/D (Hex)	TCAN (Hex)	V_{SUNLOAD} (Volts)	A/D (Hex)	TCAN (Hex)	V_{SUNLOAD} (Volts)	A/D (Hex)	TCAN (Hex)
5.1	FF	FF	4.3	D6-DF	D6-DF	3.5	AE-B5	AE-B5	2.7	86-8C	86-8C
5.0	F9-FF	FF	4.2	D1-DA	D1-DA	3.4	A9-B0	A9-B0	2.6	81-86	81-86
4.9	F4-FF	FF	4.1	CC-D5	CC-D5	3.3	A4-AB	A4-AB	2.5	7C-81	7C-81
4.8	F4-F9	FF	4.0	C7-CF	C7-CF	3.2	9F-A6	9F-A6	2.4	77-7C	77-7C
4.7	F4	FF	3.9	C2-CA	C2-CA	3.1	9A-A0	9A-A0	2.3	72-77	72-77
4.6	E5-F3	E5-F3	3.8	BD-C5	BD-C5	3.0	95-9B	95-9B	2.2	6D-71	6D-71
4.5	E0-EA	E0-EA	3.7	B8-C0	B8-C0	2.9	90-96	90-96	2.1	68-6C	68-6C
4.4	DB-E4	DB-E4	3.6	B3-BB	B3-BB	2.8	8B-91	8B-91	2.0	63-67	63-67

V _{SUNLOAD} (Volts)	A/D (Hex)	TCAN (Hex)	V _{SUNLOAD} (Volts)	A/D (Hex)	TCAN (Hex)	V _{SUNLOAD} (Volts)	A/D (Hex)	TCAN (Hex)	V _{SUNLOAD} (Volts)	A/D (Hex)	TCAN (Hex)
1.9	5E-62	5E-62	1.3	40-42	40-42	0.7	22-23	22-23	0.1	04	FF
1.8	59-5D	59-5D	1.2	3B-3D	3B-3D	0.6	1D-1E	1D-1E	0.0	00	FF
1.7	54-57	54-57	1.1	36-38	36-38	0.5	18-19	18-19			
1.6	4F-52	4F-52	1.0	31-33	31-33	0.4	13	13			
1.5	4A-4D	4A-4D	0.9	2C-2E	2C-2E	0.3	0E-0A	0E-0A			
1.4	45-48	45-48	0.8	27-28	27-28	0.2	09	FF			



8.11 Courtesy Lamp, Police Mode, Taxi Mode

8.11.1 Courtesy Lamp Operation with Illuminated Entry and Exit Delays

While the ignition switch (S01) is in the accessories, ignition or start position, and any door (including boot for wagon/E265) ajar status (S08, S09, S10, S11, S7) indicates that a door is open, then the courtesy lamp output will activate (VOL) with a short linear fade-in (T13). With the door (including boot for E265/Wagon) closed, it will deactivate (VOH) with a short linear fade-out (T13).

While the ignition switch is in the off position, and while a door (including boot for E265/Wagon) remains open, the courtesy lamp will remain active until the battery saver timeout has expired (T11 in Normal Mode and T12 in plant mode or intermediate plant mode). When all open doors (including boot for E265/Wagon) are closed, an exit delay shall occur, during which the courtesy lamp will remain active for a short time of T14. It will then fade linearly to the deactivated state over a period of T15. This exit delay shall be overridden by an open door (including boot for E265/Wagon), turning the ignition out of the off position, or by a remote key activation, or pressing of the dome lamp switch.

If the ignition key is removed from the ignition switch (i.e.: removed from the transponder magnetic field) within a time (T09) of turning the ignition switch to the off position and with no door (including boot for E265/Wagon), having been opened, then an exit delay shall occur. This function may be disabled at RBAU plant. If disabled this will override transponder related dome lamp functions.

While the ignition switch is in the off position, activation of the panic alarm by momentarily pressing the remote key 'Panic' (S05) button shall activate the courtesy lamps. Activation of the remote 'Door Unlock' (S02), 'Boot Release' (S04), or deactivation of the panic function by either the remote key or by time out shall turn on courtesy lamps causing an entry delay to occur, during which the courtesy lamp will remain active for a short time of T14. It will then fade linearly to the deactivated state over a period of T15. This delay shall be overridden by an open door (including boot for E265/Wagon), turning the ignition out of the off position, or by a remote key activation, or pressing of the dome lamp switch.

While the ignition switch is in the off position, momentarily pressing the remote key 'Door Lock' (S03) button to lock the doors, while they are closed, it will deactivate (VOH) with a short linear fade-out (T13).

If no remote key has been taught (i.e.: in plant mode or intermediate plant mode); then the courtesy lamp will deactivate after a shorter time than usual, T12, when the doors (including boot for E265/Wagon) are open, regardless of the position of the ignition switch.

When the courtesy lamp turns on, it will have a short linear fade-in (T13). Likewise, when lamp turns off, it will have a linear fade-out (T13). If any fade-in or fade-out period is interrupted by a change of state, then the fade shall start from the current brightness level and only last the time it takes to complete the action.

The courtesy lamp switch via CAN (S23) can be used to turn on and off the courtesy lamp, independent of other conditions. BEM will act on positive edge of the courtesy lamp switch.

8.11.2 Battery Saver Supply

When the battery saver supply is active (VOES), it allows the courtesy lamps, the boot lamp and the glovebox lamp to illuminate. Also the TILA supply shall also be active to ensure that the keyhole illumination. Battery saver also provides power to sun load sensor and antenna for audio.

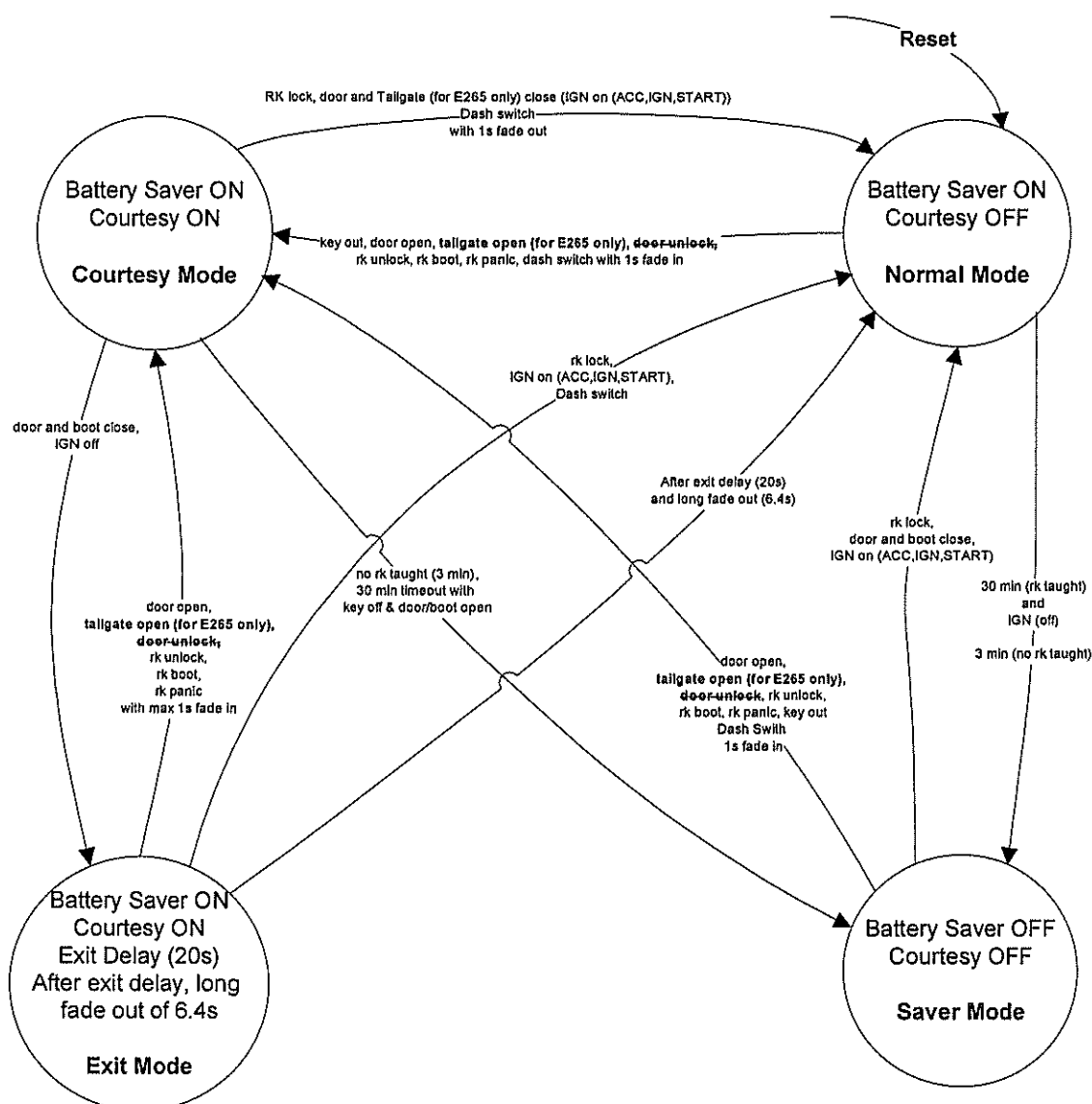
The battery saver supply is activated when the ignition switch (S01) is in the accessories, ignition or start position. The battery saver supply is also activated by the opening or closing a door/boot (S08, S09, S10, S11, S7), or pressing the remote key 'Door Unlock' (S02), 'Door Lock' (S03), 'Boot' (S04) or 'Panic' (S05) button. The battery saver supply is deactivated (VOUS) after a time (T11) with the ignition switch being turned to the off position as long as the last event of opening or closing a door/boot or pressing any remote key button has occurred.

If no remote key taught (i.e.: in plant mode or intermediate plant mode), the battery saver supply will deactivate after a shorter time period of T12 regardless of the position of the ignition switch. The battery saver timeout is re-enabled whenever the ignition switch is turned between the off and accessories positions, a door/boot is opened or closed, or a remote key door or boot release functions are operated.

If the Battery saver supply is deactivated, and Audio Keyless Play is activated, then if BEM is not in remote key plant mode, the Battery Saver can be re-enabled by bit 39 in CAN message 307 from audio module.

The table below shows six states for the courtesy lamp and battery saver supply.

State	Summary of State
Normal Mode	Battery Saver ON, Courtesy OFF
Saver Mode	Battery Saver OFF, Courtesy OFF
Courtesy Mode	Battery Saver ON, Courtesy ON
Exit Mode	Battery Saver ON, Courtesy ON. This consists of T14 exit delay and T15 fade out.
Audio Keyless Play	Battery Saver ON, Courtesy OFF or ON
Police Mode	Depends on the configuration (refer 8.11.3.2)



State diagram for courtesy lamp and battery saver supply control
 For any discrepancies, the functional description supercedes the above state diagram

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8.11.3 Special Modes

8.11.3.1 Door Trigger Disable Modes

When this mode is activated, dome lamps will not be turned on by :

- Any Door ajar
- Boot ajar (Wagon/E265)
- Transponder removal from ignition
- BEM reset

In this mode, dome lamps can be turned on by:

- World Diagnostics System (WDS)
- Remote unlock (for time out)
- Remote panic (for time out)
- Remote boot (for time out)
- Dash lamp switch

In this mode, dome lamps can be turned off by:

- World Diagnostics System (WDS)
- Remote Lock
- Remote panic (for time out)
- Dash lamp switch
- Door closing (for time out)
- Boot closing (Wagon/E265 for time out)
- Battery saver time out

The following methods can be used to enable/disable *Door Trigger Disable Mode*:

- Dash Switch is pressed and held for T17.
- Setting/Clearing *bit 5* in *Police/Taxi Mode Control Byte* by using WDS (refer section 8.11.3.5)

Courtesy lamp activation for remote key unlocks can be enabled/disabled by setting/clearing *bit 0* in *Police/Taxi Mode Control Byte* (refer section 8.11.3.5).

If *Master Door Trigger Disable Mode* is enabled by setting bit 4 in *Police/Taxi Mode Control Byte*, it will inhibit toggling bit 5 (*Door Trigger Disable Mode*) by pressing the dash switch. *Master Door Trigger Disable Mode* bit can be set/cleared only by WDS (refer section 8.11.3.5).

8.11.3.1.1 Acknowledgement for enabling/disabling Door Trigger Disable Mode

Acknowledgement for enabling/disabling *Door Trigger Mode* is done by setting *Buzzer Request bit 18* in *Message ID \$403* for T55, and pulsing courtesy lamp. Courtesy lamp pulsing is done as shown in the table below.

Action	Acknowledgement by courtesy lamp
Disabling <i>Door Trigger Disable Mode</i>	
Enabling <i>Door Trigger Disable Mode</i>	

After disabling *Door Trigger Disable Mode*, courtesy lamp will pulse and then stay on. After enabling *Door Trigger Disable Mode*, courtesy lamp will pulse and then turn off.

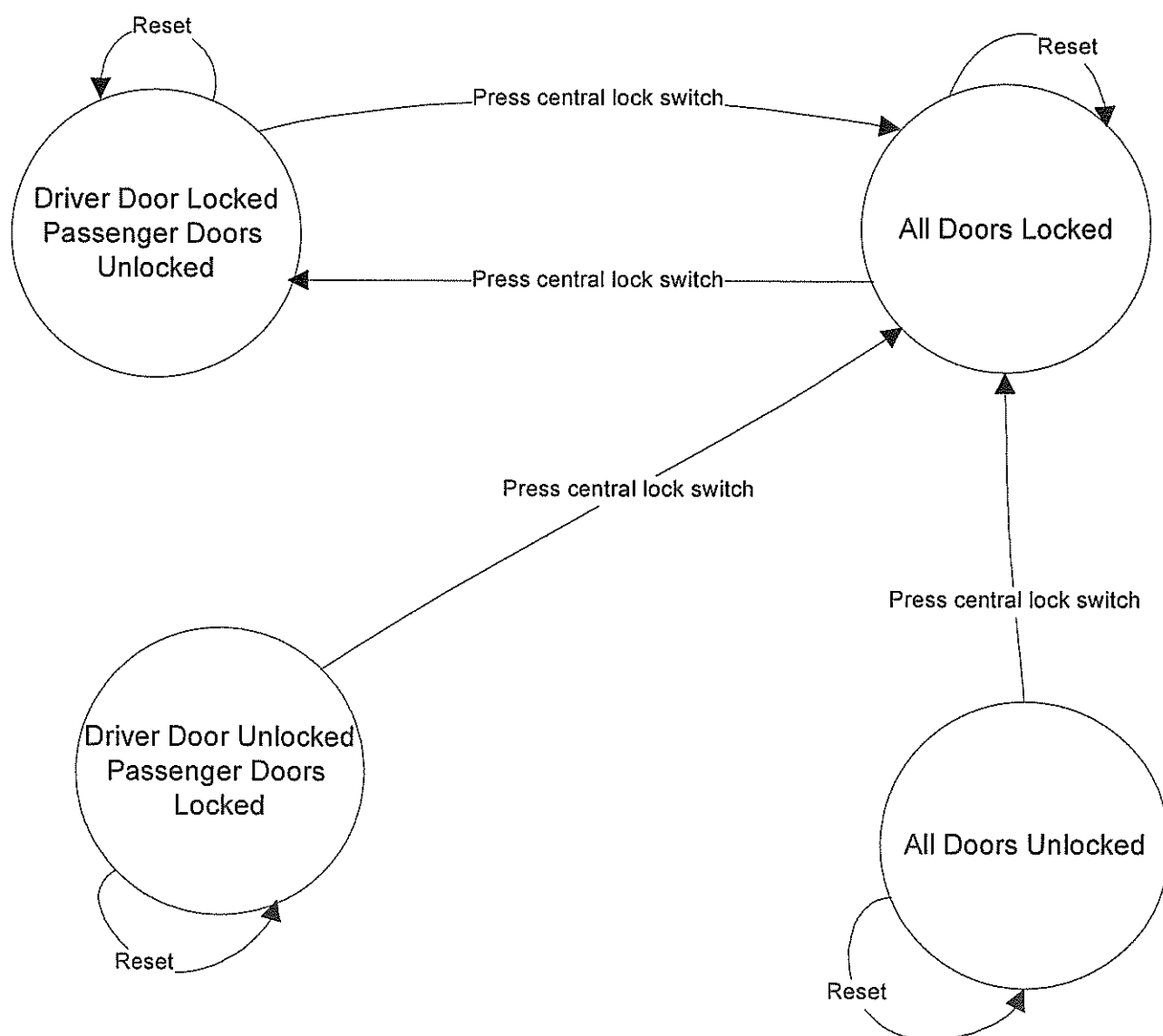
8.11.3.2 Police Mode

Police mode is a subset of the courtesy lamp and battery saver operation explained in sections 8.11.1 and 8.11.2. Any combination of courtesy lamp, battery saver, horn warning and hazard warning can be enabled/disabled by setting/clearing bits (bit 0, bit 1, bit 2, and/or bit 3) in the *Police/Taxi Mode Control Byte* (refer section 8.11.3.5). For example, both courtesy lamp and hazard warning will not activate for remote key unlock, if the courtesy lamp and hazard warning are disabled.

8.11.3.3 Taxi Mode

If any door is unlocked, one press of central door lock dash switch will lock all doors. If all doors are locked, one press of central door lock dash switch will unlock only the passenger doors. Refer state diagram for door lock status in Taxi Mode for more details. Taxi Mode can be enabled/disabled by setting/clearing bit 7 in the *Police/Taxi Mode Control Byte* (refer section 8.11.3.5).

After system reset, the BEM software checks the passenger door ajar inputs. If any passenger door is ajar, the software will assume the passenger doors are unlocked.



State diagram for door lock status in Taxi Mode

For any discrepancies, the functional description supercedes the above state diagram

8.11.3.4 Panic Alarm Disablement

Panic alarm can be enabled/disabled by setting/clearing *panic alarm bit* in the *Police/Taxi Mode Control Byte* (refer section 8.11.3.5). If it is disabled, panic button will not activate the panic alarm.

8.11.3.5 Police/Taxi Mode Control Byte

The following table defines the bits in the *Police/Taxi Mode Control Byte*.

Bit	Description	Enable	Disable	Police Mode
Bit 7	Taxi Mode	1	0	
Bit 6	Panic Alarm Inhibit	1	0	
Bit 5	Door Trigger Disable Mode.	1	0	
Bit 4	Master Door Trigger Disable Mode	1	0	
Bit 3	Hazard Warning Inhibit	1	0	☐
Bit 2	Horn Warning Inhibit	1	0	☐
Bit 1	Battery Saver Inhibit	1	0	☐
Bit 0	Courtesy Lamp Inhibit	1	0	☐

This control byte is configured using WriteDataByCommonIdentifier(\$2E) service with CommonIdentifier E11B.

With Battery Saver Inhibit is disabled, Battery Saver will be off with Ignition and Accessories off and will remain on with any Accessories and Ignition on including the Start position where Ignition is on and Accessories is off. To reset the BEM to default mode, all bits in *Police/Taxi Mode Control Byte* need to be cleared (writing 0x00 the control byte).

8.12 Central Door Locking Related Functions

8.12.1 Door and Boot Ajar Status

The BEM reports the door ajar status for all four individual doors at all times (S8, S9, S10, S11), subject to the conditions for CAN message transmission, section 8.18. The BEM also reports the boot ajar signal (S07).

8.12.2 Boot/Tailgate Release

8.12.2.1 Manual Boot/Tailgate Release

The manual boot/tailgate release switch (S12, momentary), when enabled, will activate (VOL) the boot/tailgate release solenoid for a time of T18 and then deactivate (VOH).

The switch is enabled during the following situations:

- Any door is in Unlock State (BEM determines lock state of passenger doors by tracking operation of passenger door lock/unlock activations).
- When the ignition switch (S01) is in the Accessories position.
- When battery saver is active, the manual boot/tailgate release switch will only be enabled if the vehicle speed, as determined via CAN, indicates that the vehicle is moving at a speed of no more than 12 km/h. At speeds above this, the switch is disabled.
- If there is no CAN data (speed data), the BEM defaults to 0 km/h, thereby allowing the boot to be activated. BEM will continue to monitor the CAN line for valid data. Upon receipt of valid data the valid vehicle speed will be used. No CAN data means the PCM message is missing or the PCM is sending "invalid data" (\$FF) message.
- After BEM reset with ignition on it is assumed the vehicle speed is 0km/h until a valid CAN message is received.

The switch is disabled under the following situations:

- All doors locked with battery saver timer timed out. Note: In police modes battery saver timer is still to operate, even when battery saver is disabled.

Boot or Tailgate Release Output Conditions					
Action	Activation Methods	Output		Time at V _{ous}	Time at V _{OES}
Boot/tailgate solenoid release, if switch is enabled	Press boot/tailgate release switch (S9)	Boot or tailgate release		-	T18

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8.12.2.2 Manual and Remote Boot/Tailgate Release for E265 (Proposal)

BEM controls the power to solenoids for E265 Tailgate and Rear window. Solenoids are activated by separate switches external to BEM.

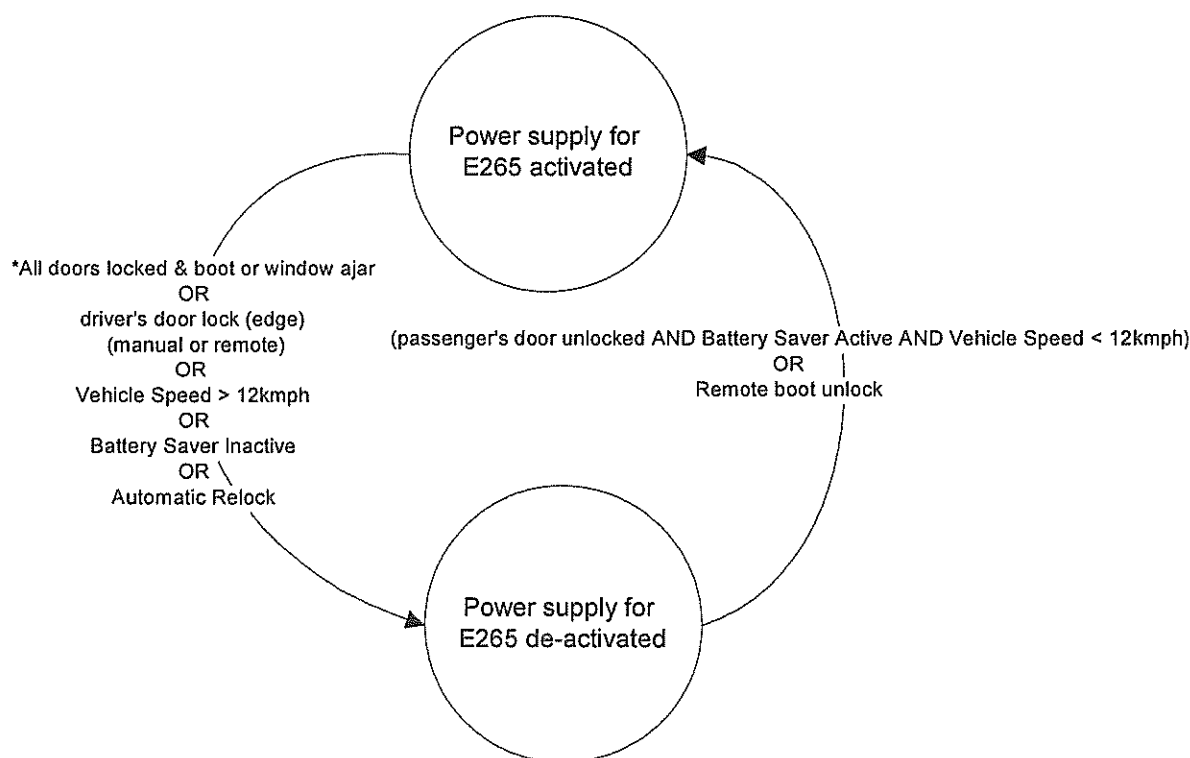
Power supply to solenoids activated due to one of the following conditions:

- Passenger door is unlocked (manually or by remote key) for time when battery saver is active and vehicle speed is less than 12kmph.
- Remote boot release key pressed.

Power supply to solenoids deactivated due to one of the following conditions:

- All doors are locked, boot/tailgate window ajar and this function enabled.
- Driver's door lock edge due to manual or remote key operation.
- Vehicle speed is greater than 12kmph.
- Battery saver timed out.
- Automatic relock after remote unlock or remote release and neither tailgate nor doors are opened for 45 seconds.

Solenoids cannot be activated by remote key. There will be no boot release input from IP to BEM on E265, but sedan and wagon will both have it.



State diagram for E265 power supply
For any discrepancies, the functional description supercedes the above state diagram

*This deactivation method is enabled by setting bit 0 of E265 Control byte. If this bit is cleared then this deactivation method is not enabled. This bit can only be set by BOSCH Plant. Current strategy is to have this deactivation method enabled.

Read body configuration using ReadDataByCommonIdentifier service (\$22) with PID equal to \$E11A.

State	Description	Comment
00	Unknown Body Type	If a door (excluding 5th door) is ajar and remote lock is requested, all doors will unlock and horn alert will occur.
01	Sedan	If a door (excluding 5th door) is ajar and remote lock is requested, all doors will unlock and horn alert will occur.
02	Wagon	If a door (excluding 5th door) is ajar and remote lock is requested, all doors will unlock and horn alert will occur.
03	Utility	
04	LWB	
05	4WD (E265)	If a door (excluding 5th door) is ajar and remote lock is requested, all doors will unlock (and power will be supplied to tailgate and window) and horn alert will sound. If the 5th door input is active when a remote lock is requested, then the horn warning will sound and other doors will remain locked.
06	Coupe	
07	Van	

Body type is configured using WriteMemoryBlock service (\$3B) for Block \$00. RBAU plant will set the body type as \$00 to identify the module as not been configured.

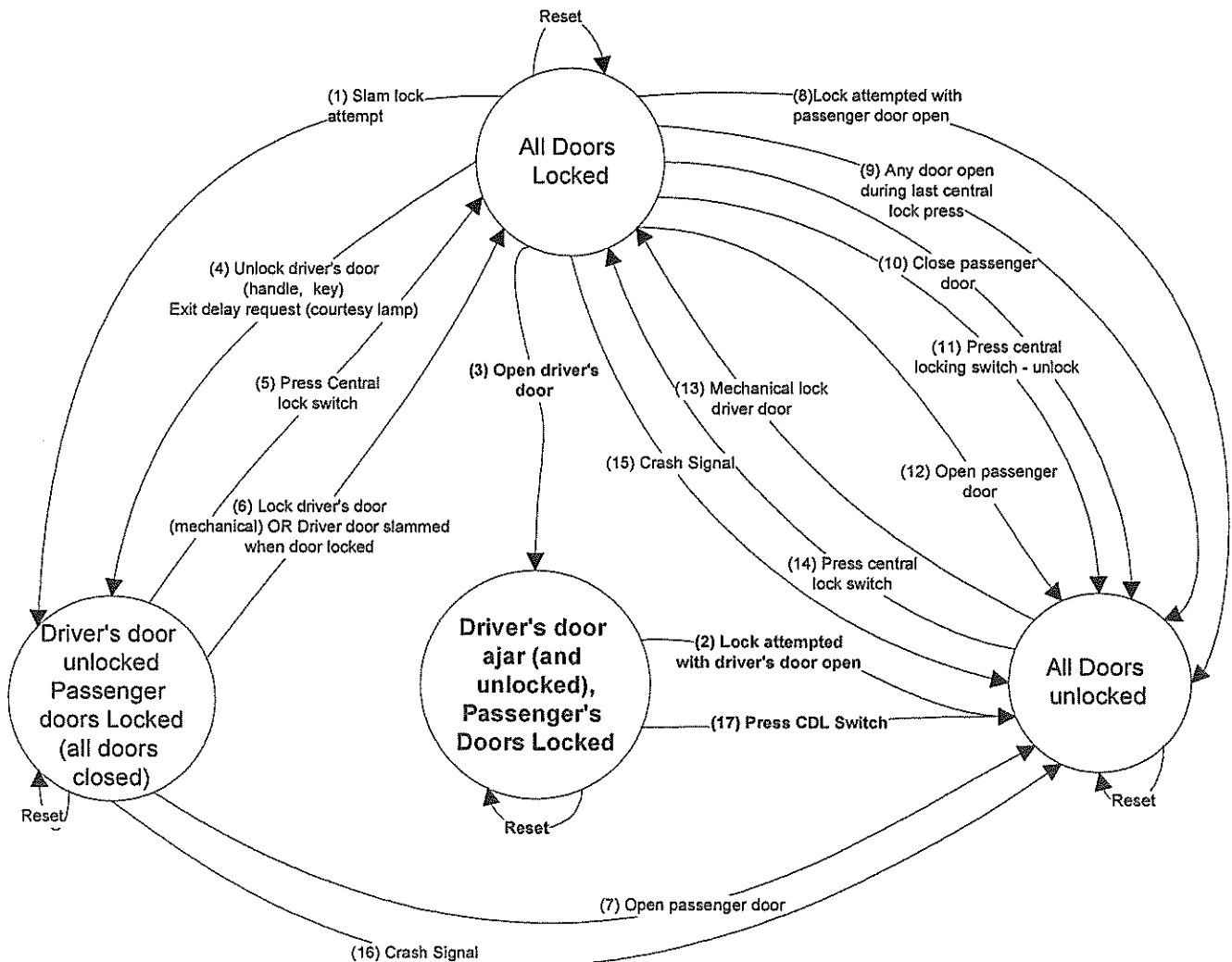
8.12.3 Manual Central Locking Functions

Manual central locking operation may be initiated by pressing the central door locking switch via CAN (S13, momentary); manually operating the driver's door lock mechanism (S06) by using a DST key, soldier button or interior door handle; or by opening a door (S08, S09, S10, S11).

A crash (identified by the air bag deployed signal via CAN with IGN on) triggers an unlock action. All door ajar switches may return erroneous information during & after a crash. Once a crash unlock has occurred, then subsequent door switch inputs shall be ignored including the manual Central Locking switch bit via CAN, until the next ignition switch cycle occurs from off to out of Ignition and Accessories off state.

When a slam door is detected, unlock driver's door only, unless a passenger door is open. The driver's door is detected as slammed, if it has been just closed and the door is locked. When driver's door is ajar and central lock switch is pressed or soldier button of driver's door is pressed, all door unlock will be activated even if passenger doors are in unlocked state.

After power up or a reset, lock status will not be changed to avoid lockout risks or doors unlocking on power up, and it is assumed that passenger doors are in locked state.



For any discrepancies, the functional description supercedes this state diagram

State diagram for door lock operation

Operation is summarised in the following table (Refer Sect 8.11.3.3 for Taxi Mode operation):

Action	Activation Methods (<i>Transitions</i>)	Output		Time at VOUS	Time at VOES
Driver's door unlocks	- Mechanically unlock driver's door (S06) (4)*. - Open driver's door (S08) when it is locked (3)*. - Lock attempted with driver's door open (2). - Driver's door slammed when door locked (16), (1).	Driver's door unlock line		—	T19
All doors unlock	- Press central door locking switch (S13) with driver's door opened (S08) and passenger door locked (17). - Press central door locking switch (S13) with driver's door locked (11). - Open a passenger's door with driver's door locked (S9, S10, S11) (7) or (12)*. - Lock attempted (driver door) with a passenger door open (S9, S10, S11) (8). - Passenger door is closed with driver's door locked (10). - Crash indicated (Air Bag deployed via CAN) (15) or (16).	Door unlock line		—	T19
		Driver's door unlock line **		—	T19
All doors lock	- Press central door locking switch (S13) with driver's door unlocked and closed (5) or (14). - Mechanically lock driver's door (S06)* (6) or (13)	Door lock line		—	T19

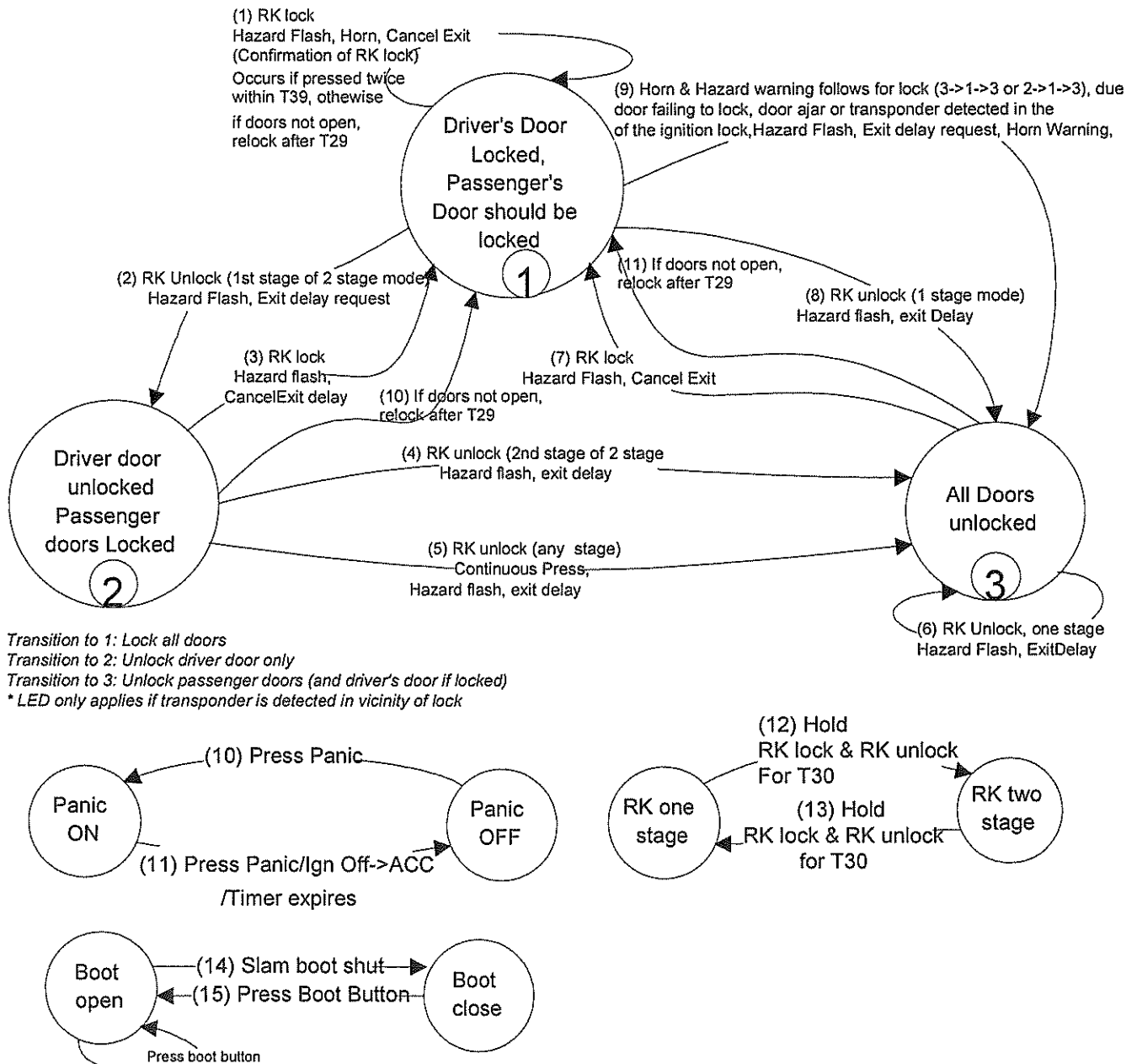
** If driver's door is already unlocked, then driver's door unlock line is not activated.

8.12.4 Remote Central Locking Functions with Boot Release

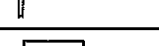
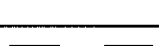
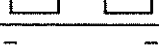
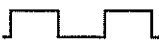
Remote entry and Panic alarm functions operate when the ignition switch (S01) is in the off position, although the RF receiver is active at all times. A routine to distribute EEPROM wear is employed for frequently changing bytes (rolling code counters).

Two-stage unlocking may be activated or deactivated for each remote key. If one-stage unlocking is active, then a single press of the 'Door unlock' button shall unlock all doors. If two-stage unlocking is active, then only the driver's door shall unlock on the first remote key press of the 'Door unlock' button. The remaining doors shall open in a second press, or if the button is still held down once the first press is complete. To toggle between one stage and two stage unlocking, press both remote key "door lock" and "door unlock" buttons simultaneously for T30. The status for each remote key is stored in EEPROM and defaults to one-stage mode. Changing modes is the only instance where multiple button presses are acknowledged.

Pressing the 'Panic' button activates the panic alarm. If pressed, the panic alarm shall operate until the ignition is switched out of the off position, the button is pressed a second time, or the panic alarm period times out (T26). Hazard lamp flashing is activated via CAN.


State diagram for remote central locking functions

For any discrepancies, the functional description supercedes the above state diagram

Action	Activation Methods (<i>Transitions</i>)	Output		Time at VOUS	Time at VOES
Remote driver door unlock	Press the remote 'Door Unlock' button (S02) when driver's door is locked with two stage unlocking active (2).	Driver's door unlock line		—	T19
		Hazard lamps via CAN		—	T20
Remote all doors unlock	Press the remote 'Door Unlock' button (S02) when driver's door is locked with one stage unlocking active (8).	Driver's door unlock line		—	T19
		Other door unlock line		—	T19
		Hazard lamps via CAN		—	T20
Remote passenger doors unlock	- Press the remote 'Door Unlock' button (S02) with the driver's door unlocked and two stage unlocking active (4), (6). - Continue to hold the remote 'Door Unlock' button (S02) after the driver door has unlocked for 2 seconds with two stage unlocking active (5), (6).	Other door unlock line		—	T19
		Hazard lamps via CAN		—	T20
Remote door lock	Press the remote 'Door Lock' button (S03) when the driver's door is unlocked (3), (7).	Door lock line		—	T19
		Hazard lamps via CAN		T20	T20
Confirmation of remote door lock	Press the remote 'Door Lock' button (S03), when all doors are locked. Horn occurs only on second press within T39. (1).	Door lock line		—	T19
		Hazard lamps via CAN		T20	T20
		Horn relay		—	T24
Boot/Tailgate solenoid release	Press remote 'Boot' button (S04) (15).	Boot release		—	T18
		Hazard lamps via CAN		—	T21
Panic alarm activation	Press the remote 'Panic' button (S05) (10).	Hazard lamps via CAN		T27	T27
		Horn relay		T27	T27
Panic alarm deactivation	- Press the remote 'Panic' button (S05) (11). - Turn the DST ignition key out of off position (S01) (11). - Panic alarm timer expires (T26) (11).	Hazard lamps via CAN		—	—
		Horn relay		—	—
Toggle between one stage and two stage unlocking	Press both 'Door Lock' and 'Door Unlock' buttons simultaneously for T30 (12), (13)	Hazard lamps via CAN		T20	T20

Note: Multiple remote key presses are ignored except for mode changes.

8.12.5 Remote Lock Horn and Hazard Warning

If the remote key is used to lock the vehicle (S03, momentary), a number of checks are made to ensure it is secured. A warning shall sound if the BEM detects an open door (S08, S09, S10, S11), an unlocked driver's door (S06), or if a transponder is in the vicinity of the steering lock. This last cause would mean that either a DST ignition key had been left in the lock, or that a transponder had been left near the lock for the purposes of theft while the vehicle is unattended. If the warning is due to a transponder detected by the steering lock, then the CAN message (See Section 11), to activate the LED, will also remain on. The CAN message, activating the LED, reverts back to normal operation if a subsequent remote lock is successful, or if the ignition switch is turned out of the off position.

Activation Method	Output	Time at VOUS	Time at VOES
Remote door lock where a door is ajar (open), a door fails to lock or a transponder is detected in the vicinity of the ignition switch (Transition 9)	Door lock line	—	T19
	Door unlock lines (a door ajar or door fails to lock)	—	T19
	Door unlock line(driver) (transponder detected)	—	T19
	Hazard lamps via CAN	T20, T23, T23	T20, T20, T23, T23
	Horn relay	T25	T24
	CAN message to activate LED output (if BEM detects a transponder)	—	—
	Courtesy lamp output (if BEM detects a transponder)	—	Exit delay

8.12.6 Automatic Door Re-lock

If the remote key is used to unlock the vehicle (S02, momentary) and then no door is opened (S08, S09, S10, S11), the driver's door is not locked (S06), or the ignition switch (S01) is not turned, then the doors will automatically re-lock after a period of time, T29.

8.13 RF Remote Control Related Functions

The main function of this module is to recognise a control message from a trained remote key, and pass on the recognised key values and key edges to other software modules. In addition, it also provides key training function for matching with the right remote key and resynchronisation criteria if the trained remote key falls out of the synchronized range.

8.13.1 RF Control Features

RF signals are received by the BEM via the Barra Receiver Module (BRM), and responds to remote keys that conform to the Shark protocol (as referred to in section 3.2). In addition to detecting button presses for remote entry functions, remote key training and resynchronisation features are included.

The RF receiver is powered at all times, although the BEM will only respond to a trained remote key when ignition is OFF, or during remote key training process when the ignition switch (S01) is in the accessory position.

The rolling code counters incorporated within the Shark protocol are used to establish the legitimacy of the remote key transmission and to determine new presses (edges) in environments where packets are not correctly received during a single transmission. The counter bytes are stored in non-volatile memory once in 32 remote key presses, as suggested in the Shark protocol. Intermediate counters are in RAM.

The remote entry receiver is capable of passively re-synchronising to a transmitter in one of two ways if they become out of synchronisation. The transmitter is considered out of synchronisation if it is more than 1024 counts greater than the receiver is.

A receiver may passively resynchronise to a single programmed transmitter in two ways:

- If the transmitter's counter is less than 45,055 counts greater than that of the receiver, the TIC matches and three consecutive transmissions are received in a row.

- If the transmitter that is out of synchronisation transmits within T52 seconds of another transmitter that is trained to the BEM and is valid.

In plant mode, the BEM responds only to a master remote key (TIC code \$A0 00 00 00 with rolling code disregarded). Additional remote keys cannot be taught to the BEM whilst in plant mode. After exiting from plant mode normal remote key teaching can be performed. Once the master remote key is erased from memory, it can never be re-taught to a BEM. Master remote keys are only produced for the use of Bosch, Sumitomo and Ford plants.

The Barra BEM only responds to 5 different control messages from a trained 433MHz RF remote key, which sends out SHARK protocol. They are

- Door Lock
- Door unlock
- Tailgate/Boot release
- Panic
- (Door lock) + (Door unlock)

RF receiver is being polled at a rate of 24ms ON/ 40ms OFF (See Section 5.6 for Standby current) at all time. BEM will only respond to trained key when ignition is OFF, or during key training process when ignition is in the accessories position.

8.13.2 RF Controlling Codes

All control codes only operate when the ignition switch is at the off position.

Lock all doors.

- Door lock (control code = 0x10)

Unlock all doors if one stage unlocking is activated. Only unlock driver's door from all door locked if two stage unlocking is activated, the next press to the unlock key (or holding unlock key for 2 seconds) will unlock all doors.

- Door Unlock (control code = 0x20)

Release tailgate/boot.

- Tailgate/boot release (control code = 0x40)

Initiate/stop car alarm system.

- Panic (control code = 0x08)

Toggle between 1 stage door unlocking system or 2 stages door unlocking system.

- Door Lock key together with Door unlock key (control code = 0x30)

Table A

Key Icon	Control code in hexadecimal form	Control code in binary form
Lock	0x10	0001 0000
Unlock	0x20	0010 0000
Tailgate/Boot Release	0x40	0100 0000
Panic	0x08	0000 1000
Lock + Unlock	0x30	0011 0000

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8.13.3 Counter Bytes

The counter is two bytes of codes that will be incremented with each separate button activation. The bits of these two bytes will be interleaved to form the rolling code bytes used for encryption. The counter bytes provide the functions of

- ensuring 1 action per button pressed (if required)
- encrypting the RF message

With continuously updating to this counter, it is the best option to use this counter as a random number source for message encryption. Refer to document RKE Protocol SDS – Release 1.3 for description of data encryption.

The counter bytes are written to the EEPROM of the BEM. To avoid wearing out of the EEPROM, the counter byte will be written to EEPROM once in 32 remote key presses. Within these 32 presses, the counter bytes are written to the volatile RAM of the microprocessor.

The BEM has the ability to remember the details of the last command received, and is also able to tell which of the trained remote keys it is. This information can be obtained from BEM via diagnostic command (Refer to Engineering Specification FOA Barra Body Electronics Module High and Low series F005 VP 0058, section 10 for details).

8.13.4 Remote Key Teaching

In other cases (plant mode, intermediate plant mode and normal operating mode), switching ignition switch from the off position to the accessories position activates a short time window (T32). If three momentary activations of the rear demist switch are detected in this time, Remote Key Teach Mode is entered.

Acknowledgment that the system has entered Teach Mode is via the mode acknowledgment sequence (see section 8.17.2 for further details). Dealer may use a diagnostics command that is available to simulate the pressing of the switch. To use this command, the ignition switch must be placed in the accessories position before the command is sent.

Once teach mode has been entered, all remote key codes stored in the BEM's non-volatile memory will be cleared. Upon receipt of a valid remote key code, the system will write this code to non-volatile memory and acknowledge receipt of this code is also discussed in section 8.17.1. Receipt of an already acknowledged code will result in no action. A maximum of eight unique remote keys may be taught. Teach mode will be exited if the ignition switch is turned out of accessories into the off or ignition position. Acknowledgement that training mode has been exited is via the mode acknowledgement sequence (section 8.17.1).

8.13.5 Plant Mode

In Plant Mode, the BEM responds only to a Master Remote Key (code \$A0 00 00 00 as TIC), additional Remote Keys cannot be taught to the BEM whilst in Plant Mode. As there is only one working remote key allowed in Plant mode, EEPROM byte "RFKeyCount" is set to 0x55 indicating that it is in plant mode. Upon entering key training mode, this EEPROM byte will be set to 0x80.

8.13.6 Master Remote Key

Master remote key only operates when the BEM is in plant mode. After exited from plant mode, the master remote key cannot operate any function of the BEM, nor can the Master remote key be trained as an operating remote key. The TIC of the Master remote key (\$A0 00 00 00) is considered to be unique for Master remote key and will not be used for any other remote keys. Master remote key does not need to be re-synchronised. BEM will respond with an action, to a signal from a Master remote key, with a different rolling code.

8.13.7 Receiver Resynchronization

The remote entry receiver is capable of passively resynchronizing to a remote key in one of two ways if they become out of synchronization. The remote key is considered out of synchronization if it is more than 1024 counts greater than the receiver. A receiver may passively resynchronize to a single programmed remote key if the remote-key's counter is less than 45,055 counts greater than that of the receiver, the TIC matches and three transmissions are received in a row, with the transmitted counter incrementing only one time with each transmission. A receiver may also passively resynchronize if a recognized out of synchronization remote key sends a transmission within 30 seconds of a different valid remote key's transmission.

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8.13.8 Intermediate Plant Mode

In the first stage plant mode, a Master Remote Key will be able to control all vehicles installed with a BEM set to plant mode. To prevent Master Remote Key access to a plant-mode BEM installed vehicle, an intermediate plant-mode is set up. In this mode, access to the car has to be done through the mechanical key. The BEM will enter intermediate plant-mode state by setting of a diagnostic PID via the CAN line. Once this intermediate plant-mode state is entered, it can never be reversed back to the first state plant mode.

8.13.8.1 Entering Intermediate Plant Mode

When BEM receives diagnostic command WriteDataByCommonIdentifier (\$2E) with PID \$C901 set to \$01 via the CAN bus, the BEM will then enter this intermediate plant mode. PID \$C901 cannot be set to any other values by WDS. PID \$C901 will be ignored during normal operating mode. After Intermediate plant mode is entered, EEPROM address

0x820 = 0x80

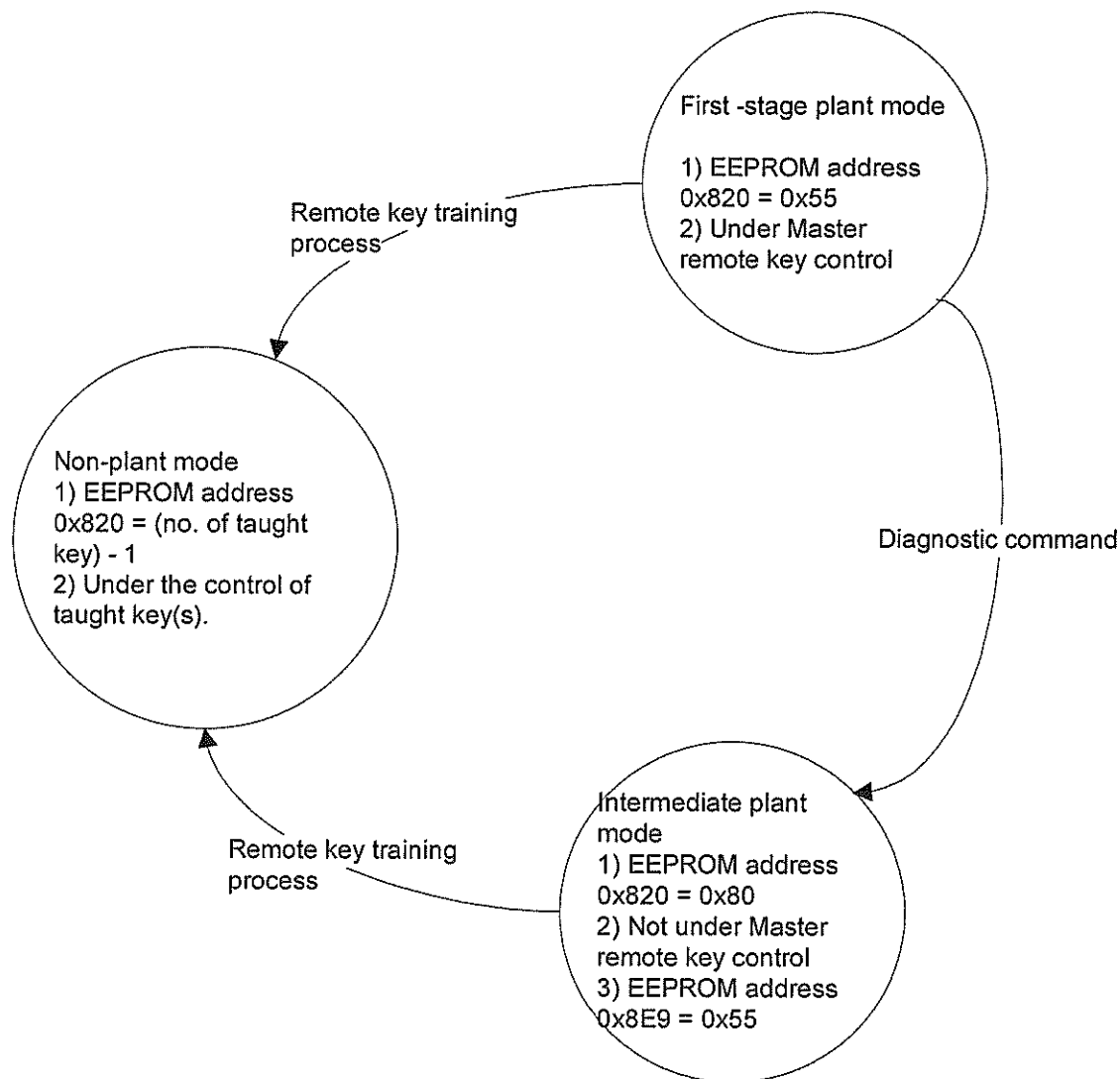
0x8E9 = 0x55

8.13.8.2 Differences between first stage Plant Mode and Intermediate Plant Mode

In the intermediate plant mode, BEM is no longer under Master Remote Key control, all other plant mode features remain unchanged. Refer to Engineering Specification FOA Barra Body Electronics Module High and Low series (F005 VP 0058) for features in the first stage plant mode (Section 8.16.1) Same as Grizzly with addition of seat memory communications.

8.13.8.3 Exiting Intermediate Plant Mode

To exit the intermediate plant mode, normal remote keys (not the Master Remote key) training mode has to be involved. Remote key can be trained to the BEM using the standard training procedure (Refer to Engineering Specification FOA Barra Body Electronics Module High and Low series F005 VP 0058, section 8.13.4 for details) or through diagnostic control (Refer to Engineering Specification FOA Barra Body Electronics Module High and Low series F005 VP 0058, section 10 for details). Once the teaching process is completed, the BEM is considered to have come out of the intermediate plant mode and entered normal operating mode. The same key teaching process will also get the BEM to come out of plant mode from first stage plant mode, thus skipping the intermediate plant mode process. After entering key training process, BEM will not get back to Intermediate plant mode with the use of WDS tools alone. Training zero remote key to the BEM will also bring the BEM out of intermediate plant mode or normal plant mode.

**Intermediate Plant Mode State diagram****For any discrepancies, the functional description supercedes the above state diagram**

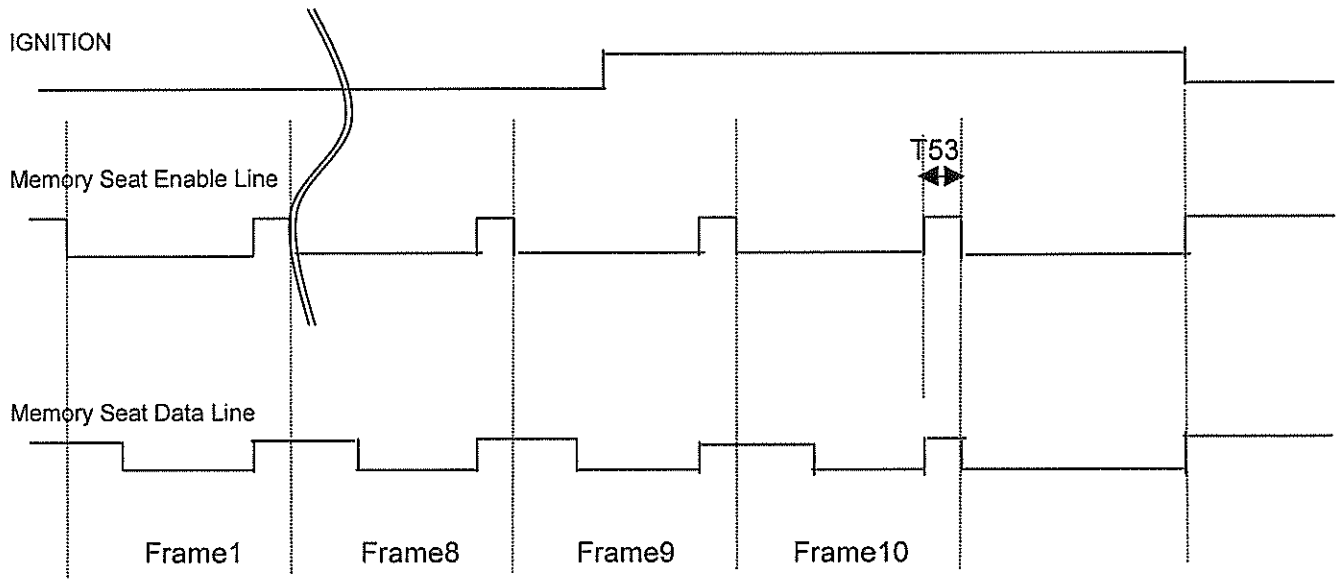


Figure showing Multiple Frames sent – Ignition goes low after 10 frames sent

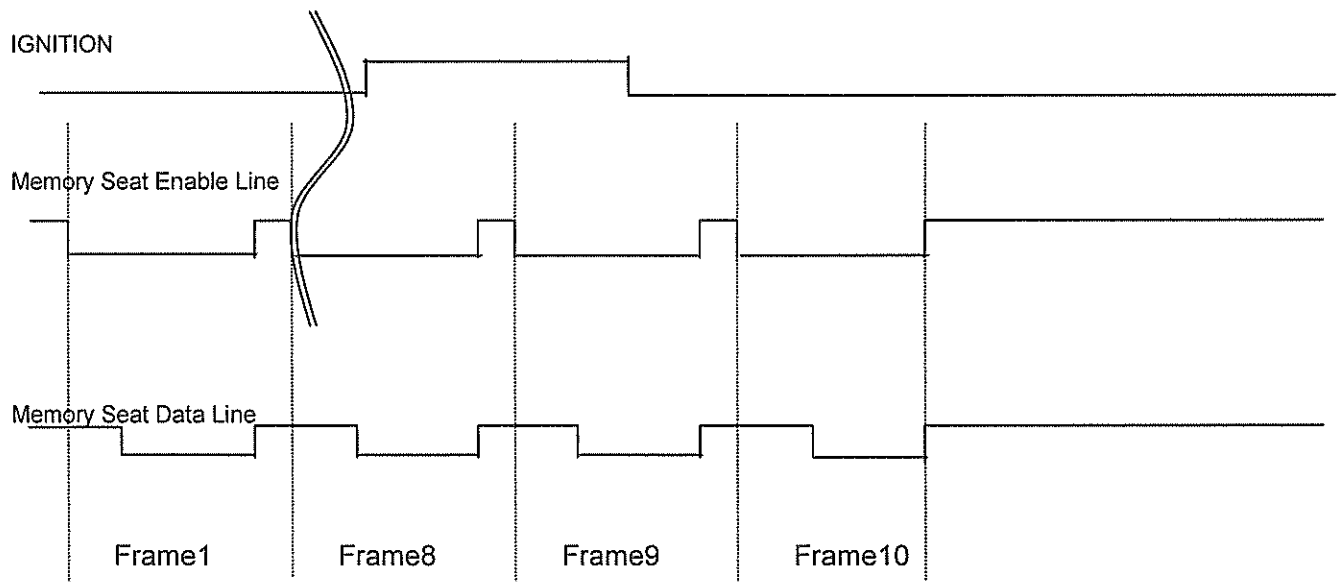


Figure showing Multiple Frames sent – Ignition goes low before 10 frames sent

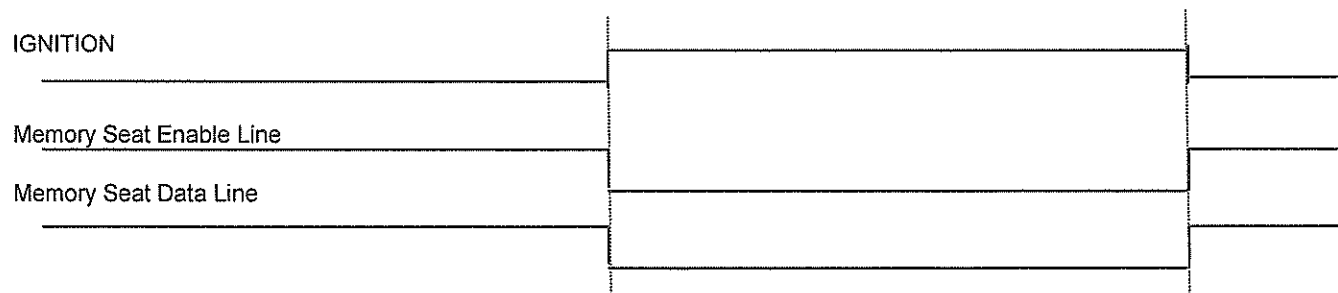


Figure showing signal when no memory seat data is being transmitted.

Message \$403 BEM

Signal Name	Bit used	After Reset	After first remote key unlock press
Priority key 1	49	0	1
Priority key 2	50	0	0
Priority key 3	51	0	0
Priority key 4	52	0	0
Priority key 5	20	0	0
Priority key 6	21	0	0
Priority key 7	22	0	0
Priority key 8	23	0	0
Master Remote Key/Plant Mode	None	None	None

8.14.2 Signalling in Remote Key Plant Mode

While the vehicle responds to the master remote key, no priority key data shall be sent over CAN. If the ignition switch is turned to the ignition or start positions then memory seat frames corresponding to the master remote key shall be transmitted continuously. Once the ignition switch is turned to accessories or off, then the last frame shall be completed and communication shall stop. No frames shall be transmitted in response to an unlock command from a master remote key.

8.15 Automatic Headlamps with Twilight Sensing and Delayed Turn-off (High Series)

When the ignition key (S01) is turned to the ignition or accessories positions, setting the lamp switch to the automatic headlamps position (S21) can enable automatic headlamps control. Various factors like twilight sensor status, ignition on/off and driver door status (S08) will decide headlamps on/off when lamp switch is at "AUTO" position. If "AUTO" is not selected, the mentioned factors can only affect the BEM headlamp control output, but not the headlamp on/off itself. Table A below summarises the conditions contributing to BEM headlamp output decision

The ambient brightness level is determined by reading the frequency signal from the twilight sensor. To prevent unnecessary activation and deactivation of the headlamps, hysteresis and delay times are used. The BEM has provision for eight different sets of ambient light thresholds in EEPROM to be set in the Bosch plant and configured by Ford or dealers.

Table A

Case	Twilight sensor status	Ignition or Accessory (S1)	Driver Door Ajar (S8)	BEM Headlamp relay control output
1	Light	On	N/A	Off
2	Light to dark	On	N/A	Delayed off to on (T35)
3	Dark to light	On	N/A	Delayed on to off (T36)
4	Light	On to off	N/A	Off
5	Dark	On to off	Not triggered	Delayed on to off (T37)
6	Dark	On to off	Triggered	Delayed on to off (T38)
7	Light	Off	N/A	Off
8	Light to dark	Off	N/A	Off
9	Dark	Off (less than T37 from On)	Closed to open	If already on then T38 Delay from on to off
10	Dark	Off	N/A	Off
11	Dark to light	Off	N/A	Off
12	Light	Off to on	N/A	Off

Note: T38 will become 0 seconds when DC headlamp control driver fails, with only AC headlamp control driver working.
T37 will become 0 second when DC headlamp control driver fails, with only AC headlamp control driver working.

From the results shown on the table:

- When ignition is on, driver door status does not have any effect to BEM Auto-headlamps output.
- Whether Auto-headlamp is selected on the lamp switch or not, auto-headlamp circuit will function to give a BEM headlamp control. The BEM headlamp control will run the headlamp output when lamp switch is put to the "AUTO" position.

Assumptions:

1. Twilight sensor will supply an accurate pulse-width-modulated (PWM) signal with different frequency at different light intensity conditions.
2. Twilight sensor will supply accurate PWM signal in no more than 0.5 seconds after supplying power to the sensor.
3. Humidity, temperature or EMC does not effect the twilight sensor's performance.

8.15.1 Delay Response periods

There are four delay response periods programmed in the EEPROM at position 87Ah to 87Eh. Depending on market requirement and customer expectation, the delay periods can be adjusted accordingly. We shall consider time period **T35**, **T36**, **T38** and **T37**. The delay response period is important in preventing unnecessary flashing of the headlamps.

When the BEM headlamp driver output is enabled, and the ignition is turned to the off position, the BEM headlamp driver output will remain ON for an additional **T38** seconds after the driver door opens. If the driver door remains closed all the time, headlamp driver output will be disabled after **T37** seconds.

When ignition is at the run or accessory position, the BEM headlamp driver output will be enabled/disabled based on ambient light level. The delay response period for the BEM headlamp driver output to react to a bright condition is **T36** seconds, while the delay response period for the BEM headlamp driver output to react to a dark condition is **T35** seconds.

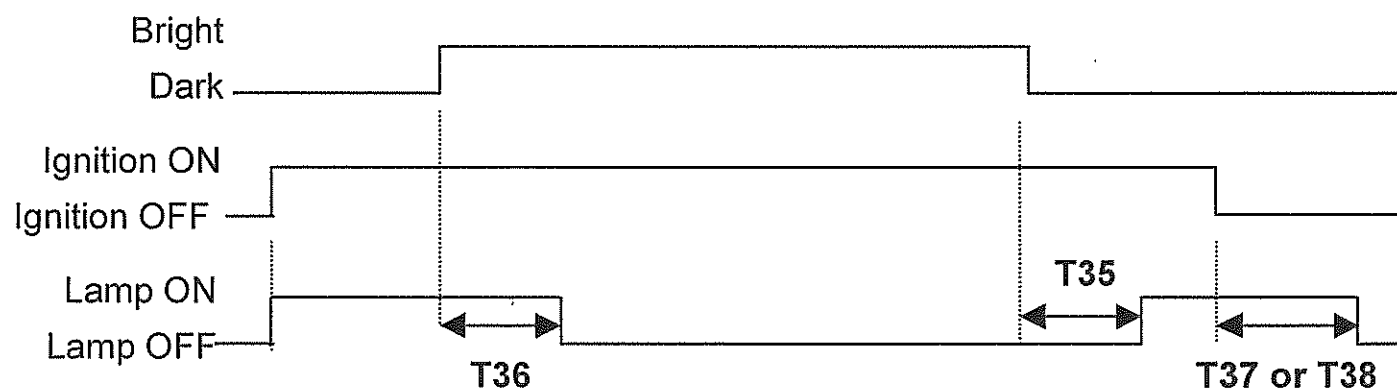


Figure A

Note: If delay turned-on time (**T35**) value as read from the EEPROM is outside the positive tolerance, ie, excessively long, a default failsafe mode will apply. During this mode, the auto-headlamp output will be enabled when ignition key is in the Accessories or Ignition ON position, and will be disabled when ignition key is in the Ignition OFF position.

8.15.2 Hardware And Software Control

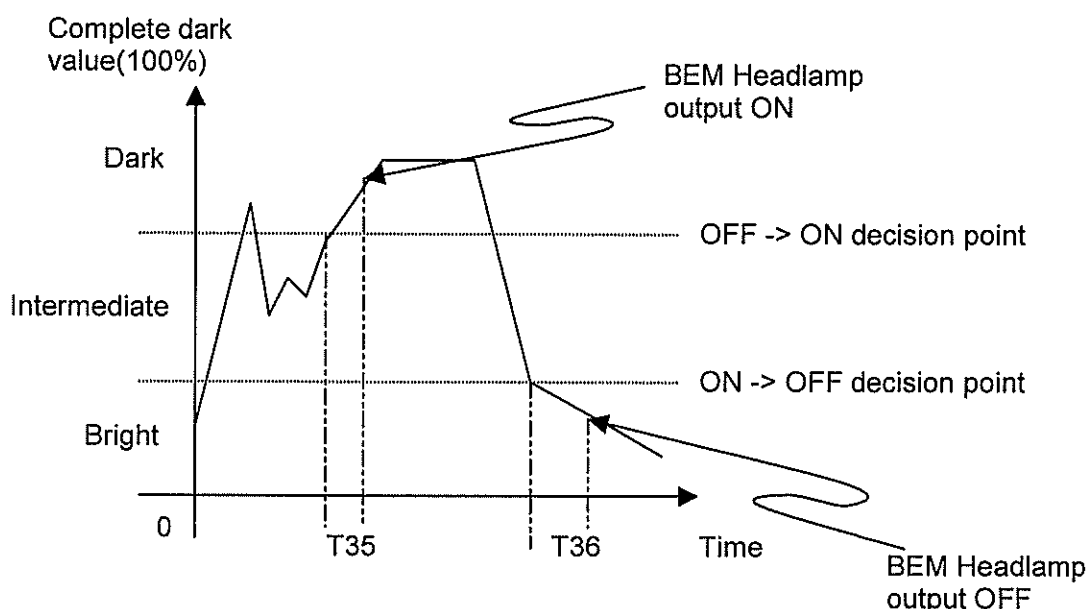


Figure B: Condition: Ignition or Accessories On

There are two control signals from the microprocessor and 1 feedback signal to the microprocessor for the automatic headlamp control system. The 2 control signals are a DC output and an AC output. When the ignition switch is at the run or accessories position under a dark condition, the microprocessor will set its DC output to be logic "HIGH" level and its AC output to be inactive, to enable the BEM headlamp control output. If condition is bright and the BEM headlamp output needs to be disabled, the microprocessor will set its DC output "LOW" and AC output will be set to be active. Figure B shows the decision points and timing for the BEM control outputs to respond at various conditions. Both outputs perform the same function and are in support of one another, such that the auto-headlamp feature will still be functioning if one of the output drivers fails. Table C below shows the relationship between headlamp response to the DC and AC drivers, assuming the headlamp switch is at "AUTO" position.

Note that there is an intermediate section where there is no change in the BEM control output. There are 8 EEPROM programmable values for each of OFF to ON and ON to OFF decision points. Manufacturer can control the EEPROM programmed values, while the dealer can only select which of the programmed values to be used. In addition, there is a set of default values that cannot be changed. This set of default values is being applied when the BEM leaves the factory. All 18 values, as supplied by the customer, will be referred to frequency in Hz obtained from the twilight sensor.

Selection from the 8 pre-programmed values will be done by diagnostic commands with diagnostic tools. Table D shows the values set.

Table C:

Direct "DC" Signal At Microprocessor	Pulsed "AC" signal At Microprocessor	Ignition Switch Position	Headlamp output
LOW	Inactive	Off	Off
LOW	Inactive	Ignition or Accessories	On
LOW	Active	Off	Off
LOW	Active	Ignition or Accessories	Off
HIGH	Inactive	Off	On
HIGH	Inactive	Ignition or Accessories	On
HIGH	Active	Off	On
HIGH	Active	Ignition or Accessories	On

The table below shows the switch on points and the switch off points of the Auto Headlamps defining various settings for dusk to determine when Headlamp relays are to be enabled. Likewise various settings for dawn to determine when the headlamp relays are to be disabled. There is an intermediate section where there is no change in the BEM control output. Eight EEPROM programmable values for each of OFF to ON transitions are selectable through diagnostics (ie 0 to 7). A further eight EEPROM programmable values for each of the ON to OFF transitions are also selectable through diagnostics (ie 0 to 7). Bosch control the EEPROM programmed values at their manufacturing plant. Only the dealer can select the programmed values to be used. All 16 values will be referred to as a frequency in Hertz, obtained from the twilight sensor.

Selection from the 8 pre-programmed values will be done by diagnostic commands with diagnostic tools.

Table D

Settings selection (located in EEPROM)	PID \$7141		Frequency (Hz)	Input captured value		Saved values in EEPROM (hex)	Tolerance (%)
	Byte 1	Byte 2		Decimal	Hexadecimal		
			Headlamp output OFF to ON:				
0x55 (default value)	0x00	X	340	368	0x170	0x170 (located in ROM)	±0.5%
0x00	0xFC	X	380	329	0x149	0x149	±0.5%
0x01	0xFD	X	370	337	0x151	0x151	±0.5%
0x02	0xFE	X	360	347	0x15B	0x15B	±0.5%
0x03	0xFF	X	350	357	0x165	0x165	±0.5%
0x04	0x01	X	330	379	0x17B	0x17B	±0.5%
0x05	0x02	X	320	391	0x187	0x187	±0.5%
0x06	0x03	X	310	403	0x193	0x193	±0.5%
0x07	0x04	X	300	417	0x1A1	0x1A1	±0.5%
			Headlamp output ON to OFF:				
0x55 (default value)	X	0x00	150	833	0x341	0x341 (located in ROM)	±0.5%
0x00	X	0xFC	230	543	0x21F	0x21F	±0.5%
0x01	X	0xFD	210	595	0x253	0x253	±0.5%
0x02	X	0xFE	190	658	0x292	0x292	±0.5%
0x03	X	0xFF	170	735	0x2DF	0x2DF	±0.5%
0x04	X	0x01	130	962	0x3C1	0x3C1	±0.5%
0x05	X	0x02	110	1136	0x470	0x470	±0.5%
0x06	X	0x03	90	1388	0x56C	0x56C	±0.5%
0x07	X	0x04	70	1786	0x6FA	0x6FA	±0.5%

Notes:

- 1) Tolerance refers to software timing error only.
- 2) Frequencies values shown in table D are reference values as supplied by the twilight sensor supplier. BEM software always works with the input capture values received.
- 3) Factory Setting selections are (0x07, 0x55) after the final tester at RBAU and (0x04, 0x00) if read by WDS using \$22 command on PID \$7141 at FoA. This corresponds to input-captured values of 0x1A1 for the headlamps to turn ON, and 0x341 to turn OFF (highlighted in the above table in grey).
- 4) When a dealer selected a threshold level, the BEM will cross check that the selected OFF to ON threshold is always at a darker condition to the selected ON to OFF threshold. If it is not the case, the BEM will respond to the \$2E message from the WDS with a negative response code \$22 "condition not correct".
- 5) Setting selections are compared to fixed ROM values 0x55 and 0x08. If the setting selection is not equal to 0x55 or it is not smaller than 0x08, it is considered as an Out-Of-Range case.
- 6) Selected hexadecimal Input Captured value for the Headlamp OFF to ON thresholds are compared with fixed ROM values of 0x00FF and 0x03FF. If the selected value does not fall between the fixed ROM values, it is considered as an Out-Of-Range case.

- 7) Selected hexadecimal Input Captured value for the Headlamp ON to OFF thresholds are compared with fixed ROM values of 0x00FF and 0x08FF. If the selected value does not fall between the fixed ROM values, it is considered as an Out-Of-Range case.
- 8) Selected Headlamp ON to OFF threshold is compared with the selected Headlamp OFF to ON threshold. When the selected Headlamp OFF to ON threshold has a higher Input Captured value than that of the selected Headlamp ON to OFF threshold, it is considered as an Out-Of-Range case.
- 9) All Out-Of-Range cases trigger the failsafe feature as described in section 8.15.3.4.

The single feedback from the driver/relay circuit is for diagnostic purpose. This feedback is connected to an analogue-to-digital input of the microprocessor, so as to confirm loading of various components in the driver/relay circuit.

8.15.3 Diagnosis

Diagnosis involves the detection of:

- AC headlamp control driver or Connector A failure
- DC headlamp control driver or Connector B failure
- Twilight sensor input short circuit
- Twilight sensor input open circuit

When Auto-headlamps feature is selected while ignition is at the run or accessories position, a frequency counting input and an analogue-to-digital input on the microprocessor are used to collect the diagnostic information as mentioned above. The analogue-to-digital input is connected to the driver/relay circuit and the frequency counting input is connected to the twilight sensor circuit.

With Auto-headlamp feature selected and ignition already at the run or accessories position, auto-headlamp and twilight sensor diagnosis will perform every time the Auto-headlamps are required to switch on. In addition, auto-headlamp and twilight sensor diagnosis will also be done just after the headlamps are required to turn off. This additional diagnosis is used to check if the headlamps are turned on unnecessarily due to a circuit failure. However, the headlamp off diagnostics will not be performed when Ignition switch is at Ignition OFF position.

The twilight sensor diagnostic check will perform before the driver/relay diagnostic check.

8.15.3.1 Twilight sensor diagnosis

The twilight sensor circuit feedback will supply a reading under any light condition.

CASE A: Received PWM signal having frequency above the darkest condition. (420 Hz)

Possible error:

- 1) Faulty sensor.
- 2) BEM has internal circuitry failure.

Error message reported: Out of range. (DTC \$9790 set)

CASE B: Received PWM signal having frequency below the brightest conditions (1 to 47 Hz).

Possible errors:

- 1) Faulty sensor.
- 2) BEM has internal circuitry fault.

Error message reported: Out of range. (DTC \$9790 set)

CASE C: No PWM signal is received (0 Hz).

Possible errors:

- 1) External short to battery causes input HIGH.
- 2) External short to ground causes input LOW.
- 3) Sensor has failed open-circuited.
- 4) Sensor has failed short-circuited.
- 5) BEM has internal circuitry fault pulling input HIGH.
- 6) BEM has internal circuitry fault pulling input LOW.
- 7) Sensor is not loaded onto the circuit.

Error message reported: Open circuit or short circuit to ground. (DTC \$9791 set)

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Note: With the above faulty situations (case A, B and C) detected on the twilight sensor portion, as well as an error detected on the sunload sensor portion, BEM output for Auto-headlamps is defaulted ON when ignition key is in the Ignition ON or Accessories position, and output is off when ignition key is in the Ignition OFF position, until the fault is not detected any longer.

8.15.3.2 Driver and Relay Diagnosis

When BEM output is turning the Auto-headlamp on.

- Turn on one output (either AC (Note 2) or DC) at a time, reverse the output sequence next time turning on the headlamps
- Read the feedback value after turning on the 1st output.
- Turn on second output once feedback has been checked.
- Next time the other output (DC or AC) is tested.

Right after BEM output turned off the Auto-headlamps, when Ignition switch is not at Ignition OFF position

- Read the feedback line for the expected value.

With this diagnosis, we can determine if

- Headlamp circuit shorts to a voltage source.

Error message: Auto-headlamp Driver Circuit Failure. (DTC \$9693 set)

- Headlamp circuit shorts to ground.

Error message: Auto-headlamp Driver Circuit Failure. (DTC \$9693 set)

- One of the driver circuits is not performing. (Note 1)

Error message: Auto-headlamp Driver Circuit (DTC \$9693 set)

- One of the driver connectors is disconnected. (Note 1)

Error message: Auto-headlamp Driver Circuit (DTC \$9693 set)

- Feedback line is shorted internally.

Error message: Auto-headlamp Driver Circuit (DTC \$9693 set)

We cannot test the following in normal operation, as the BEM cannot be sure of whether the "AUTO" position is selected.

- headlamp relay is loaded or not. (Should be able to test by the micro. controlling the headlamp)
- lamp switch is loaded or not.
- broken wiring or connection.

Note 1: One of the drivers is assumed to be working.

Note 2: Headlamp ON test for AC driver must be done when the ignition switch is not in the Ignition OFF position.

8.15.3.3 EEPROM settings Diagnosis

To get a meaningful headlamp output ON to OFF, and OFF to ON threshold value, the EEPROM settings must lie within range. Headlamp output will cease to perform and behave as if it is in a failed sensor condition when:

- Any of the two threshold-selection bytes (EEPROM location 0x8A0, 0x8A1) is not between 0x00 to 0x07, or the default setting of 0x55.
- Any OFF to ON threshold value lies outside 0x0100 to 0x02FF.
- Any ON to OFF threshold value lies outside 0x0100 to 0x08FF.
- Cross-check that OFF to ON threshold is always at a darker condition to ON to OFF threshold.

Note* Checks are carried out every time an EEPROM settings is called up to be used by the software.

Under these failed conditions, the headlamp output will be defaulted on when ignition key is in the Accessories and Ignition ON position, while the headlamp output will be defaulted off during Ignition OFF position.

8.15.3.4 Failsafe Feature

The 2 drivers (AC and DC) design provides control to the Auto-headlamp feature even if 1 driver fails. The other driver will take over the job and keep control on the Auto-headlamp function. The same applies to the connectors.

If one connector drops off, the remaining connector will provide 1 driver for Auto-headlamps control. During the time when AC driver circuits is faulty, DC driver will take up the responsibility to control all auto-headlamp functions. When only DC headlamp driver is faulty, AC driver will provide all auto-headlamp functions, except the delayed turn-off feature at ignition off. This difference in behaviour will also give the indication of an AC driver failure.

In situation when the microprocessor fails and both drivers are inoperable, the pull-ups at the AC driver input will make sure that headlamps will stay ON, providing that the ignition switch is either at the ignition position or the accessories position. Otherwise, headlamps will be off when ignition is at the off position. Delay response feature will not operate if the microprocessor fails. The same applies to the situation where the connector powering the microprocessor drops off.

Under the situation when the twilight sensor is not loaded and "AUTO" is selected on the lamp switch, the default is having the headlamps on all the time when ignition switch is at the run or accessories position, and the headlamps off when ignition is off.

During a power reset situation where sensor data is not readily available, BEM will assume a dark condition and turn the auto-headlamp output ON, until sufficient data is collected from the sensor to determine the light condition. Upon power reset, the BEM will make sure that sufficient twilight sensor information can be collected within 2 seconds. If the light condition is "BRIGHT", the BEM will respond by setting a delayed turn-off situation to the auto-headlamp output, with a delay time of T36. During cranking, if the BEM should reset (due to low battery voltage), then the auto-headlamp drivers will be turned on as per power reset scenario.

Similar situation applies to stop mode. When the car is left untouched for over an hour, the BEM will enter stop mode. If the stop mode is exit by turning ignition key to ignition ON position, BEM will still need the required time (no more than 2 seconds) to collect twilight sensor information. During the data collecting time, the auto-headlamp output will not respond until the new light condition data is received.

Below is a summary of conditions where the failsafe feature applies. With this failsafe feature applied, the auto-headlamp output will be enabled when ignition key is in the Accessories or Ignition ON position, while the auto-headlamp output will be disabled during the Ignition OFF position.

- 1) Microprocessor obtained an out-of-range threshold values from the EEPROM. (Section 8.15.3.3)
- 2) Microprocessor obtained an out-of-range threshold selection settings from the EEPROM. (Section 8.15.3.3)
- 3) Microprocessor obtained threshold values from the EEPROM so as to enable auto-headlamp output at a brighter light condition than to disable the auto-headlamp output.
- 4) Microprocessor failed to supply an auto-headlamp output. (DTC \$9693 set)
- 5) Failure conditions are detected on both auto-headlamps output. (DTC \$9693 set)
- 6) Components failure on both auto-headlamp output circuits. (DTC \$9693 set)
- 7) Faulty twilight sensor. (Section 8.15.3.1) (DTC \$9790 set)
- 8) Faulty sun load sensor. (Section 8.15.3.1) (DTC \$9258 set)
- 9) Twilight/Sun load sensor disconnected. (Section 8.15.3.1) (DTC \$9259, \$9791 set)
- 10) Delayed turn-on time is excessively long. (Section 8.15.1)

8.15.3.5 Warning Buzzer

When there is a BEM headlamp output failure detected, a Buzzer warning bit will be set for 3 seconds when ignition is just turned from ON to OFF position, and from OFF to ON position. This Buzzer warning bit is the Buzzer Request bit (See CAN Message List Section 11.4 for details) Message ID to a logic HIGH for 3 seconds. Under certain condition when one of the two headlamp output drivers failed, for example, at a situation where the FET at the DC driver output fails, the auto-headlamp will still perform correctly. However, since a failed output is detected, warning buzzer will still sound until the fault has been fixed.

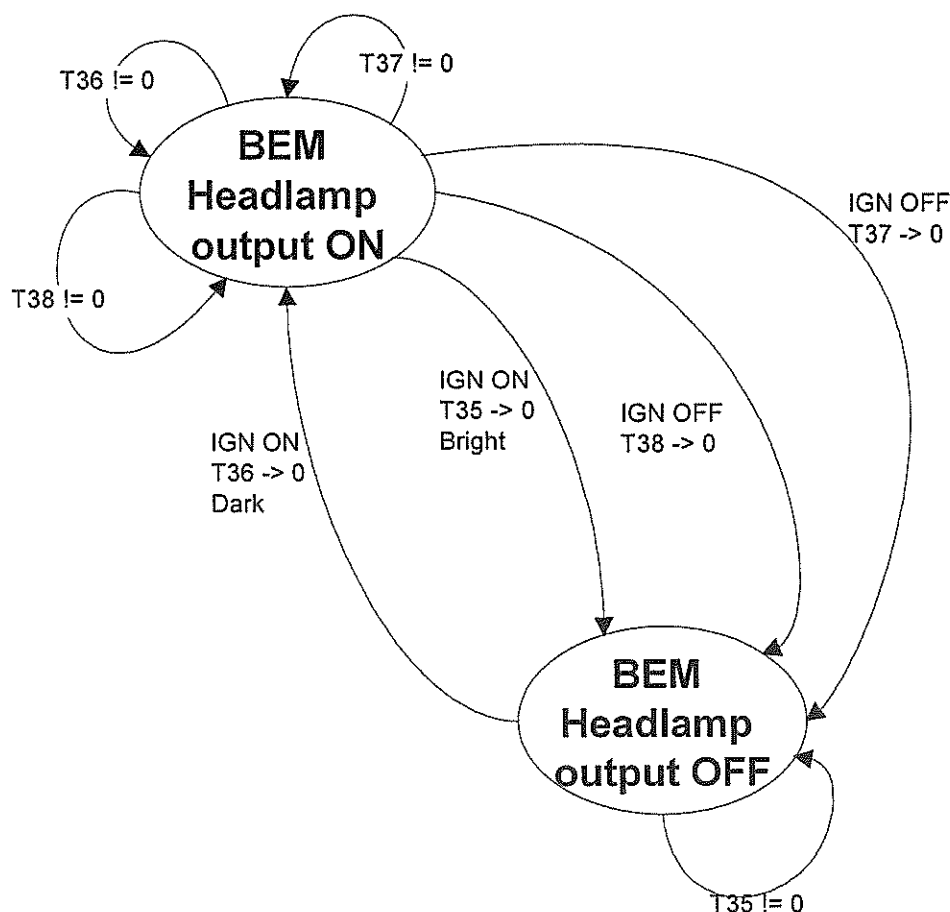
8.15.3.6 State diagram

T35: Delayed BEM headlamp output-enable time.

T36: Delayed BEM headlamp output-disable time, excluding conditions for T3 and T4.

T38: Delayed BEM headlamp output-disable time with output already enabled, ignition just turned off and driver's door triggered.

T37: Delayed BEM headlamp output-disable time with output already enabled, ignition just turned off and driver's door not triggered.



State diagram for auto-headlamp functions

For any discrepancies, the functional description supercedes the above state diagram

8.16 Electronic Immobilisation System

This document complements the Ford design specification "Barra EEC/BEM Immobilisation Code Communications", revision preliminary 2 and should be read in conjunction with this document (See Section 3.3.3). The BEM serves as an integral part of the vehicle immobilisation system. The associated components are:

- Ignition key with embedded TIRIS Digital Signature Transponder (DST). The DST ignition key serves as an authority to start and drive the vehicle using traditional mechanical encoding and electronic transponder technology. The DST, TIRIS part number RI-TRP-BRHP, is housed in a 23 mm glass package and utilises challenge response technology. For the BEM to mobilise, the transponder must respond, when requested, with a unique 24-bit identification code, that the BEM has learnt, and a correct 24-bit response to a 40-bit challenge issued by the BEM. The correct response can only be generated when the DST has been taught the 40-bit encryption key by the BEM. Ford must gain concurrence with Bosch regarding the position of the DST within the key head.
- Transceiver Ignition Lock Assembly (TILA). The TILA, in addition to the steering lock and ignition and starter switch functions, serves as an interface between the DST and the BEM through the use of a transceiver circuit. This consists of an electronic circuit and an antenna in a self-contained housing that attached to the side of the steering lock. Under BEM control, the transceiver uses a RF magnetic field to transmit and receive data with the DST. For further TILA details, refer to Engineering Specification (FOA Barra Body Electronics Module High and Low Series) F005VP0058, section 3.2.
- Body Electronics Module (BEM). The BEM is responsible for verifying that the DST ignition key is one that is recognised as a known DST key, by verifying that its fixed code and challenge response match. Since each BEM uses a unique encryption key, programmed at the factory, a transponder will only be capable of mobilising the vehicle (ie: BEM) to which it was last taught. Once this is satisfied, the BEM will mobilise, accept a 32 bit challenge from the PCM module over the CAN bus and send a 24 bit response that is based on an encryption key it has learnt from the PCM, also over the CAN bus.
- Powertrain Control Module (PCM). The PCM is responsible for allowing the vehicle to start, send a challenge to the BEM and, once it has received a correct response to its challenge, then continue running. The PCM should allow at least 2.0 seconds for a correct BEM response before acting to immobilise the engine.

8.16.1 Plant Mode and PCM Security ID Training

An Original Equipment (OE) BEM is set to PCM-BEM, BEM-TILA and BEM-RF Receiver Plant Modes at the factory. At this stage, any attempt to start the vehicle shall result in the BEM attempting to train a DST ignition key. Any immobilisation communication initiated by the PCM shall result in the BEM responding with a 'Ready to Learn Seed' message. Remote entry functionality is restricted, as described in the remote entry software description.

At all times, PCM related PID commands can identify the PCM- BEM plant mode, as well as the PCM-BEM communicating status.

The training sequence in the Ford plant consists of two stages, DST ignition key teaching and learning the PCM security ID.

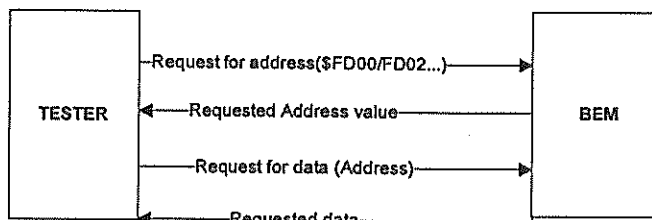
To teach DST ignition keys to the BEM, each DST key must be placed in the ignition switch (S01) and turned to the ignition ON position. For each DST key, the BEM will record the DST's unique identification number and program the DST with its own unique encryption key. Code acknowledgement from the BEM will indicate successful teaching of an ignition key. For the first DST taught, there is one acknowledgment, then there are two for the second DST, etc. In plant mode, a minimum of two and a maximum of eight DST ignition keys may be taught. The acknowledgment of a successfully trained DST ignition key shall occur within time 2.5s of inserting and turning a DST key for training. In the event that programming fails (eg: due to a faulty DST), then no acknowledgment will be produced. Monitoring the error status of the DST ignition key via the diagnostic link can also identify programming failure.

BEM-TILA plant mode (the DST ignition key teaching stage) is exited by the following events:

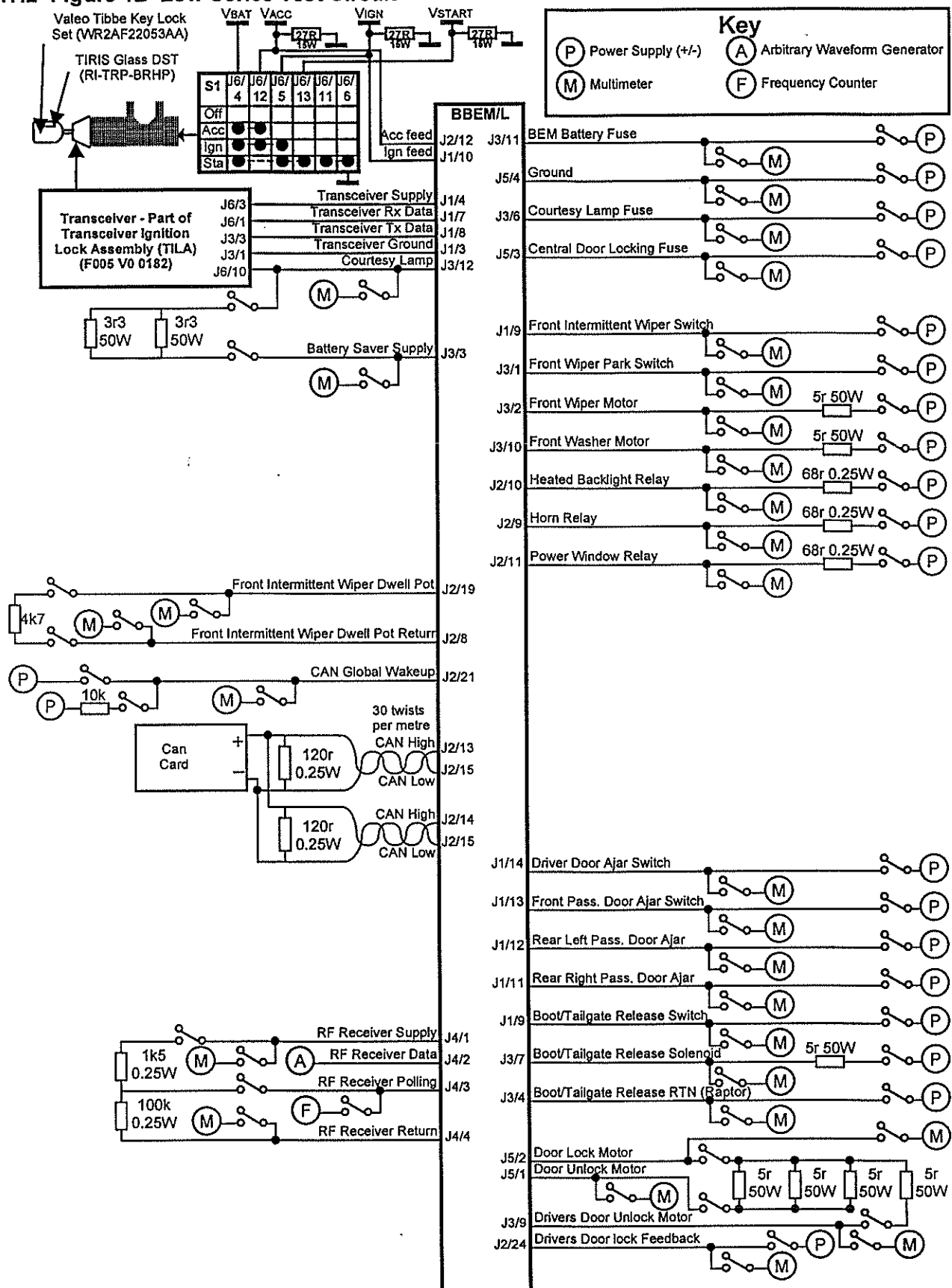
- Interruption of BEM power once two or more DST ignition keys have been taught.
- Leaving the ignition off for 60 seconds once two or more DST ignition keys have been taught.

17.3 RAM Memory Map

To obtain byte value of a variable, the tester need to first get the address (of the variable) by reading from fixed address. Then the tester uses this address to read the byte.

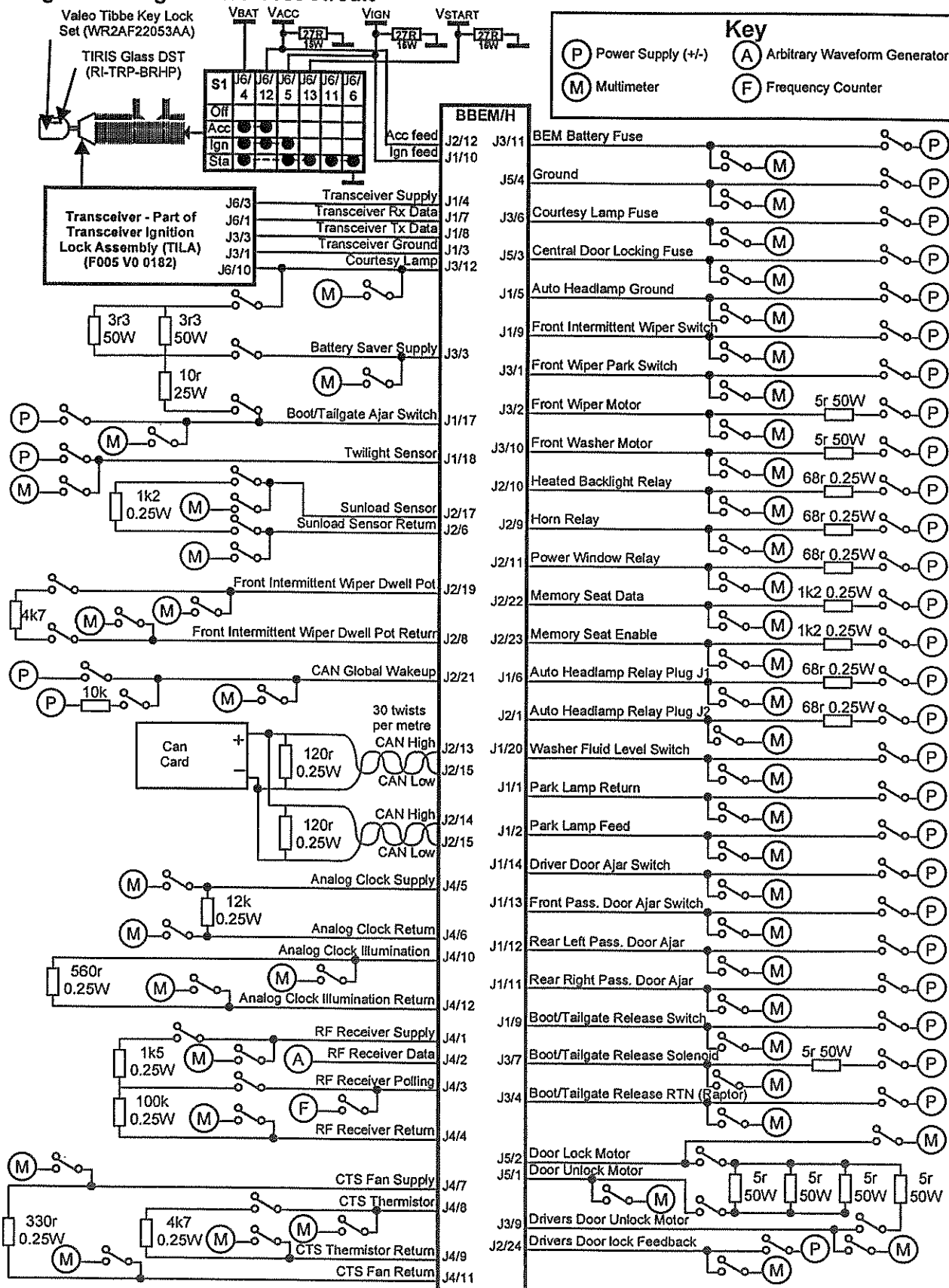


Variable	Indirect Address	Comments
PTADiagnosticFlag_ubg	FD00	Consist of switch bits to select between PTADiagnostic_ubg and standard operation output register for Port A. (1 == Select Diagnostics)
PTADiagnostic_ubg	FD02	Consist of diagnostic control bits for Port A
PTAInputBuffer_ubg	FD04	Consist of debounced input values for Port A
PTAOutputBuffer_ubg	FD06	Consist of standard operation output values for Port A
PTBDiagnosticFlag_ubg	FD08	Consist of switch bits to select between PTBDiagnostic_ubg and standard output register for Port B. (1 == Select Diagnostics)
PTBDiagnostic_ubg	FD0A	Consist of diagnostic control bits for Port B
PTBInputBuffer_ubg	FD0C	Consist of debounced input values for Port B
PTBOutputBuffer_ubg	FD0E	Consist of standard operation output values for Port B
PTCDiagnosticFlag_ubg	FD10	Consist of switch bits to select between PTCDiagnostic_ubg and standard output register for Port C. (1 == Select Diagnostics)
PTCDiagnostic_ubg	FD12	Consist of diagnostic control bits for Port C
PTCInputBuffer_ubg	FD14	Consist of debounced input values for Port C
PTCOutputBuffer_ubg	FD16	Consist of standard operation output values for Port C
PTDDiagnosticFlag_ubg	FD18	Consist of switch bits to select between PTDDiagnostic_ubg and standard output register for Port D. (1 == Select Diagnostics)
PTDDiagnostic_ubg	FD1A	Consist of diagnostic control bits for Port D
PTDInputBuffer_ubg	FD1C	Consist of debounced input values for Port D
PTDOutputBuffer_ubg	FD1E	Consist of standard operation output values for Port D
PTEDiagnosticFlag_ubg	FD20	Consist of switch bits to select between PTEDiagnostic_ubg and standard output register for Port E. (1 == Select Diagnostics)
PTEDiagnostic_ubg	FD22	Consist of diagnostic control bits for Port E
PTEInputBuffer_ubg	FD24	Consist of debounced input values for Port E
PTEOutputBuffer_ubg	FD26	Consist of standard operation output values for Port E
PTFDiagnosticFlag_ubg	FD28	Consist of switch bits to select between PTFDiagnostic_ubg and standard output register for Port F. (1 == Select Diagnostics)
PTFDiagnostic_ubg	FD2A	Consist of diagnostic control bits for Port F
PTFInputBuffer_ubg	FD2C	Consist of debounced input values for Port F
PTFOutputBuffer_ubg	FD2E	Consist of standard operation output values for Port F
PTGDiagnosticFlag_ubg	FD30	Consist of switch bits to select between PTGDiagnostic_ubg and standard output register for Port G. (1 == Select Diagnostics)
PTGDiagnostic_ubg	FD32	Consist of diagnostic control bits for Port G
PTGInputBuffer_ubg	FD34	Consist of debounced input values for Port G
PTGOutputBuffer_ubg	FD36	Consist of standard operation output values for Port G
PTHDiagnosticFlag_ubg	FD38	Consist of switch bits to select between PTHDiagnostic_ubg and standard output register for Port H. (1 == Select Diagnostics)
PTHDiagnostic_ubg	FD3A	Consist of diagnostic control bits for Port H
PTHInputBuffer_ubg	FD3C	Consist of debounced input values for Port H
PTHOutputBuffer_ubg	FD3E	Consist of standard operation output values for Port H
PTLDiagnosticFlag_ubg	FD40	Consist of switch bits to select between PTLDiagnostic_ubg and standard output register for Port L. (1 == Select Diagnostics)
PTLDiagnostic_ubg	FD42	Consist of diagnostic control bits for Port L
PTLOutputBuffer_ubg	FD44	Consist of standard operation output values for Port L
PTTInputBuffer_ubg	FD46	Consist of debounced input values for Port L
CrashKey_uwl	FD48	Write 0x55AA to here to trigger special tests

17.2 Figure 1B Low Series Test Circuit


17. APPENDIX

17.1 Figure 1A High Series Test Circuit



Memory Location Name	Address	Data to be programmed
RFKeyCodes[8] [6]	☠ 0940	Store 8 Remote Key Codes of 6 bytes (4 ID, 2 Rolling Code)
BOSCH_DTC[10]	☠ 0980	Master Error Log – not cleared by FoA
VIN_Number[8]	II 0990	FF (all 8 bytes) – VIN numbers can be stored once by FoA
Spare[24]	☠ 0998	Unassigned
Module_Serial_Number[24] (Reference Part 2 Diagnostics Specification, F005VP0058)	II 09B0	For first final test, store running serial number in 09B0 – 09B3 as packet BCD. Eg serial number 12345 would be 00 01 23 45. If module has been tested before and already has a number do not change. Address 09B4 – 09C8 can be left as is without clearing to FF. Serial number stored by RBAU.
Bootloader_Switch (Flasher 6)	☠ 09FB	Enable Bootloader function (default) = \$BE, Disable ≠ \$BE
PutPlantMode	II 09FD	BE- BEM into Plant Mode by writing to EEprom with 55 indicating download complete.
ClearDTCs	II 09FE	BE- Clearing DTC's by writing to EEprom with 55 indicating download complete
CheckDownLoad	II 09FF	BE- Downloading default data (see*) including Plant Mode by writing to EEprom with 55 indicating download complete

* To Be checked by ICT or Final Tester

II To be programmed and checked by ICT or Final Tester

☠ Restricted area - exclusive use only by BEM

16. Document References

Description	High Series	Low Series
Product Numbers for Barra BEM OE	F005V00222	F005V00207
Product Numbers for Barra BEM P&A	F005V00276	F005V00274
RBAU sales drawing	F005 VS 0006	
RBAU, SWS and Ford part numbers	Please refer to sales drawing	
RBAU vehicle wiring diagram	Please refer to sales drawing	
RBAU product brief	PB1600	PB1604
RBAU diagnostic specification (Global Diagnostic Specification Part II)	F005 VP 0058	
RBAU system FMEA	SFMEA1600	
RBAU design FMEA	DFMEA1600	
RBAU control plan	CP1600	

Memory Location Name	Address	Data to be programmed
OpCodeResetCount	08A7	Op Code Reset Count
BadAddressResetCount	08A8	Bad Address Reset Count
LVIResetCount	08A9	Low Voltage Reset Count
DSTKeyCodesA[8][3]	08AA	FF (all 24 bytes) – 8 Codes of 3 bytes each from individual DST's
DSTKeyCodesB[8][3]	08C2	FF (all 24 bytes) – Same as DSTKEYCodesA
DSTKeyCodesChecksum[8]	08DA	FF (all 8 bytes) – Checksum of A and B of each of the 8 DST's
Spare[9]	08E2	Unassigned
FTMaster	08EA	FF – Non Master, all others assigned to Master level BEM
FactoryMode	08EB	OE: 'o' – for Original Equipment BEM's (ascii \$6F) P&A: 'p' – for Parts and Accessories BEM's (ascii \$70)
FTADRWindowTime	08EC	Final Tester Time in seconds
CountryCode	08ED	Australia: 'A' (ascii \$41)
FTLastResult	08EE	Pass: 'P' (ascii \$50) Fail: 'F' (ascii \$46)
FTPAssCount	08EF	Number of times test has passed 0-FE (FF=never tested)
FTFailCount	08F0	Number of times test has failed 0-FE (FF=never tested)
FTFirstTest_Day	08F1	Day of first final test (0-31) in BCD format
FTFirstTest_Month	08F2	Month of first final test (0-12) on BCD format
FTFirstTest_Year	08F3	Year of first final test (0-99) in BCD format
FTLastTest_Day	08F4	Day of latest final test (0-31) in BCD format
FTLastTest_Month	08F5	Month of latest final test (0-12) in BCD format
FTLastTest_Year	08F6	Year of latest final test (0-99) in BCD format
ICTLastResult	08F7	Pass: 'P' (ascii \$50) Fail: 'F' (ascii \$46)
ICTPassCount	08F8	Number of times test has passed 0-FE (FF=never tested)
ICTFailCount	08F9	Number of times test has failed 0-FE (FF=never tested)
ICTFirstTest_Day	08FA	Day of first IC test (0-31) in BCD format
ICTFirstTest_Month	08FB	Month of first IC test (0-12) on BCD format
ICTFirstTest_Year	08FC	Year of first IC test (0-99) in BCD format
ICTLastTest_Day	08FD	Day of latest IC test (0-31) in BCD format
ICTLastTest_Month	08FE	Month of latest IC test (0-12) in BCD format
ICTLastTest_Year	08FF	Year of latest IC test (0-99) in BCD format
DTC_Count_ms[10]	0900	Most Significant Byte of Ignition Cycle Counter to clear DTC's
DTC_Count_h[10]	0910	High Byte of Ignition Cycle Counter to clear DTC's
DTC_Count_l[10]	0920	Low Byte of Ignition Cycle Counter to clear DTC's
DTC_DISABLE_BYTE[3]	0930	\$FF - Selective enable of extra functions and/or DTC's for low and high series (default for production) - Bit 0 – Washer Motor Relay (Enable = 1, Disable = 0) - Bit 1 – AutoHeadlamp - Bit 2 – Memory Seat - Bit 3 – CTS - Bit 4 – Sunload Sensor - Bit 5 - Unused - Bit 6 – Send only Priority Key 1 (1), Send 8 Priority Keys (0) - Bit 7 – No diode fitted (1), Diode fitted with Alarm Option (0)
Body Type	* 0933	00 – Set as Unknown by RBAU, Set by FoA at Plant
POLICE/TAXI Mode Control Byte[3]	* 0934	00 (all 3 bytes) – Enables default function
E265 Config Byte	* 0937	01 – Enabled for E265 Tailgate Release
Rom_Check_Sum[2]	* 0938	C73C – Mask Checksum
Boot Release Time	* 093A	08 – 800ms Boot Release On Time (01 to 0A)
Central Door/Unlock Time	* 093B	02 – 200ms Central Locking Motor Time (01 to 04)
Spare[2]	093C	Unassigned
Version No	093F	EEPROM data checked against firmware version number (in ROM). If different, firmware version number will be written to EEPROM and checksum will be updated automatically.

Memory Location Name	Address	Data to be programmed
EECEncryptionKeyA[4]	II 0871	FF (all 4 bytes) – Encryption key from EEC
EECEncryptionKeyB[4]	II 0875	FF (all 4 bytes) – Same as EECEncryptionKeyA
EECEncryptionKeyChecksum	II 0879	FF – Checksum of all 4 EECEncryptionKey bytes for both A and B
Headlamp_Uplmt[8]	* 087A	02 – Delayed headlamp turn on time
Headlamp_off1_t	* 087B	0F – Delayed headlamp turn off time with Accessories and Ignition on
Headlamp_off2_t	* 087C	14 – Delayed headlamp turn off time with Headlamps on, ignition just turned off and driver's door opened
Headlamp_off3_t0	* 087D	00 – 1 st byte of Delayed headlamp turn off time with Headlamps on, Ignition just turned off and driver's door closed
Headlamp_off3_t1	* 087E	0C – 2 nd byte of Delayed headlamp turn off time with Headlamps on, Ignition just turned off and driver's door closed
Headlamp_on2_t	* 087F	00 – Delayed turn on time, Ignition just goes to Accessory or Ignition position
Hlamp_Uplmt[8][2]	* 0880	01 – 1 st byte of Auto headlamp turn off threshold 1
	* 0881	49 – 2 nd byte of Auto headlamp turn off threshold 1
	* 0882	01 – 1 st byte of Auto headlamp turn off threshold 2
	* 0883	51 – 2 nd byte of Auto headlamp turn off threshold 2
	* 0884	01 – 1 st byte of Auto headlamp turn off threshold 3
	* 0885	5B – 2 nd byte of Auto headlamp turn off threshold 3
	* 0886	01 – 1 st byte of Auto headlamp turn off threshold 4
	* 0887	65 – 2 nd byte of Auto headlamp turn off threshold 4
	* 0888	01 – 1 st byte of Auto headlamp turn off threshold 5
	* 0889	7B – 2 nd byte of Auto headlamp turn off threshold 5
	* 088A	01 – 1 st byte of Auto headlamp turn off threshold 6
	* 088B	87 – 2 nd byte of Auto headlamp turn off threshold 6
	* 088C	01 – 1 st byte of Auto headlamp turn off threshold 7
	* 088D	93 – 2 nd byte of Auto headlamp turn off threshold 7
	* 088E	01 – 1 st byte of Auto headlamp turn off threshold 8
	* 088F	A1 – 2 nd byte of Auto headlamp turn off threshold 8
Hlamp_Lowlmt[8][2]	* 0890	02 – 1 st byte of Auto headlamp turn on threshold 1
	* 0891	1F – 2 nd byte of Auto headlamp turn on threshold 1
	* 0892	02 – 1 st byte of Auto headlamp turn on threshold 1
	* 0893	53 – 2 nd byte of Auto headlamp turn on threshold 1
	* 0894	02 – 1 st byte of Auto headlamp turn on threshold 1
	* 0895	92 – 2 nd byte of Auto headlamp turn on threshold 1
	* 0896	02 – 1 st byte of Auto headlamp turn on threshold 1
	* 0897	DF – 2 nd byte of Auto headlamp turn on threshold 1
	* 0898	03 – 1 st byte of Auto headlamp turn on threshold 1
	* 0899	C1 – 2 nd byte of Auto headlamp turn on threshold 1
	* 089A	04 – 1 st byte of Auto headlamp turn on threshold 1
	* 089B	70 – 2 nd byte of Auto headlamp turn on threshold 1
	* 089C	05 – 1 st byte of Auto headlamp turn on threshold 1
	* 089D	6C – 2 nd byte of Auto headlamp turn on threshold 1
	* 089E	06 – 1 st byte of Auto headlamp turn on threshold 1
	* 089F	FA – 2 nd byte of Auto headlamp turn on threshold 1
Hlamp_Hselect	* 08A0	\$07 – Threshold 7 from pick of 0 to 7 thresholds for Auto headlamp turn off thresholds (Note Auto EEPROM download will write \$55, the Final tester must re-write this to \$07 and verify accordingly)
Hlamp_Lselect	* 08A1	\$55 – Default threshold 0 from Pick 0 to 7 thresholds from 0 to 7 for Auto headlamp turn on thresholds
TempSensor_Calibration_Factor	☠ 08A2	FF – Set by Sumitomo by referring to Calibration Factor on CTS
BEMPCMTrainingStep	☠ 08A3	1st byte of 2 byte Training Step
PowerOnResetCount	☠ 08A4	Power on Reset Count
ExternalResetCount	☠ 08A5	External Reset Count
WatchdogResetCount	☠ 08A6	Watchdog Reset Count

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15. EEPROM Memory Map

The EEPROM is programmed according to the following table:

Memory Location Name	Address	Data to be programmed
EEprom-2 [512 bytes]		
Spare[512]	0600	Unassigned
EEprom-1 [512 bytes]		
FordPartNumberPrefix[4]	* 0800	For OE and P&A: 33 52 32 33 (ascii for 3R23). Note: External housing P&A label will use BA18.
FordPartNumberSuffix[2]	* 0804	For OE and P&A: High Series: 02 01 (Denotes Part Number Suffix: BB) Low Series: 00 01 (Denotes Part Number Suffix AB) Note: External housing P&A label will use Suffix B for High Series and Suffix A for Low Series.
Final Tester ID[3]	II 0806	Final Tester Inventory Number IT "0553" in BCD
Spare	0809	Unassigned
BEMPCMTrainingStep1	* 080A	FF – 2nd byte of 2 byte Training Step
ANTI_SCAN_Active	080B	55 – Anti-scan Active for approximately 20 sec (00-Inactive otherwise)
Spare[4]	080C	Unassigned
SpareKeyProgramEnable	* 0810	55 – Control permission of customer key training
RBSpareKeyProgOverride	* 0811	55 – Control permission to change above setting at Bosch
DSTContinuousChallengeMode	* 0812	FF – Useful for EMC tests
DSTRetryLimit	* 0813	FF – Number of retries before giving up on immobilisation (FF=infinite)
KeyOutDomeMode	* 0814	5F – Select dome lamp activation with Key lamp off or Key removal
Spare[2]	0815	Unassigned
BootReleaseEnableSpeed	* 0817	0C – Speed at which boot switch on dashboard is inhibited (12km/h)
EEpromErrorAddress_ube[8]	0818	Address – If EEPROM cannot be programmed, each address is recorded (maximum 4 addresses)
RFKeyCount	* 0820	55 - Plant Mode for Remote Key, otherwise out of plant mode details the number of keys programmed.
RFKeyUnlockMode	* 0821	\$FF - 1 Stage unlocking for all 8 Remote Keys (default for production), \$00 – 2 Stage unlocking for all 8 Remote Keys eg Bit 0 – Remote Key 1 (1 =1 Stage, 0 =2 Stage) Bit 1 –Remote Key 2, etc
Spare[29]	0822	Unassigned
DSTPlantMode	* 0840	OE: 55 P&A: 00 Software Initiated EEPROM download will write \$55. Therefore, for P&A parts, at the end of self-down load, this byte need to be written and verified by tester for the value \$00.
DSTEncryptionKeyA[5]	II 0841	Do after the serial number. Convert the serial number as per section 17.5
DSTEncryptionKeyB[5]	II 0846	Same as DSTEncryptionKeyA[5]
DSTEncryptionKeyChecksum	II 084B	Checksum of all 5 DSTEncryptionKey bytes (LSB) for A = Checksum of all 5 DSTEncryptionKey bytes (LSB) for B
Spare[4]	084C	Unassigned
DSTKeyCount	* 0850	00 – Number of Ignition keys currently taught to the system
DSTTotalKeyCount	* 0851	00 – Number of Ignition keys ever taught
Spare[30]	0852	Unassigned
EECPlantMode	* 0870	OE: 55 P&A: 00 Software Initiated EEPROM download will write \$55. Therefore, for P&A parts, at the end of self-down load, this byte need to be written and verified by tester for the value \$00.

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BOSCH  ROBERT BOSCH (AUSTRALIA) PTY. LTD. A.B.N. 48 004 315 628	PRODUCTION TEST SPECIFICATION FOA Barra Body Electronics Module High and Low Series	F005 VP 0067 Page 19 of 29
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13. TILA Functional Test

The BEM must be set to plant mode, and the ignition must be high to perform this test. If the transponder is in the presence of the TILA, the transponder will be taught to the BEM within 5 seconds. The steps required to perform this test are as follows:

- Download bytes into EEPROM to put the BEM into plant.
- Set the ignition feed to high.
- Using the CCP upload command, poll the BEM for up to 5 seconds. If the address DSTKeyCount bit 0 is set to 1, the transponder has been taught to the BEM and the TILA test is successful. If the bit is still 0 after 5 seconds, fail the test.

14. RF Receiver Functional Test

Applying data directly to the input pin of the BEM and then reading an EEPROM memory location tests the RF Receiver input.

The BEM must be set to plant mode, and Ignition/Accessories/Start must be low to perform this test.

- Apply a SHARK protocol (Refer to Barra BEM Engineering Specification) Master Key Data to RF_Receiver_Data (J4/2).
- Confirm memory location a_mgbuff_ubg[4]=10

- Turning to the ignition position with an already trained DST ignition key once two or more ignition keys have been taught.
- Once eight DST ignition keys have been taught.

Once BEM-TILA plant mode is exited, then the BEM, still in PCM plant mode, is ready to learn the PCM's security ID. If a DST key that is known by the BEM is inserted in the ignition lock and turned to the ignition ON position, then the BEM shall mobilise (due to a successful challenge/response between BEM and TILA) and be ready to accept the security ID from the PCM via CAN. The PCM needs to transmit the 'learn seed' message with the security ID continuously to allow the BEM to read and confirm the data. The BEM will indicate that it has tentatively stored the encryption key by responding with the 'learnt seed' message reflecting the learnt code via the CAN line back to the PCM. The PCM must then acknowledge that the BEM has correctly received the key by sending the 'acknowledge learnt seed' message until the ignition is turned off. Upon receipt of this code, the BEM shall start storage of the security ID within 0.2s and then shall perform a code acknowledgment action after the completion of the storage, and the PCM-BEM plant mode is exited. Under normal conditions, this acknowledgment will begin within time 2.5s after the DST ignition key is turned to the ignition position. Note that in some rare circumstances, other BEM functions may require EEPROM availability, may result in an increase in seed learning time. The BEM shall not accept a challenge from the PCM until the ignition switch has been cycled out of the ignition position. Both PCM-BEM and BEM-TILA plant modes can be exited together when an ignition key with a trained DST is put to ignition ON position to get out of BEM-TILA plant mode. Under this situation, BEM will try learning PCM seed in the same action, and will exit PCM-BEM plant mode, providing that PCM is also in plant mode.

If a BEM has not learnt a security ID and the PCM is not in plant mode, then the BEM will respond with the 'Ready to Learn Seed' message. No acknowledgment will occur in this case. To remedy the situation, the PCM would need to be replaced with one in plant mode, changed back into plant mode or be instructed to perform a parameter reset operation to allow the BEM to learn the security ID.

8.16.2 Normal (P&A) Mode

A parts and accessories (P&A) BEM is set to normal (armed) mode at the factory. A BEM in this state will not allow the plant mode operations to be performed. To install a P&A BEM into a vehicle, both DST ignition key teaching (section 8.16.6, 0) and PCM parameter reset operations (section 8.16.8) will need to be performed. Armed mode and PCM security ID status can be identified by the BEM's mobilisation state and current status of PCM communications PIDs respectively.

8.16.3 Normal Operation – BEM Mobilisation

When the ignition switch (S01) is in the off position, the BEM is in the armed state. Turning the DST ignition key to accessories, ignition ON or start positions will cause the BEM to commence a challenge of the ignition key's DST. The challenge created by the BEM is randomly determined by the BEM. The BEM's software ensures the randomness of the challenge, even after reset or power up events. For a new challenge, the BEM must also calculate the expected response for the purposes of verifying the DST's answer. In the event that a new challenge is not ready (such as having to perform two challenges in succession), then the old challenge may be reused. When a challenge is used, it is flagged as being 'stale', and initiation of the calculation of a new challenge is begun immediately. The BEM initially sends a randomly generated 40-bit challenge. The DST will only be able to respond correctly to the challenge if it knows the BEM's encryption key. The BEM also requests the fixed ID code from the transponder, which will identify whether it is one that is known by the BEM.

Successful completion of these two steps will change the BEM's mobilisation status to the mobilised state. Failure of the challenge will result in the mobilisation state changing to immobilised. An error shall be logged and the transponder ignition key status PID will identify the type of error. If an error is detected, the BEM will continue challenging the transponder until it passes or the ignition switch is turned off, except in an Anti-Scan situation. The number of retries can be configured by Bosch, and is currently set to 'infinite'. To trigger an Anti-Scan situation, BEM must receive a DST response with the right data format but incorrect reply. BEM will then stop challenging in the next 20 seconds. BEM will resume challenging after this 20 seconds period and continue challenging until either

- Mobilised state is achieved
- Ignition switch is turned to Ignition off position
- Another Anti-Scan situation is triggered

8.16.4 Normal Operation – PCM Enabling

Once the ignition switch (S01) is in the ignition or start positions, the BEM shall look for a 'pre-challenge' message from the PCM, and respond with a 'request challenge' message. Any challenges received from the PCM shall cause the BEM to respond with a 'not ready, reading transponder' message. Once the BEM has mobilised, it shall request a challenge to the PCM and, wait for the challenge message from the PCM to appear on the CAN bus. The BEM shall then use the security ID, initially learnt from the PCM, to calculate a response and transmit it as a 'response code'. The BEM shall respond in this way to each 'challenge' message received. A 'challenge acknowledge' message from the PCM shall indicate to the BEM that the response has been accepted. Diagnostic interrogation of the BEM may be used to monitor the status of PCM communications.

8.16.5 Challenges from PCM after Normal Start Sequence

In the event that the BEM receives a 'pre-challenge' from the PCM after it has seen a 'challenge acknowledge' message, it will assume a re-challenge situation. The BEM shall request PCM challenge and determine the correct response, but will initially respond with the 'not ready, reading transponder' message. It will simultaneously change to the armed state and challenge the DST inside the ignition key. If the DST ignition key is successfully validated, then the BEM will mobilise again and respond to the PCM's challenge with its 'response code'. An invalid DST will result in immobilisation as in the normal operation case.

8.16.6 Dealer DST Key Teach Mode

This mode may be used by dealers to train DST ignition keys to a new P&A BEM, or for general DST ignition key erasing and teaching.

NOTE: As a security measure, entering dealer teach mode shall erase the BEM's copy of the security ID from memory. This necessitates that the PCM be instructed to perform a parameter reset following completion of dealer teach mode, for the vehicle to be successfully started.

It is recommended that this procedure be performed with the operator sitting inside the vehicle with the doors closed to correctly distinguish mode and code acknowledgments.

Dealer teach mode is entered by setting state = \$11 within \$C125 PID. The ignition switch (S01) must be in the off position to enter this mode. Dealer teach mode can only be activated during a time window of T49 just after Ignition is turned to the off position. Entry into dealer teach mode and deletion of the PCM security ID is confirmed by a mode acknowledgment. A BEM in plant mode shall also accept a dealer DST ignition key teach mode request.

To teach DST ignition keys to the BEM, each DST ignition key must be placed in the ignition switch (S01) and turned to the ignition position. For each DST ignition key, the BEM will record the DST's unique identification number and program the DST with its own unique encryption key. A sequence of code acknowledges will indicate successful teaching of the current DST ignition key. For the first DST ignition key taught, there is one acknowledgment, then there are two for the second DST ignition key, etc. In dealer teach mode, any number of DST ignition keys up to eight may be taught. The acknowledgment of a successfully trained DST ignition key shall normally occur within time 2.5s of inserting and turning a DST ignition key for training. In the event that programming fails (eg: due to a faulty DST), then no acknowledgment will be produced. Monitoring the error status of the DST ignition key via the diagnostic link can also identify programming failure.

Dealer teach mode is exited by the following events:

- Interruption of BEM power.
- Leaving at ignition off for 10 seconds.
- Turning to the ignition position with an already trained DST ignition key once two or more DST ignition keys have been taught.
- Once eight DST ignition keys have been taught.

Exiting of dealer DST ignition key teach mode is confirmed by another mode acknowledgment.

Please also refer to the CAN Message List identified in Appendix 11

8.16.7 Customer Spare DST Key Teach Mode

Customer spare DST ignition key teach mode allows the vehicle user to train and delete DST ignition keys on a limited basis. The PCM security ID is not disturbed during this process.

For security reasons, availability of this mode can be set at the Bosch plant, and optionally by the Ford plant and the dealers.

By using a suitable diagnostic tool (eg. WDS), customer spare DST ignition key programming access can be enabled or disabled through a diagnostic command. However, Bosch is also able to control the dealers access to this command. In summary, Bosch may supply the following configurations:

Customer ignition spare key teach mode enable	Dealer override enable	Outcome
Enabled	Enabled	Customer can train his own DST ignition keys, but dealer can change customer's access rights
Disabled	Enabled	Customer can't train his own DST ignition keys, but dealer can change customer's access
Enabled	Disabled	Customer can train his own DST ignition keys and dealer can not change this
Disabled	Disabled	Customer can't train his own DST ignition keys and dealer can not change this

It is recommended that this procedure be performed with the operator sitting inside the vehicle with the doors closed to correctly distinguish mode and code acknowledgments.

To enter customer spare DST ignition key teach mode, the operator must have two DST ignition keys that are already trained to the BEM. One of the DST ignition keys must be inserted into the ignition switch (S01), turned to the accessories position and then removed. Within time 5 seconds, the other trained DST ignition key must be inserted into the ignition switch, and turned to the ignition position. At this point a mode acknowledgment shall occur to confirm entry into the teach mode.

Once teach mode has been entered, the user may teach new DST ignition keys or delete any other DST ignition keys known by the system.

When deletion of DST ignition keys is performed, only the DST ignition keys that are not used in the teach sequence are deleted. The two DST ignition keys that are used in the sequence are not deleted. To delete up to six DST ignition keys, the second DST key must be cycled to the ignition position a second time and then removed, followed by the first DST key, which must also be turned to the ignition position. At this point, the BEM will perform two code acknowledgments to indicate that only two DST ignition keys are now taught (the two that were just used to perform the sequence). A mode acknowledgment will then indicate completion of the teaching mode.

If the user intends to teach new DST ignition keys to the BEM, then after entering teach mode, up to six more DST ignition keys can be inserted into the ignition switch and turned to the ignition position. If teaching is successful, then code acknowledgments corresponding to the total number of DST ignition keys taught will occur.

Customer spare DST ignition key teach mode is exited by the following events:

- Interruption of BEM power.
- Leaving at ignition off for 10 seconds.
- Turning to the ignition position with an already trained DST ignition key, except the second DST key for entering Customer DST teaching mode. The second DST is used for starting the deletion procedure.
- Once eight DST ignition keys have been taught.

8.16.8 PCM Parameter Reset

A PCM may be instructed to perform a parameter reset on the BEM. This may be used when a BEM or PCM is replaced in the field; the dealer DST ignition key teach mode is performed; or the BEM is otherwise not ready to receive a security ID from a PCM that is in plant mode.

The BEM must first be mobilised by turning the ignition switch (S01) to the ignition position with a taught DST. The BEM will then be in a state where it is ready to receive a challenge from a PCM. (In the event that the BEM is still waiting for a security ID from the PCM, this procedure can still be used). The BEM must not have received a challenge response acknowledgment sequence from the PCM on this ignition cycle; otherwise, it will ignore any attempt to do a parameter reset. The ignition switch must be turned off and back to ignition to clear this situation.

The PCM must have been instructed to perform a parameter reset using the appropriate diagnostic tool and commands. Instead of a challenge, the PCM is to send the 'learn seed' message to the BEM. In the event that the BEM has previously learnt a seed (or is a P&A BEM), it shall respond with a 'seed learnt previously' message. If the PCM then sends the correct 'erase seed' message, then the BEM shall then respond with a 'ready to learn seed' message. Once at this state, the remainder of the parameter reset procedure follows the same steps as the plant procedure does.

8.16.9 Transceiver Diagnostics

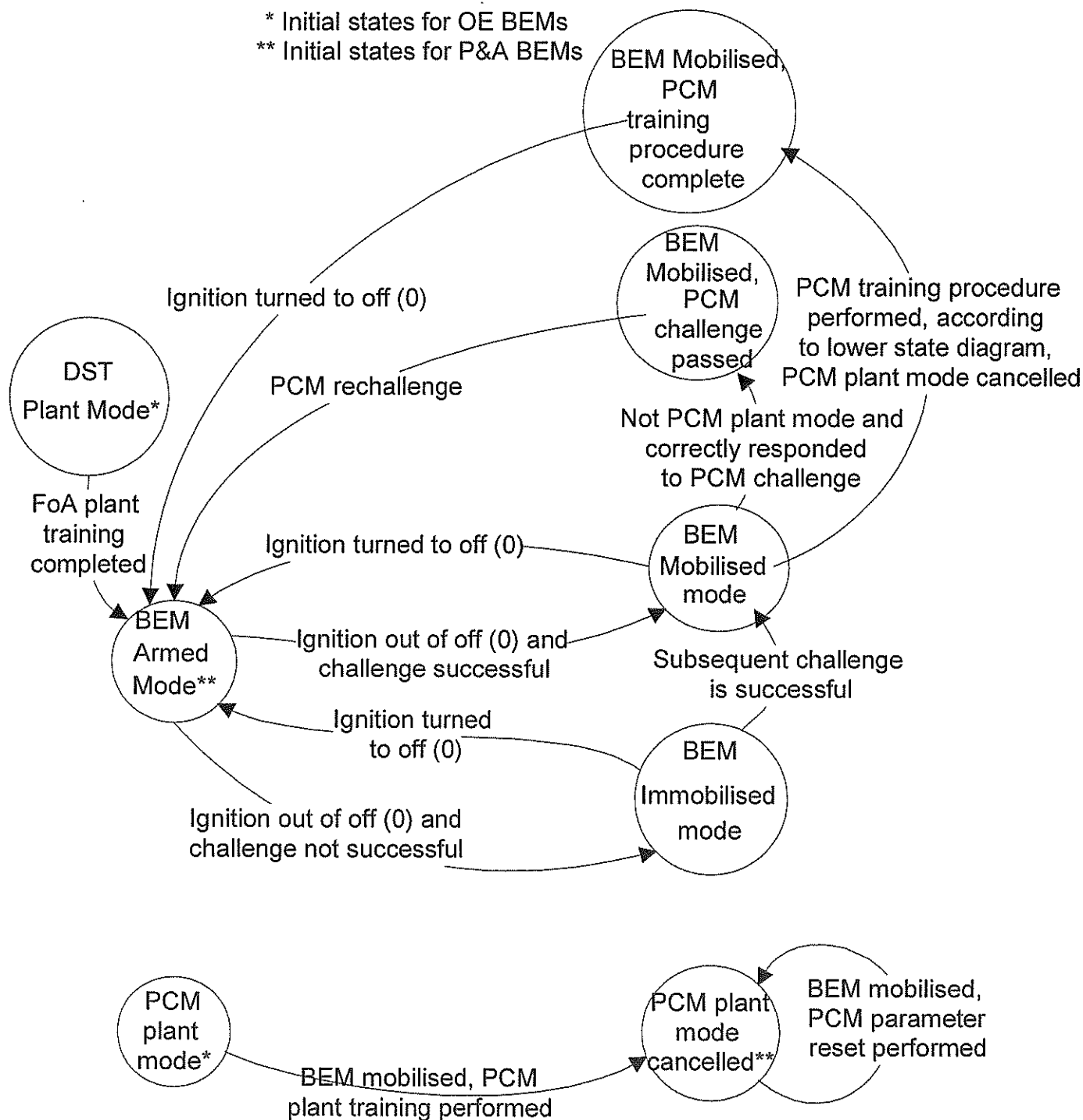
Theoretical analysis shows that the following individual error events may be detected by exercising the TILA transceiver with a DST ignition key inserted. For a complete listing, refer to Section 10).

- BEM transceiver outputs open circuit or short to ground: (DTC \$FFF8)
 - ☐ Transceiver supply open circuit.
 - ☐ Transceiver transmit data open circuit.
 - ☐ Transceiver supply short circuit to ground.
 - ☐ Transceiver transmit data short circuit to ground.
- BEM transceiver outputs short to supply: (DTC \$FFF7)
 - ☐ Transceiver supply short circuit to supply.
 - ☐ Transceiver transmit data short circuit to supply.
- Transceiver internal hardware error (note that interface hardware errors shall fall within one of the above scenarios): (DTC \$A103)
 - ☐ Antenna driving circuit.
 - ☐ Antenna receiving circuit.
- Less than two DST programmed to vehicle (DTC \$9213)
- Faulty transponder or read only type transponder detected (CRC check fails, suggesting data is not formatted correctly). (DTC \$9602)
- Transponder responds incorrectly to challenge response or DST ID unknown. (DTC \$9601)

BEM Immobilisation System

* Initial states for OE BEMs

** Initial states for P&A BEMs



State diagram for BEM immobilisation functions

For any discrepancies, the functional description supercedes the above state diagram

8.17 Code And Mode Acknowledgment Sequences

8.17.1 Code Acknowledgment Sequence

The following sequence is used as a system acknowledgment that an item (DST ignition key or remote key) has been taught successfully. The door locks are toggled into one position and then back again, with an intermediate delay time. Refer to the following table:

Action	Output	Intermediate Time at VOUS	Time at VOES
Condition 1: Lock to unlock to lock (S06)	Door Unlock Output 	T31	T19
	Door Lock Output 	T31	T19
Condition 2: Unlock to lock to unlock (S06)	Door Unlock Output 	T31	T19
	Door Lock Output 	T31	T19

8.17.2 Mode Acknowledgment Sequence

The following sequence will acknowledge entry into or exit from both training modes (DST ignition keys or R. remote keys). The door locks are switched to the opposite state. Note that if the doors are open (S08, S09, S10 or S11), then a door unlock action will immediately follow a lock action to prevent a lockout. Keeping doors closed will assist in identifying the difference between a code acknowledgment and a mode acknowledgment. Refer to the following table:

Action	Output	Time at VOUS	Time at VOES
Condition 1: Lock to unlock (S06)	Door Unlock Output 	—	T19
Condition 2: Unlock to lock (S06)	Door Lock Output 	—	T19

8.18 CAN Communications

Apart from the points mentioned in this document, Barra BEM CAN system shall agree with the Ford document ES-XS4T-12K529-CB (See Section 3.3.1). Also refer to CAN Message Section 11:

8.18.1 Activation of BEM's CAN Transceiver

The BEM will activate the BEM's CAN transceiver in any of the following situations:

- The accessories line is activated (ignition switch positions (I) or (II)).
- The ignition line is activated (ignition switch positions (II) or (III)).
- The CAN global wake up line is activated by another module.
- The CAN global wake up line is activated by the BEM.

8.18.2 Activation of the CAN global wake up line

The BEM will activate the CAN global wake up line in the following situations:

- The accessories line is activated (ignition switch positions (I) or (II)).
- The ignition line is activated (ignition switch positions (II) or (III)).
- The BEM has changed CAN data, to be transmitted with the ignition switch in the off (0) state.
- For time T49 following the BEM's last changed CAN transmission.
- For time T49 following the ignition switch being turned to the off (0) position
- When the BEM needs to send the first status message after power-up or reset.

8.18.3 Receiving CAN messages

The BEM will receive CAN data from the CAN bus, subject to the message ID filtering criteria, whenever the CAN transceiver is active. It will also act upon received CAN data, assemble and transmit CAN data as defined in Ford document ES-XS4T-12K529-CB (See Section 3.3.1).

8.18.4 Transmitting CAN messages

- The accessories line is activated (ignition switch positions (I) or (II)).
- The ignition line is activated (ignition switch positions (II) or (III)).
- The BEM has changed CAN data, to be transmitted with the ignition switch in the off (0) state.
- For time T49 following the BEM's last changed CAN transmission.
- For time T49 following the ignition switch being turned to the off (0) position.

Note: CAN data is not transmitted if another module has activated the global wake up line, unless the BEM's global wake up line is also activated.

8.18.5 M403 message

When the ignition switch is at the Ignition Off position, this message is an event-triggered message. BEM will only transmit this M403 message to the CAN bus when the BEM senses a change of status to the previously transmitted message.

When ignition switch is at Accessories or Ignition ON position, M403 message will be sent periodically at a rate of 2 messages per second. In addition, when there is a change of status to certain priority events, BEM will immediately send another M403 message. After sending this extra message, BEM will wait for another 0.5 second before sending the next periodic M403 message. The priority events are

- 1) Left rear door ajar
- 2) Right rear door ajar
- 3) Left front door ajar
- 4) Right front door ajar
- 5) Heated backlight segment
- 6) Hazard flash request
- 7) Boot/Tailgate ajar
- 8) Buzzer request

At the worst case, a non-priority event can take up to 500ms to get transferred to the CAN bus after the ignition switch is turned from Ignition OFF to Ignition ON position. (Also refer to CAN Message Section 11 for all messages that are a non-priority event)

8.18.6 CAN Message Data

The CAN messages applicable to the BEM are as defined within the message list for the Barra Falcon. This defines the size and content of each message, along with the transmission model (eg: whether the message is sent on a polled or event driven basis).

The BEM implements a filtering strategy to minimise the demands of the CAN module on the software resources. The BEM shall only accept messages numbers that conform to the following structure (represented in binary):

xxxxxxx11x.

The BEM shall transmit messages according to the defined transmission model whenever it activates the global wake up line, or when the ignition switch is out of the off (0) position. If another module has activated the global wake up line, the BEM shall not necessarily transmit any messages, but shall follow the state diagram in Figure A. Naturally, a transmission from another module may require a response from the BEM.

8.18.7 CAN Global Wake up Control

The intention of the global wake up is to allow CAN functionality with the ignition switch in the off (0) position, while allowing the CAN system to power down during times of inactivity.

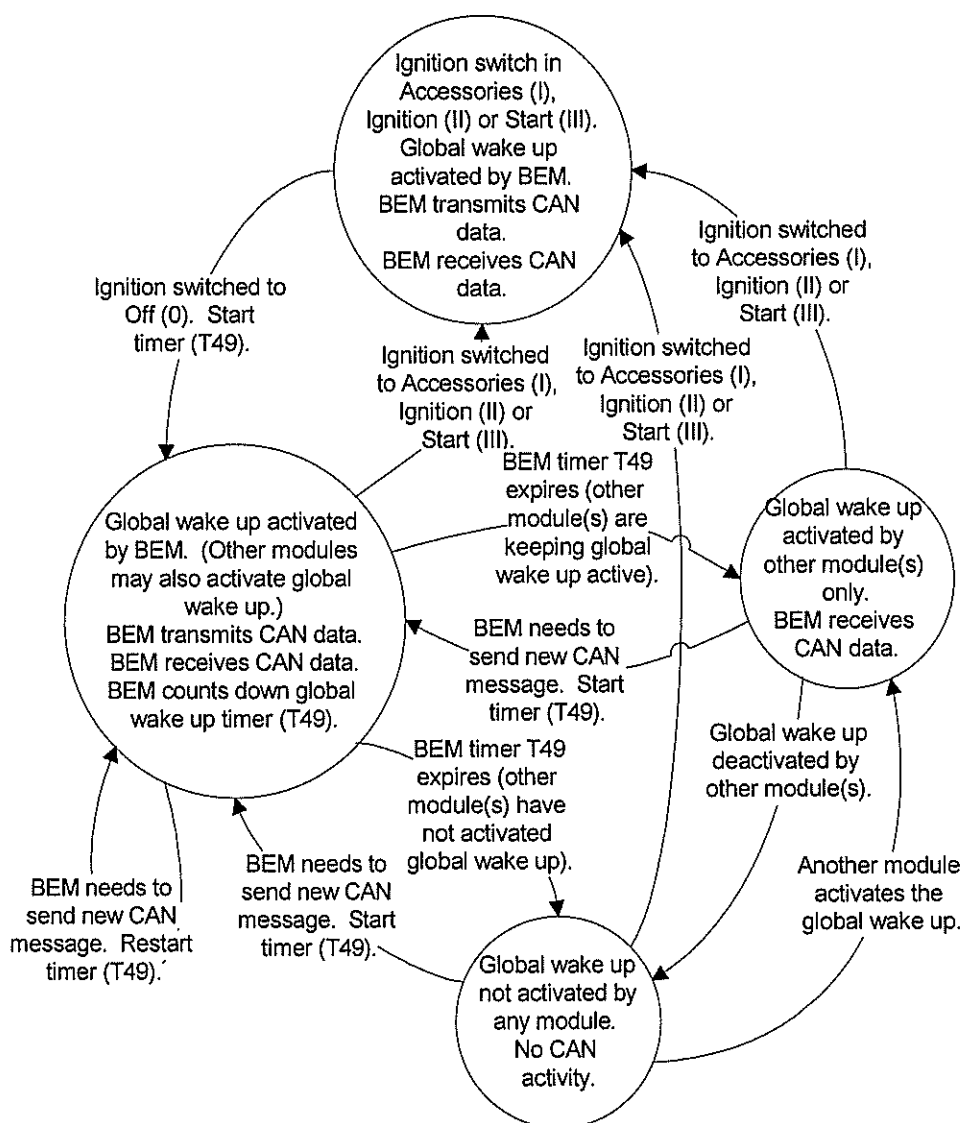
While the ignition switch is out of the off position (I, II or III), the BEM activates the global wake up line in order to keep CAN circuits active.

At ignition off position, the BEM shall begin to time a period T49, starting with either of the following events:

- The ignition switch is turned to the off (0) position.
- The BEM has to transmit changed messages.

During the period that the BEM is timing out, it shall continue to transmit periodic messages at a rate of 2 messages per second. A further change to a BEM message shall result in the timeout being restarted.

The duration of the time T49 is to allow for the fact that if one event happens, other events are likely to follow. For example, a door unlock event is likely to be followed by a person boarding the vehicle and starting it. Keeping the global wake up line active in this situation allows for faster response of systems that utilise CAN based functions.



State diagram for CAN global wake-up and CAN data transmission
For any discrepancies, the functional description supercedes the above state diagram

8.18.8 BEM Default Strategies

Message Identifier	Bit	Message Name	Functions Effected	Fault Condition	Comments (BEM Action/Assumption)
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Received by BEM

\$207	16-31	Vehicle Speed	Boot Release	No Message	• 0 km/h, boot can be released
				Invalid Data	• 0 km/h, boot can be released
					• 0 km/h if Ignition is off
			Int Wipers	No Message	• 0 km/h, no speed dependent wipers
				Invalid Data	• 0 km/h, no speed dependent wipers
\$307	61	Rear Demist Switch Status	Rear Demist	No Message	• No Demist function
			Remote Key Teach	No Message	• No Remote Key teach
	39	Illumination Battery Saver Request	Battery Saver in Ignition off	No Message	• BEM will assume no requirement to turn on Battery Saver when Ignition off after timeout. This will effect radio reception
	38	Door Lock Switch Status	Door Lock Operation	No Message	• No door lock function
	37	Dome Lamp Switch Status	Dome Lamp Control incl. Police Mode	No Message	• No dome lamp control & Police Mode
\$427	14	Air Bag Deployed	Door Unlock function	No Message	• Door s are left in their current state, no action
	8	DTC Logging (\$D900)	Logging of DTC's (\$D900)	No Message	• DTC's \$D900 not logged.
\$6F6	0-63	PCM Immobilisation transfer	Immobilisation	No Message	• No action, BEM waits for challenge
				Invalid Data	• BEM responds to challenge, up to PCM to verify response and mobilise vehicle. If challenge is incorrect format, BEM does not respond and continues to wait for correct challenge.

Transmitted by BEM

\$403	63	Right front door ajar	Cluster indicates door status	No Input	• Door always closed, 0 sent
	62	Left front door ajar	Cluster indicates door status	No Input	• Door always closed, 0 sent
	61	Right rear door ajar	Cluster indicates door status	No Input	• Door always closed, 0 sent
	60	Left rear door ajar	Cluster indicates door status	No Input	• Door always closed, 0 sent
	59	Boot/Tailgate ajar	Cluster indicates Boot/Tailgate status	No Input	• Boot/Tailgate always closed, 0 sent
	58	Rear Demist Segment	CDU indicates demist status	No Input	• No Demist jewel, 0 sent
	57	Accessories switch status	Used by other CAN modules.	No Input	• No Accessories, 0 sent
	56	Low Washer switch	Cluster warning jewel	No Input	• Washer always full, 0 sent
	55	Auto Headlamp Switch	Auto Headlamps		• Not used, 0 sent
	54	Headlamp fail warning	Buzzer function related to Auto Headlamp driver failure	BEM Feedback not working	• BEM will send 0 as a default.
	53	Hazard on request	Cluster, Remote Entry, Panic, Boot	Remote Functions not working	• Hazards always off, 0 sent
	52	Priority Key 4	Memory Seat, Audio, ACC, mirror	Remote Functions not working	• No memory seat function, 0 sent
	51	Priority Key 3	Memory Seat, Audio, ACC, mirror	Remote Functions not working	• No memory seat function, 0 sent
	50	Priority Key 2	Memory Seat, Audio, ACC, mirror	Remote Functions not working	• No memory seat function, 0 sent

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Message Identifier	Bit	Message Name	Functions Effected	Fault Condition	Comments (BEM Action/Assumption)
	49	Priority Key 1	Memory Seat, Audio, ACC, mirror	Remote Functions not working	• 1 sent unless reset occurred then 0 sent indicating No memory seat function
	48	Smart Shield jewel on (LED) request	Cluster transponder near lock warning	Transponder in Ignition warning not working	• No warning, 0 sent
	40-47	Cabin temperature	Climate Control (ACC)	Ignition off, open circuit, short circuit or exceeding 74.5°C or is less than -28°C, up to 200ms after Ignition on, thermistor stabilisation time less than 4 sec	• \$FF will be sent
	32-39	Sunload Sensor Driver	Climate Control (ACC)	Ignition off, open circuit, short circuit or exceeding 4.7V or less than 0.2V, up to 0.5 sec after ignition on	• \$FF will be sent
	24-31	Sunload Sensor Passenger	Climate Control (ACC)	Ignition off, open circuit, short circuit or exceeding 4.7V or less than 0.2V, up to 0.5 sec after ignition on	• \$FF will be sent
	23	Priority Key 8	Memory Seat, Audio, ACC, mirror	Remote Functions not working	• No memory seat function, 0 sent
	22	Priority Key 7	Memory Seat, Audio, ACC, mirror	Remote Functions not working	• No memory seat function, 0 sent
	21	Priority Key 6	Memory Seat, Audio, ACC, mirror	Remote Functions not working	• No memory seat function, 0 sent
	20	Priority Key 5	Memory Seat, Audio, ACC, mirror	Remote Functions not working	• No memory seat function, 0 sent
	18	Buzzer Request	Buzzer request identifying entry/exit to door trigger disable mode	No audio CAN message received	• BEM will not enter door disable trigger mode, 0 sent
\$6F8	0-63	BEM Immobilisation transfer	Immobilisation	Invalid data from transponder	• BEM responds to PCM with a 'Not Ready, Reading Transponder' message

8.19 Diagnostic Communications

Please also refer to the diagnostic specification Part II, identified in Appendix 10

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8.20 Configurable Options

A number of configurable options are available in the BEM software. These are summarised below:

Function	Factory Setting	Description	Accessibility
DST Key out dome illumination enable	Enabled	Enables or disables feature of dome lamp illumination with DST ignition key removal from ignition switch.	Bosch plant
DST Key off dome illumination enable	Disabled	Enables or disables feature of dome lamp illumination with DST ignition key turning to the off position.	Bosch plant
Headlamp delay characteristics	See Auto Headlamp	Delay times for automatic headlamps can be varied	Bosch plant
Bosch setting of auto headlamp thresholds	See Auto Headlamp	Bosch programs eight sets of thresholds for use with the auto headlamp system. Each set consists of a high threshold and a low threshold to provide hysteresis.	Bosch plant
Selection of auto headlamp threshold	See Auto Headlamp	Allows one of eight threshold settings to be selected for auto headlamps.	Dealer, Ford & Bosch plants
Customer spare DST key teach enable	Enabled	Allows or disallows customer to perform DST ignition key training.	Dealer, Ford & Bosch plants
Bosch override of Customer spare DST key teach enable	Disabled	Bosch can prevent dealer, Ford plant and Sumitomo from altering the Customer spare DST key teach enable setting.	Bosch plant
Transponder retries	Infinite	Specifies the maximum number of retries to be attempted in the event that a DST is not correctly identified in a starting sequence. May be a fixed number up to 254, or an 'infinite' setting.	Bosch plant
CTS Calibration Factor	See CTS	Specifies the offset required on the CAN data to report a more accurate temperature generated by the Cabin Temperature Sensor (CTS) Module.	Sumitomo Plant, Ford Plant, Ford Dealer
Taxi Mode	Disabled	Unlocks Passenger doors without unlocking the Drivers Door when using the Central Locking Dash Switch	Dealer, Ford & Bosch plants
Panic Alarm Inhibit	Disabled	Disables/Enables Panic Alarm	Dealer, Ford & Bosch plants
Door Trigger Disable Mode	See Door Trigger Disable Mode	Disables/Enables Door Trigger Disable Mode (alternatively can be manually enabled/disabled via holding the Dome Lamp Dash Switch)	Dealer, Ford & Bosch plants
Master Door Trigger Disable Mode	Disabled	Disables/Enables the manual toggling of Door Trigger Disable Mode using the Dome Lamp Switch	Dealer, Ford & Bosch plants
Hazard Warning Inhibit	Disabled	Disables/Enables the Hazard Lamp Flash	Dealer, Ford & Bosch plants
Horn Warning Inhibit	Disabled	Disables/Enables the Horn Warning function	Dealer, Ford & Bosch plants
Battery Saver Inhibit	Disabled	Disables/Enables Battery Saver	Dealer, Ford & Bosch plants
Courtesy Lamp	Disabled	Disables/Enables Courtesy Lamps which if enabled by default enables Battery Saver	Dealer, Ford & Bosch plants

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Function	Factory Setting	Description	Accessibility
RF Key Unlock Mode	1 stage	Selects between 1 stage or 2 stage Central Door Unlocking (alternatively can be manually selected by pressing both Lock and Unlock buttons on the RF Remote Key)	Bosch plant
RF Key Teach Mode	See RF Key Teach	Selects RF Key Teach Mode (alternatively can be manually selected by pressing Heated Backlite Switch 3 times within 5 seconds after turning Accessories on)	Dealer, Ford & Bosch plants
Washer Motor Inhibit	Disabled	Disables/Enables Washer Motor DTC	Bosch plant
Auto Headlamp Inhibit	Disabled	Disables/Enables Auto Headlamp DTC	Bosch plant
Memory Seat Inhibit	Disabled	Disables/Enables Memory Seat DTC	Bosch plant
CTS Inhibit	Disabled	Disables/Enables Memory Seat DTC and Function	Bosch plant
Sunload Sensor Inhibit	Disabled	Disables/Enables Sunload Sensor DTC and function	Bosch plant
Send only Priority Key 1 Enable	Priority Key 1	Enables Priority Key 1 versus 8 Priority Keys being sent for Memory Seat function	Bosch plant
Diode fitted Inhibit	Disabled	Enables/Disables Boot/Tailgate Release Input to be monitored, but only if external diode is fitted in vehicle wiring loom (See vehicle wiring diagram)	Bosch plant

8.21 Timing Parameters

Parameter	Description	[Note * - stored in the EEPROM]	Time (s)	Tolerance	
T01	Heated backlight relay on time		900.0	-1%	+1%
T02	Front intermittent wiper on time		0.8	-5%	+5%
T03	Variable speed dependent intermittent wiper off time		See 8.3	-2%	+2%
T04	Fixed front intermittent wiper off time		See 8.3	-2%	+2%
T05	Front washer motor on time		1.25	-10%	+10%
T06	Front wiper motor delay after washer motor on		0.48	-15%	+15%
T07	Fail-safe wipe after wash wiper motor on time		5.0	-2%	+2%
T08	Minimum intermittent wiper dwell time for wipe on take off		3.2	-2%	+2%
T09	Power window relay on time		3600.0	-1%	+1%
T10	Power window relay on time with door open		40.0	-5%	+8%
T11	Battery saver time-out (normal mode)		900.0	-1%	+1%
T12	Battery saver time-out (plant battery saver mode)		180.0	-1%	+1%
T13	Courtesy lamp short fade in and short fade out		1.0	-5%	+5%
T14	Courtesy lamp entry delay or exit delay		20.0	-2%	+2%
T15	Courtesy lamps fade out time following entry delay or exit delay		6.4	-5%	+5%
T16	Courtesy lamp activation on DST key removal activation window		3600.0	-1%	+1%
T17	Door Trigger Disable Mode switch hold time		5.0	-2%	+2%
* T18	Boot release time		0.8	-5%	+5%
* T19	Central locking or unlocking time		0.2	-10%	+10%
T20	Hazard lamp activation time for doors lock activation		0.5	-5%	+5%
T21	Hazard lamp activation time for boot release		1.0	-5%	+5%
T22					
T23	Hazard lamp activation time during horn warning		0.2	-10%	+10%
T24	Horn warning on time		0.01	-10%	+10%
T25	Horn warning off time		0.08	-10%	+10%
T26	Panic alarm duration		30.0	-2%	+2%
T27	Panic alarm pulse width		0.2	-10%	+10%
T28	Manual boot release security time-out		600.0	-1%	+1%
T29	Automatic central door re-lock time		40.0	-2%	+2%
T30	Remote key mode change time		3.0	-2%	+2%
T31	Code acknowledgment intermediate delay		0.2	-10%	+10%
T32	Remote Key Teach Mode Access Time Window		5.0	-2%	+2%
T33	Time slice used for BEM to Seat Module communication		0.04	-20%	+20%
T34	Time delay for transition		0.02	-20%	+20%
* T35	Auto headlamp relay coil delayed turn on time		2.0	-2%	+2%
* T36	Auto headlamp relay coil delayed turn off time		15.0	-2%	+2%
* T37	Auto headlamp delayed turn off time with Headlamps on, Ign off, driver's door open		120.0	-1%	+1%
* T38	Auto headlamp delayed turn off time with Headlamps on, Ign off, driver's door closed		20.0	-2%	+2%
T39	Second press Warning delay		3.0	-2%	+2%
T40	BEM DST ignition key training time		1.0	-50%	+150%
T41	Plant mode maximum time between DST key teaches		60.0	-2%	+2%
T42	BEM security ID storage time after PCM acknowledgment		0.2	-75%	+0%
T43	BEM PCM training code acknowledgment activation delay		0.5	-5%	+5%
T44	Total PCM BEM training time		2.5	-50%	+150%
T45	Dealer DST ignition key teach mode max time between DST key teaches		10.0	-2%	+2%
T46	Customer DST ignition key teach mode entry time window		5.0	-2%	+2%
T47	Customer DST ign key teach mode max time between DST key teaches		10.0	-2%	+2%
T48	Ignition on to PCM challenge response completion (maximum)		1.5	-5%	+5%
T49	CAN global wake up deactivation delay time		120.0	-5%	+5%
T50	Low washer fluid notification delay time		20.0	-2%	+2%
T51	Power Down Mode Entry Delay		2.0	-100%	+0%
T52	Remote Entry Receiver Passive Resynchronisation time		30.0	-2%	+2%
T53	Minimum Inter-frame period		0.20	-10%	+10%
T54	Dealy from RK unlock to send data frame		0.20	-10%	+10%
T55	Duration of buzzer for enabling/disabling Door Trigger Disable Mode		2.5	-5%	+5%
T56	Duration of courtesy lamp pulse		0.5	-5%	+5%
T57	CTS Trigger high duration		25.4	-5%	+5%
T58	CTS Trigger off duration		0.2	-2%	+2%

9. Test Sequences

9.1 BEM Endurance Test Sequence Program

The sequence is intended to exercise all inputs and outputs to simulate normal operation of the module.

Barra Body Electronics Module test sequence program with monitoring																											
State	Time (sec)	Comment	Input Switch/Signal Position																								
			Ignition switch	Remote Door Unlock	Remote Door Lock	Remote Boot	Remote Panic	(Driver Door Lock)	(Pass Door Lock)	Boot/Tailgate Ajar Switch	Driver door ajar switch	Front passenger door ajar switch	Rear left door ajar switch	Rear right door ajar switch	Boot Release	Door Lock Switch	Front Intermittent Wiper Switch	Wiper Dwell Pot	(Wiper Park)	Front wash motor/switch	Washer fluid level switch	Graphics illumination feed	CAN global wakeup	Park Lamps	Heated Backlight		
1	0 - 2	Initial state	Off	Off	Off	Off	Off	Unlock*	Unlock*	Closed	Closed	Closed	Closed	Closed	Off	Off	Off	Long	X	Off	Not low	Off	Off	Auto	Off		
2	2 - 4	Graphics on (clock illumination)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	On	—	—	—	—		
3	4 - 6	Sunload/twilight sensor bright	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	Off	—	—	—	—		
4	6 - 8	Graphics off (clock illumination)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
5	8 - 10	Accessories on (TILA challenge)	Acc	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	Park*	—	—	—	—	—	—		
6	10 - 12	Ignition on (sensor data)	Ign	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
7	12 - 14	Start on (power window off)	Sta	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
8	14 - 16	Ignition on (power window resume)	Ign	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	Low	—	—	—	—		
9	16 - 18	Washer fluid low	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
10	18 - 20	Intermittent wiper on (long delay)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	On	—	1 Pulse	—	—	—	—	—	On		
11	20 - 22	Heated backlight switch on	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	Park	—	—	—	—	—	Off		
12	22 - 24	Heated backlight switch off	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
13	24 - 26	Wiper dwell short (wipe occurs)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	Short	Pulsing	—	—	—	—	—	—		
14	26 - 28	Sunload/twilight sensor dark	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
15	28 - 30	Central locking switch on (lock)	—	—	—	—	—	Lock*	Lock*	—	—	—	—	—	—	On	Off	—	—	—	—	—	—	—	—		
16	30 - 32	Central locking switch off	—	—	—	—	—	—	—	—	—	—	—	—	—	Off	—	—	—	—	—	—	—	—	—		
17	32 - 34	Spare	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
18	34 - 36	Driver's door open (unlock door)	—	—	—	—	—	Unlock	—	—	Open	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
19	36 - 38	Boot release switch on	—	—	—	—	—	—	—	—	—	—	—	On	—	—	—	—	—	—	—	—	—	—	—		
20	38 - 40	Boot release switch off	—	—	—	—	—	—	—	—	—	—	—	Off	—	—	—	—	—	—	—	—	—	—	—		
21	40 - 42	Washer fluid not low	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	Not low	—	—	—	—		
22	42 - 44	Driver's door closed	—	—	—	—	—	—	—	—	Closed	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
23	44 - 46	Boot open	—	—	—	—	—	—	—	—	Open	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
24	46 - 48	Boot closed	—	—	—	—	—	—	—	—	Closed	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
25	48 - 50	Front passenger door open	—	—	—	—	—	Unlock	—	—	Open	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
26	50 - 52	Front passenger door closed	—	—	—	—	—	—	—	—	Closed	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
27	52 - 54	Intermittent wiper off	—	—	—	—	—	—	—	—	—	—	—	—	—	Off	—	—	Park	—	—	—	—	—	—		
28	54 - 56	Washer on	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	Pulsing	On	—	—	—	—	—		
29	56 - 58	Washer off	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	Off	—	—	—	—	—	—		
30	58 - 60	Rear left door open	—	—	—	—	—	—	—	—	Open	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
31	60 - 62	Spare	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	Park	—	—	—	—	—	—		
32	62 - 64	Ignition off	Acc	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
33	64 - 66	Accessories off	Off	—	—	—	—	—	—	—	—	—	—	—	—	—	—	X	—	—	—	—	—	—	—		
34	66 - 68	Rear left door closed	—	—	—	—	—	—	—	—	—	—	Closed	—	—	—	—	—	—	—	—	—	—	—	—		
35	68 - 70	RF lock on (lock, warning)	—	—	On	—	—	Lock	Lock	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
36	70 - 72	RF lock off	—	—	Off	—	—	Unlock	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
37	72 - 74	Rear right door open	—	—	—	—	—	—	Unlock	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
38	74 - 76	Rear right door closed	—	—	—	—	—	—	—	—	—	—	Open	Closed	—	—	—	—	—	—	—	—	—	—	—		
39	76 - 78	RF boot on	—	—	—	On	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
40	78 - 80	RF boot off	—	—	—	Off	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
41	80 - 82	Central locking switch on (lock)	—	—	—	—	—	Lock	Lock	—	—	—	—	—	—	On	Off	—	—	—	—	—	—	—	—		
42	82 - 84	Central locking switch off	—	—	—	—	—	—	—	—	—	—	—	—	—	Off	—	—	—	—	—	—	—	—	—		
43	84 - 86	RF unlock on	—	On	—	—	—	Unlock	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
44	86 - 88	RF unlock off	—	Off	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
45	88 - 90	RF unlock on	—	On	—	—	—	—	Unlock	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
46	90 - 92	RF unlock off	—	Off	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
47	92 - 94	Wiper dwell long	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	Long	—	—	—	—	—	—	—		
48	94 - 96	Accessories on (TILA challenge)	Acc	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	Park	—	—	—	—	—	—		
49	96 - 98	Ignition on	Ign	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
50	98 - 100	Ignition off	Acc	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
51	100 - 102	Accessories off	Off	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	X	—	—	—	—	—	—		
52	102 - 104	Accessories on (TILA challenge)	Acc	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	Park	—	—	—	—	—	—		
53	104 - 106	Ignition on	Ign	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
54	106 - 108	Ignition off	Acc	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
55	108 - 110	Accessories off	Off	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	X	—	—	—	—	—	—		
56	110 - 112	Accessories on (TILA challenge)	Acc	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	Park	—	—	—	—	—	—		
57	112 - 114	Ignition on	Ign	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
58	114 - 116	Ignition off	Acc	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
59	116 - 118	Accessories off	Off	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	X	—	—	—	—	—	—		
60	118 - 120	RF panic on (every 200th cycle)	—	—	—	—	On**	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
61	120 - 148	RF panic off	—	—	—	—	Off	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
62	148 - 148	26 second delay	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
63	148 - 150	(Panic alarm expires)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		
Operations per cycle			4	1	1	1	1/10	3	3	1	1	1	1	1	1	1	2	1	1	3	1	1	0	0	1		
Operations per hour			96	24	24	24	24	72	72	24	24	24	24	24	24	24	48	24	24	72	24	24	0	0	24		
Operations per 1000 active hours			96000	24000	24000	24000	24000	72000	72000	24000	24000	24000	24000	24000	24000	24000	48000	24000	24000	72000	24000	24000	0	0	24000		
Legend			X	Don't care																							
			*	Undefined for the first cycle																							
			**	Every 200th cycle																							
			—	Same as previous state																							

9.2 BEM Immunity Test Functional Sequence

A simplified sequence is intended to exercise some inputs and monitor all outputs in a short period of time for tests where one cycle is to be tested at many steps.

Activity	Monitoring
• Test dwell time: 8 seconds/frequency. • Clear all DTCs before test and after a set DTC has been recorded. • All switches in the default positions, as per test circuit (including locked doors), apart from the following exceptions. • Cycle S1 from Off for 2 seconds to Ignition position for balance of dwell time. • Twilight sensor switched between 'dark' and 'light' settings.	• Monitor all outputs for inadvertent operation. • Monitor CAN bus activity and all messages for inadvertent operation. • Monitor all DTCs, including internal DTCs, for errors. • Monitor auto headlamp output • Verify Transceiver mobilisation. • Check for reset activity. • Monitor CAN bus activity.

Figure 1A High Series Test Circuit

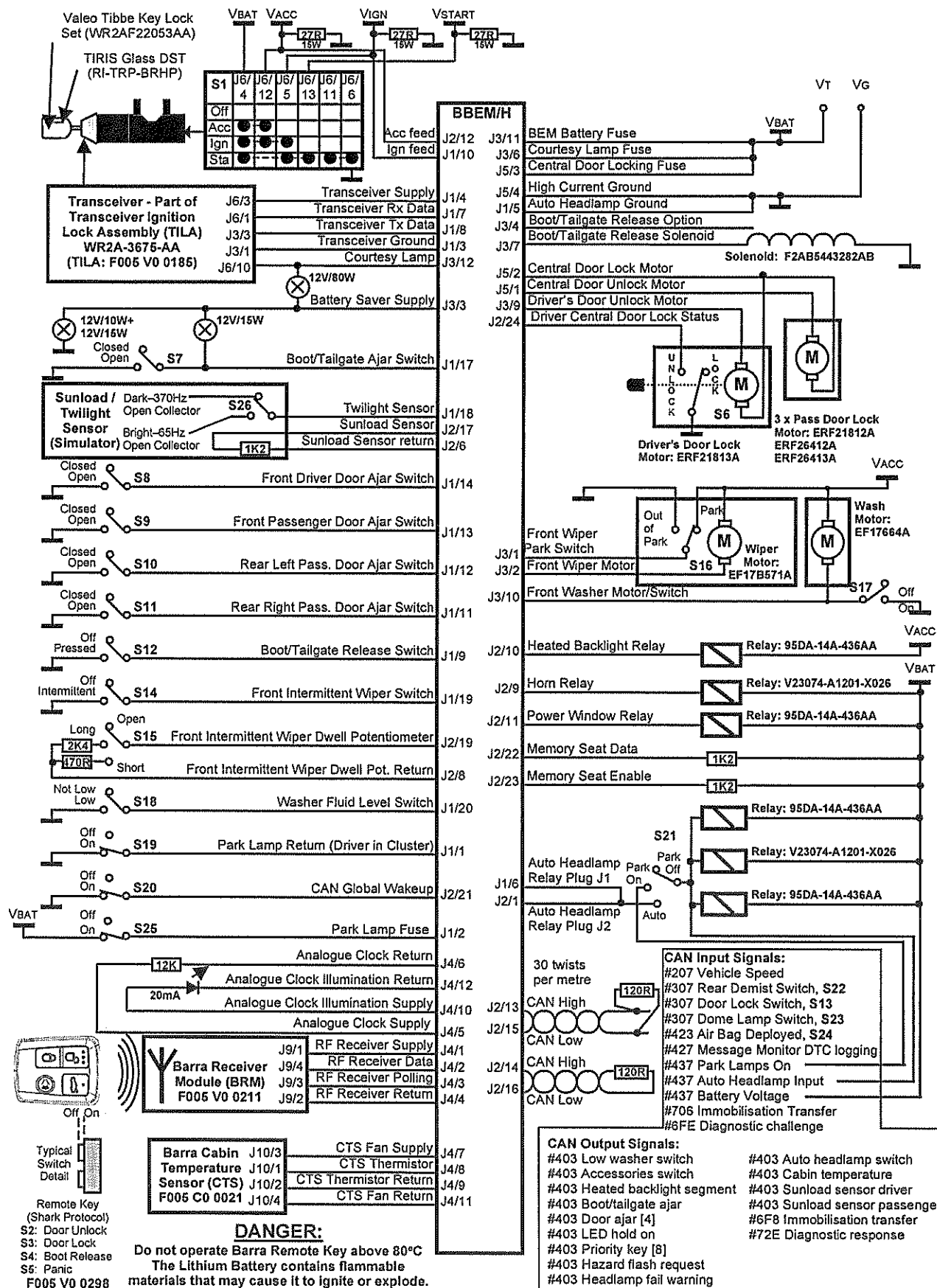


Figure 1B Low Series Test Circuit

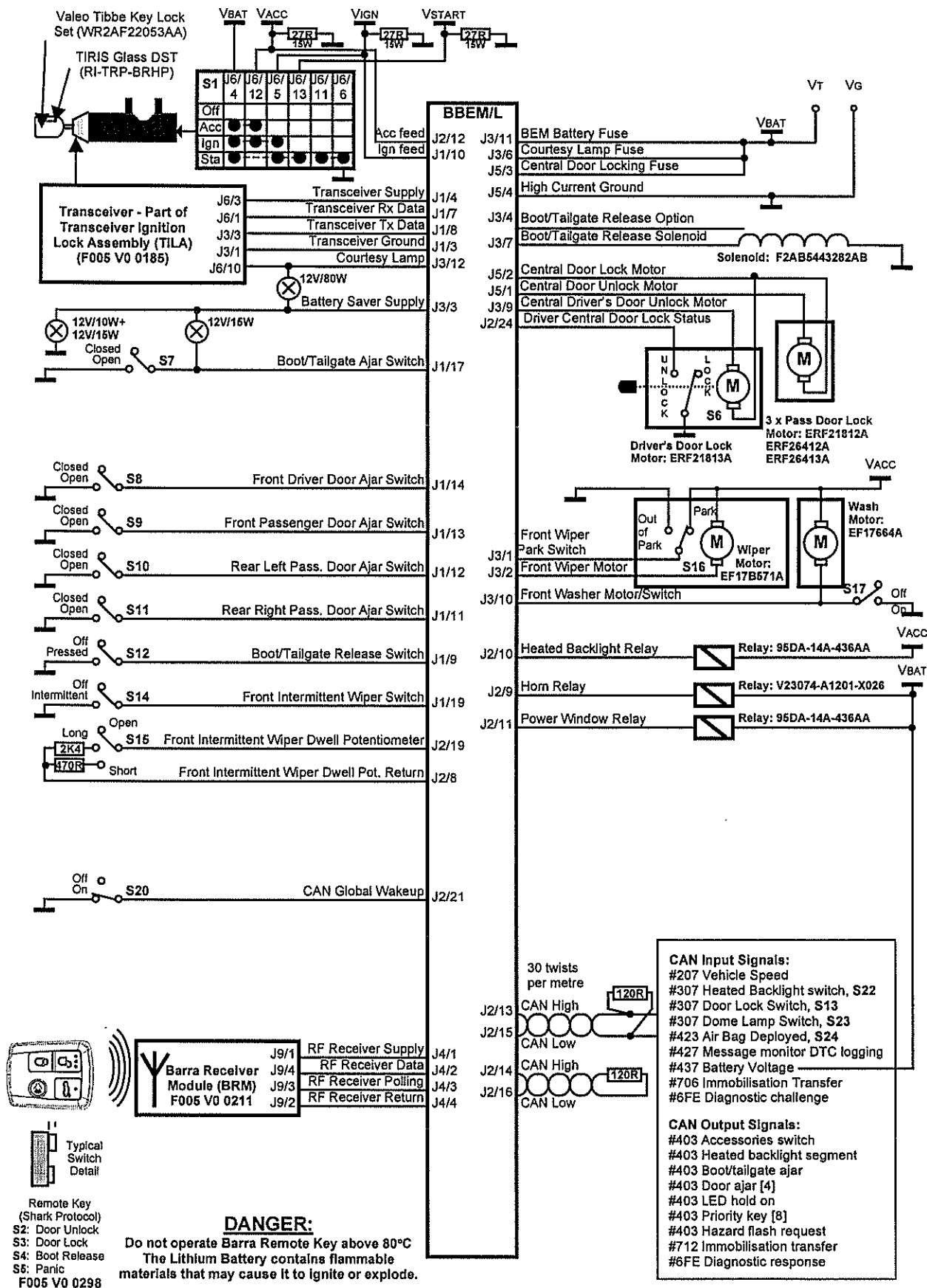


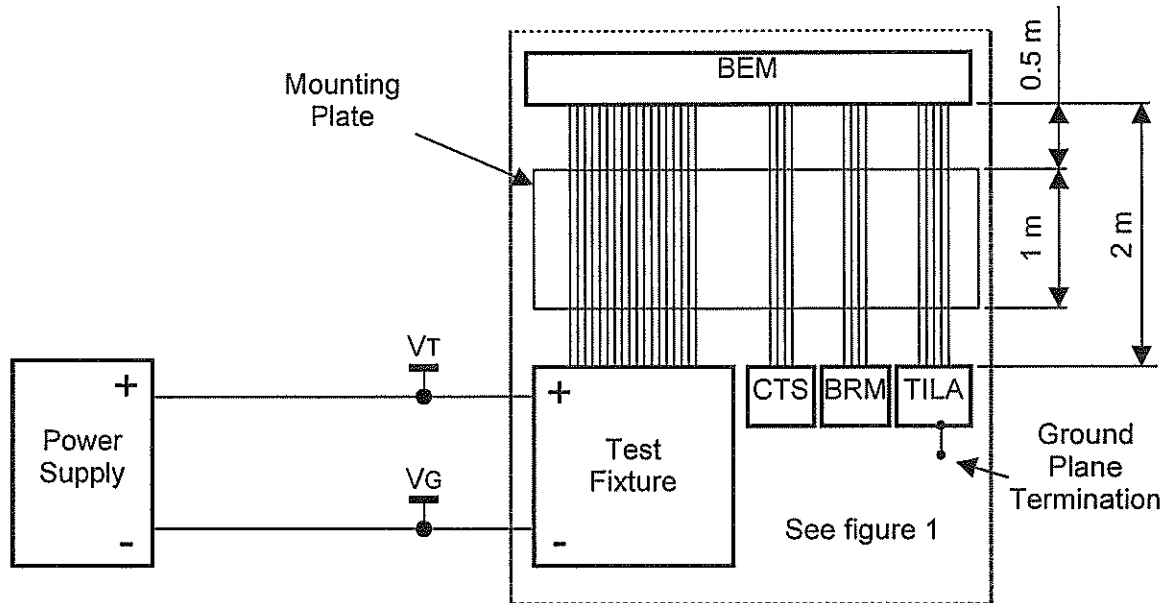
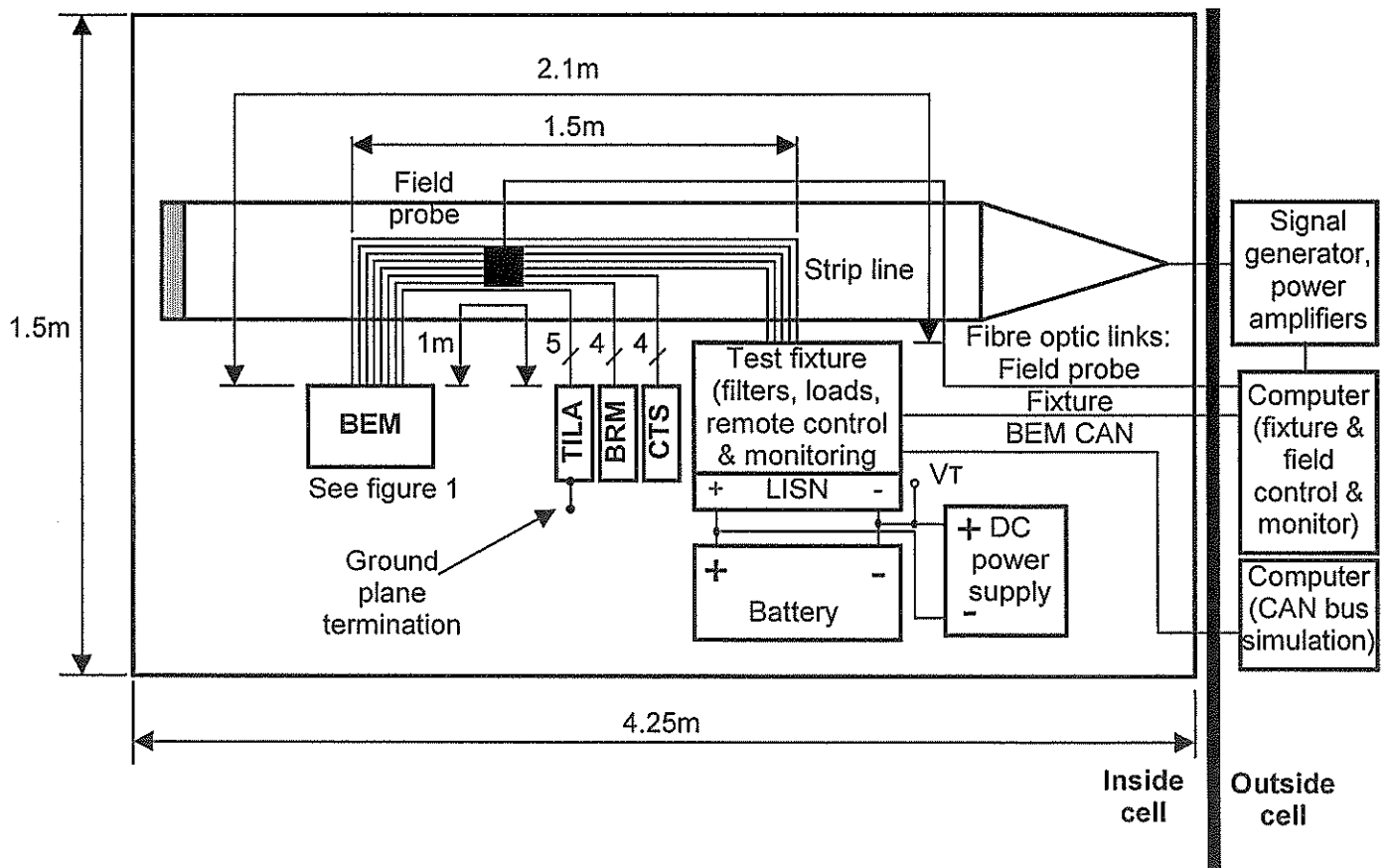
Figure 2E Parallel Wire Miscellaneous Noise Test Additional Information

Figure 2F Stripline Test Set Up Additional Information


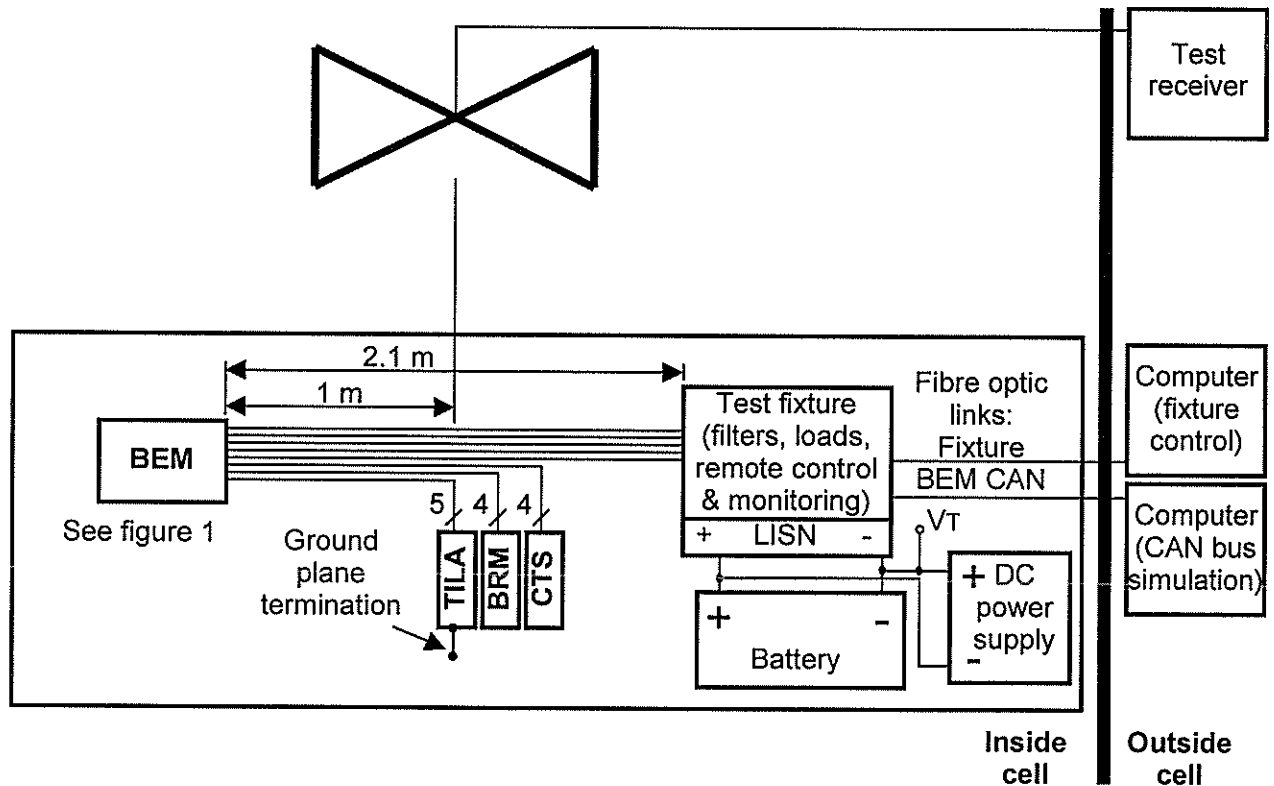
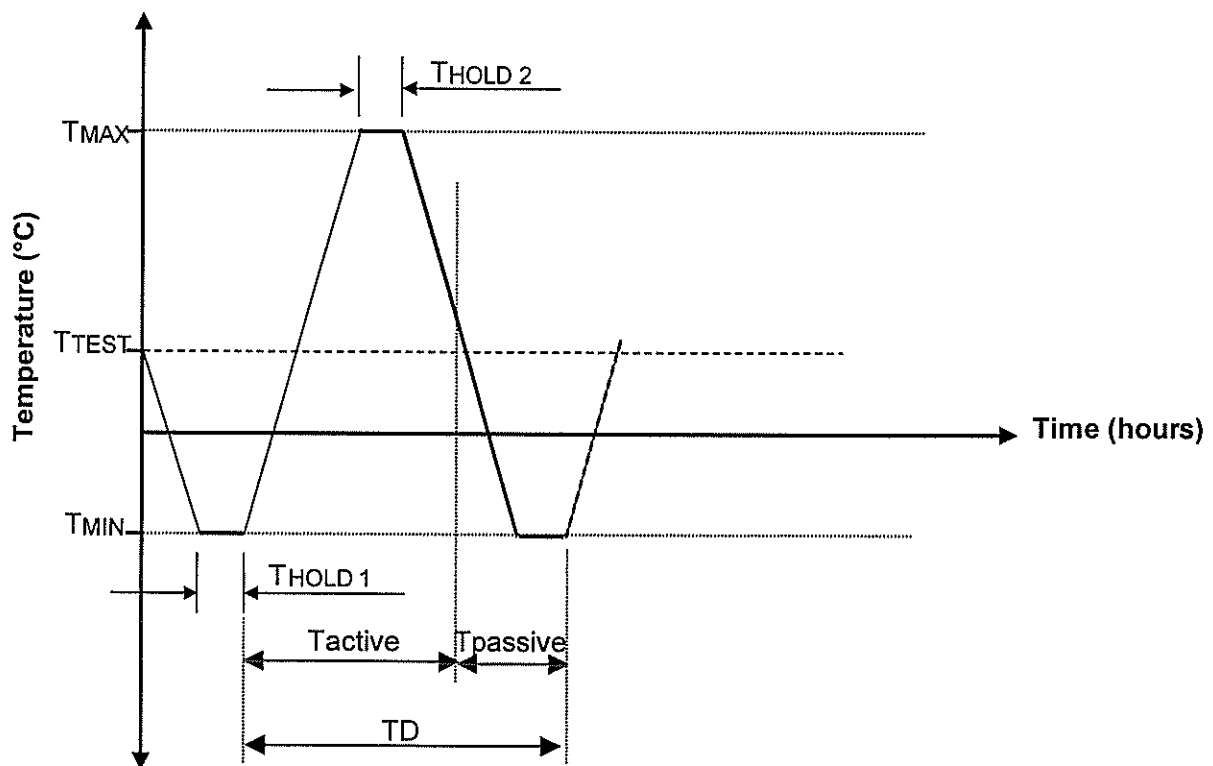
Figure 2G Narrowband Electric Field Test Set Up Additional Information

Figure 3 Powered Thermal Cycle Test Cycle


Figure 4 Temperature Life Test Cycle

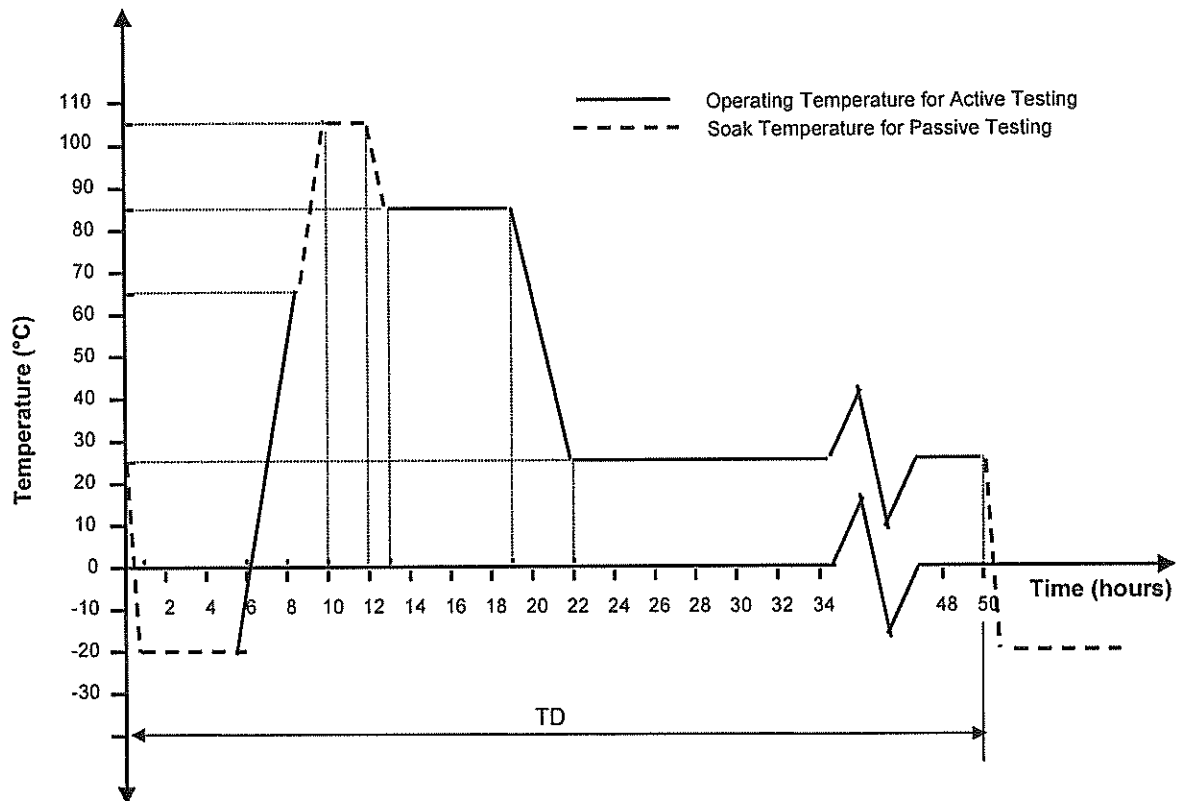


Figure 5A ESD Test CI 280-A Setup Additional Information

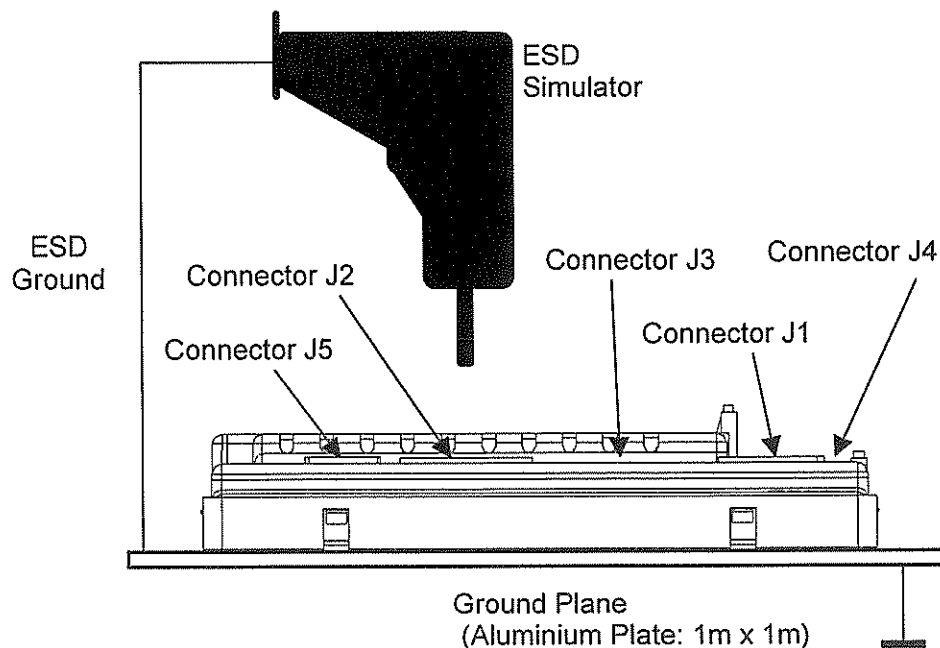


Figure 5B ESD Test CI 280-C, CI 280-D Setup Additional Information

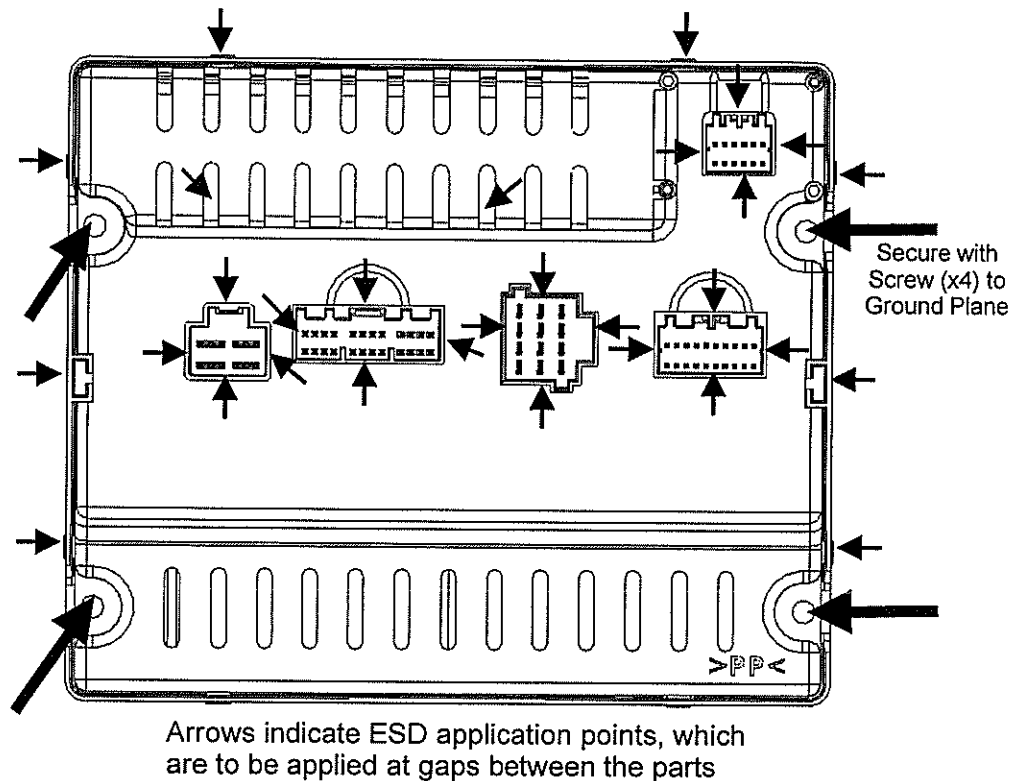


Figure 6 Mounting Position

